

Сетевые технологии

Лабораторная работа №8

Тойчубекова Асель Нурлановна

2025-12-21

Содержание I

1. Информация
2. Цель работы
3. Задание
4. Теоретическое введение
5. Выполнение лабораторной работы
6. Выводы

Раздел 1

1. Информация

1.1 Докладчик

► Тойчубекова Асель Нурлановна

1.1 Докладчик

- ▶ Тойчубекова Асель Нурлановна
- ▶ Студент 3 курса

1.1 Докладчик

- ▶ Тойчубекова Асель Нурлановна
- ▶ Студент 3 курса
- ▶ факультет физико-математических и естественных наук

1.1 Докладчик

- ▶ Тойчубекова Асель Нурлановна
- ▶ Студент 3 курса
- ▶ факультет физико-математических и естественных наук
- ▶ Российский университет дружбы народов им. П. Лумумбы

1.1 Докладчик

- ▶ Тойчубекова Асель Нурлановна
- ▶ Студент 3 курса
- ▶ факультет физико-математических и естественных наук
- ▶ Российский университет дружбы народов им. П. Лумумбы
- ▶ 1032235033@rudn.ru

Раздел 2

2. Цель работы

2.1 Цель работы

Целью данной лабораторной работы является изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.

Раздел 3

3. Задание

3.1 Задание

1. Требуется настроить динамическую маршрутизацию по протоколам RIP OSPF.

3.1 Задание

1. Требуется настроить динамическую маршрутизацию по протоколам RIP OSPF.
2. Требуется организовать туннель IPv6 поверх IPv4, позволяющий передавать данные из одной IPv6-сети в другую IPv6-сеть через сеть IPv4.

3.1 Задание

1. Требуется настроить динамическую маршрутизацию по протоколам RIP OSPF.
2. Требуется организовать туннель IPv6 поверх IPv4, позволяющий передавать данные из одной IPv6-сети в другую IPv6-сеть через сеть IPv4.
3. Выполнить задание для самостоятельного выполнения.

Раздел 4

4. Теоретическое введение

4.1 Теоретическое введение

При организации сетей передачи данных важную роль играет корректное понимание функций и различий между устройствами DTE (Data Terminal Equipment) и DCE (Data Circuit-terminating Equipment), а также принципов синхронизации и правил планирования сетевой инфраструктуры.

Оконечное оборудование данных (DTE) представляет собой устройства, обеспечивающие интерфейс с пользователем и выполняющие прикладные программы. К ним относятся персональные компьютеры, терминалы, сетевые узлы, которые формируют и обрабатывают информационные потоки. Оконечное оборудование канала данных (DCE) служит для подключения DTE к каналам связи, обеспечивая преобразование, усиление и передачу сигналов. В современных сетях маршрутизаторы с последовательными интерфейсами могут выполнять функции как DTE, так и DCE, поэтому корректнее говорить о типе интерфейса, а не устройства.

4.2 Теоретическое введение

Важным аспектом является организация сигнальных цепей. Для интерфейсов типа DTE передающий сигнал TxD является выходным, а принимающий RxD — входным; для DCE — наоборот. Это определяет особенности соединения устройств: при связке DTE–DCE сигналы соединяются напрямую, а при соединениях однотипных устройств — перекрёстно.

Режимы передачи могут быть асинхронными и синхронными. В асинхронном режиме синхронизация обеспечивается старт-битами, тогда как в синхронном используется внешняя синхронизация с отдельными сигналами для передачи и приёма данных (ST и RT). Источником синхронизации обычно выступает устройство DCE, однако возможны различные варианты организации синхросигналов, включая использование сигнала TT и компенсацию фазовых перекосов.

4.3 Теоретическое введение

Наряду с вопросами интерфейсов и синхронизации ключевым этапом построения сетевой инфраструктуры является её планирование и документирование.

Проектирование сети предполагает разработку схем трёх уровней модели OSI: физического (L1), канального (L2) и сетевого (L3). Необходимо определить топологию подключения, структуру VLAN, схему адресации и план назначения интерфейсов. Чёткое и детализированное документирование обеспечивает корректную эксплуатацию сети, упрощает администрирование и позволяет эффективно решать задачи модернизации и диагностики.

Раздел 5

5. Выполнение лабораторной работы

У нас задана сеть, адреса сегментов сети, адреса, назначаемые интерфейсам устройств.



5.2 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Таблица адресов сетей

Устройства	Сеть IPv4	Сеть IPv6
PC1 – gw-01	10.0.10.0/24	2001:10::/64
PC2 – gw-03	10.0.11.0/24	2001:11::/64
gw-01 – gw-02	10.0.1.0/24	2001:1::/64
gw-02 – gw-03	10.0.2.0/24	2001:2::/64
gw-03 – gw-04	10.0.3.0/24	2001:3::/64
gw-04 – gw-01	10.0.4.0/24	2001:4::/64

Рисунок 2: Таблица адресов сетей

5.3 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Таблица адресации

Устрой-ство	Интерфейс	Адрес IP/префикс	Шлюз по умолчанию	Следующее устройство
gw-01	eth0	10.0.10.1/24	n/a	PC1
gw-01	eth0	2001:10::1/64	n/a	PC1
gw-01	eth1	10.0.1.1/24	n/a	gw-02
gw-01	eth1	2001:1::1/64	n/a	gw-02
gw-01	eth2	10.0.4.2/24	n/a	gw-04
gw-01	eth2	2001:4::2/64	n/a	gw-04
gw-02	eth0	10.0.1.2/24	n/a	gw-01
gw-02	eth0	2001:1::2/64	n/a	gw-01
gw-02	eth1	10.0.2.1/24	n/a	gw-03
gw-02	eth1	2001:2::1/64	n/a	gw-03
gw-03	eth0	10.0.11.1/24	n/a	PC2
gw-03	eth0	2001:11::1/64	n/a	PC2
gw-03	eth1	10.0.2.2/24	n/a	gw-02
gw-03	eth1	2001:2::2/64	n/a	gw-02
gw-03	eth2	10.0.3.1/24	n/a	gw-04
gw-03	eth2	2001:3::1/64	n/a	gw-04
gw-04	eth0	10.0.3.2/24	n/a	gw-03
gw-04	eth0	2001:3::2/64	n/a	gw-03
gw-04	eth1	10.0.4.1/24	n/a	gw-01
gw-04	eth1	2001:4::1/64	n/a	gw-01
PC1	NIC	10.0.10.10/24	10.0.10.1	gw-01
PC1	NIC	2001:10::a/64	n/a	gw-01
PC2	NIC	10.0.11.10/24	10.0.11.1	gw-03
PC2	NIC	2001:11::a/64	n/a	gw-03

5.4 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Изобразим схематично моделируемую сеть. Построим схемы L1 и L3.

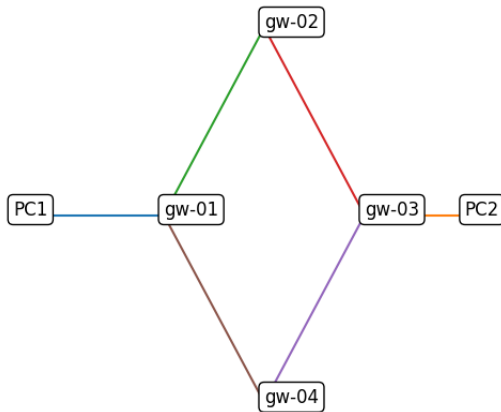


Рисунок 4: Схема L1

5.5 Настройка динамической маршрутизации в сетях IPv4 и IPv6

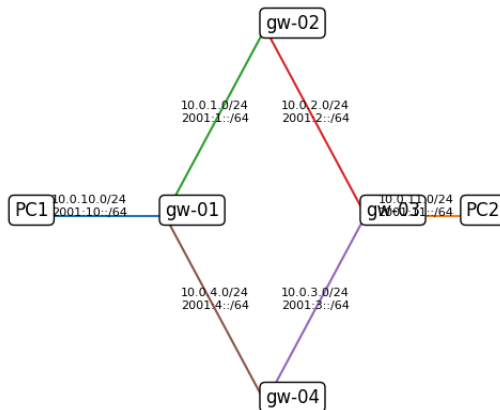


Рисунок 5: Схема L3

5.6 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Запустим GNS3 VM и GNS3. Создадим новый проект. В рабочем пространстве разместим и соединим устройства в соответствии с топологией, указанной в лабораторной работе. Используем маршрутизаторы FRR. Измените отображаемые названия устройств. Включим захват трафика на соединении между коммутатором sw-01 и маршрутизатором gw-01, а также между коммутатором sw-02 и маршрутизатором gw-03.

5.7 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Присвоим IPv4-адреса оконечным устройствам PC1 и PC2

```
PC1> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
PC1 : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ip

NAME           : PC1[1]
IP/MASK         : 10.0.10.10/24
GATEWAY         : 10.0.10.1
DNS             :
MAC             : 00:50:79:66:68:00
LPORT          : 20016
RHOST:PORT      : 127.0.0.1:20017
MTU             : 1500
```

Рисунок 6: IPv4 в PC1

5.8 Настройка динамической маршрутизации в сетях IPv4 и IPv6

```
PC2> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
PC2 : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> show ip

NAME       : PC2[1]
IP/MASK     : 10.0.11.10/24
GATEWAY     : 10.0.11.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 20018
RHOST:PORT  : 127.0.0.1:20019
MTU         : 1500
```

Рисунок 7: IPv4 в PC2

5.9 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Настроим IPv4-адреса на интерфейсах маршрутизаторов.

msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01 - PuTTY
frr# configure terminal
frr(config)# hostname msk-antoychubekova-gw-01
msk-antoychubekova-gw-01(config)# interface eth0
msk-antoychubekova-gw-01(config-if)# ip address 10.0.10.1/24
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth1
msk-antoychubekova-gw-01(config-if)# ip address 10.0.1.1/24
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth2
msk-antoychubekova-gw-01(config-if)# ip address 10.0.4.2/24
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
% Unknown command: write memory
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-antoychubekova-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
 exit
```

5.10 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02 - PuTTY
frr(config)# hostname msk-antoychubekova-gw-02
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# ip address 10.0.1.2/24
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth1
msk-antoychubekova-gw-02(config-if)# ip address 10.0.2.1/24
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-antoychubekova-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
exit
!
interface eth1
```

5.11 Настройка динамической маршрутизации в сетях IPv4 и IPv6

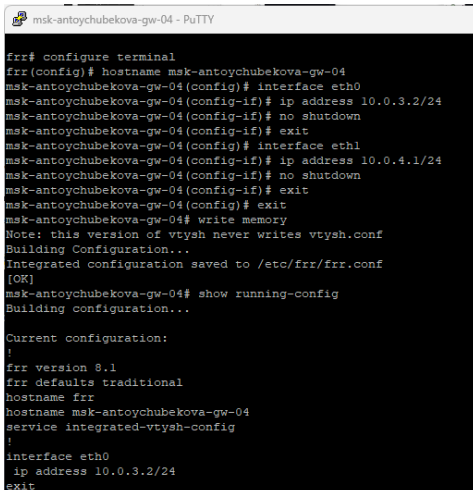
msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03 - PuTTY
frr# configure terminal
frr(config)# hostname msk-antoychubekova-gw-03
msk-antoychubekova-gw-03(config)# interface eth0
msk-antoychubekova-gw-03(config-if)# ip address 10.0.11.1/24
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth1
msk-antoychubekova-gw-03(config-if)# ip address 10.0.2.2/24
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth2
msk-antoychubekova-gw-03(config-if)# ip address 10.0.3.1/24
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-antoychubekova-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
```

5.12 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-04.



```
msk-antoychubekova-gw-04 - PuTTY

frr# configure terminal
frr(config)# hostname msk-antoychubekova-gw-04
msk-antoychubekova-gw-04(config)# interface eth0
msk-antoychubekova-gw-04(config-if)# ip address 10.0.3.2/24
msk-antoychubekova-gw-04(config-if)# no shutdown
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# interface eth1
msk-antoychubekova-gw-04(config-if)# ip address 10.0.4.1/24
msk-antoychubekova-gw-04(config-if)# no shutdown
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-antoychubekova-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 exit
```

5.13 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Присвоим IPv6-адреса оконечным устройствам PC1 и PC2.

```
PC1> ip 2001:10::a/64
PC1 : 2001:10::a/64

PC1> save
Saving startup configuration to startup.vpc
. done

PC1> show ipv6

NAME                : PC1[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 2001:10::a/64
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:00
LPORT                : 20016
RHOST:PORT           : 127.0.0.1:20017
MTU                  : 1500

PC1> 
```

Рисунок 12: IPv6 в PC1

5.14 Настройка динамической маршрутизации в сетях IPv4 и IPv6

```
PC2> ip 2001:11::a/64
PC1 : 2001:11::a/64

PC2> save
Saving startup configuration to startup.vpc
. done

PC2> show ipv6

NAME                : PC2[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                  :
ROUTER LINK-LAYER   :
MAC                  : 00:50:79:66:68:01
LPORT                : 20018
RHOST:PORT           : 127.0.0.1:20019
MTU                  : 1500

PC2> 
```

Рисунок 13: IPv6 в PC2

5.15 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Настроим IPv6-адреса на интерфейсах маршрутизаторов.

msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# configure terminal
msk-antoychubekova-gw-01(config)# ipv6 forwarding
msk-antoychubekova-gw-01(config)# interface eth0
msk-antoychubekova-gw-01(config-if)# ipv6 address 2001:10::1/64
msk-antoychubekova-gw-01(config-if)# no ipv6 nd suppress-ra
msk-antoychubekova-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth1
msk-antoychubekova-gw-01(config-if)# ipv6 address 2001:1::1/64
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth2
msk-antoychubekova-gw-01(config-if)# ipv6 address 2001:4::2/64
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-antoychubekova-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
 ipv6 address 2001:10::1/64
 ipv6 nd prefix 2001:10::/64
 no ipv6 nd suppress-ra
exit
!
```

5.16 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# ipv6 forwarding
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth1
msk-antoychubekova-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-antoychubekova-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
 ipv6 address 2001:1::2/64
exit
!
```

5.17 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# ipv6 forwarding
msk-antoychubekova-gw-03(config)# interface eth0
msk-antoychubekova-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-antoychubekova-gw-03(config-if)# no ipv6 nd suppress-ra
msk-antoychubekova-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth1
msk-antoychubekova-gw-03(config-if)# ipv6 address 2001:2::1/64
msk-antoychubekova-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth2
msk-antoychubekova-gw-03(config-if)# ipv6 address 2001:3::1/64
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
```

Рисунок 16: Настройка IPv6 в msk-antoychubekova-gw-03

5.18 Настройка динамической маршрутизации в сетях IPv4 и IPv6

```
msk-antoychubekova-gw-03# show running-config
Building configuration...
```

```
Current configuration:
```

```
!
```

```
frr version 8.1
```

```
frr defaults traditional
```

```
hostname frr
```

```
hostname msk-antoychubekova-gw-03
```

```
service integrated-vtysh-config
```

```
!
```

```
interface eth0
```

```
ip address 10.0.11.1/24
```

```
ipv6 address 2001:11::1/64
```

```
ipv6 nd prefix 2001:11::/64
```

```
no ipv6 nd suppress-ra
```

```
exit
```

```
!
```

```
interface eth1
```

```
ip address 10.0.2.2/24
```

```
ipv6 address 2001:2::2/64
```

```
exit
```

```
!
```

```
interface eth2
```

```
ip address 10.0.3.1/24
```

```
ipv6 address 2001:3::1/64
```

```
exit
```

```
!
```

5.19 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# ipv6 forwarding
msk-antoychubekova-gw-04(config)# interface eth0
msk-antoychubekova-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-antoychubekova-gw-04(config-if)# no shutdown
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# interface eth1
msk-antoychubekova-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-antoychubekova-gw-04(config-if)# no shutdown
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.1
frr defaults traditional
hostname frr
hostname msk-antoychubekova-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
exit
!
```

5.20 Настройка динамической маршрутизации в сетях IPv4 и IPv6

RIP — это дистанционно-векторный протокол динамической маршрутизации, который использует в качестве метрики количество переходов (hops). Максимальная длина маршрута — 15 переходов, при 16 маршрут считается недоступным. По умолчанию маршруты обновляются каждые 30 секунд; если обновления не поступают 180 секунд — маршрут становится неактивным, а через 240 секунд удаляется.

RIPv1 — поддерживает только IPv4, не передает маску сети (классовая маршрутизация). RIPv2 — улучшенная версия для IPv4, использует многоадресную рассылку 224.0.0.9 и передает маску (CIDR). RIPv6 — версия для IPv6, использует multicast-адрес FF02::9. Маршрутизация включается на уровне интерфейсов, каждый RIPv6-процесс имеет локальное имя и может быть несколько процессов на одном маршрутизаторе.

5.21 Настройка динамической маршрутизации в сетях IPv4 и IPv6

На маршрутизаторах настроим RIP в качестве протокола динамической маршрутизации.

msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# configure terminal
msk-antoychubekova-gw-01(config)# router rip
msk-antoychubekova-gw-01(config-router)# version 2
msk-antoychubekova-gw-01(config-router)# network eth0
msk-antoychubekova-gw-01(config-router)# network eth1
msk-antoychubekova-gw-01(config-router)# network eth2
msk-antoychubekova-gw-01(config-router)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 19: Настройка RIP в msk-antoychubekova-gw-01

5.22 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# router rip
msk-antoychubekova-gw-02(config-router)# version 2
msk-antoychubekova-gw-02(config-router)# network eth0
msk-antoychubekova-gw-02(config-router)# network eth1
msk-antoychubekova-gw-02(config-router)# network eth2
msk-antoychubekova-gw-02(config-router)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 20: Настройка RIP в msk-antoychubekova-gw-02

5.23 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# router rip
msk-antoychubekova-gw-03(config-router)# version 2
msk-antoychubekova-gw-03(config-router)# network eth0
msk-antoychubekova-gw-03(config-router)# network eth1
msk-antoychubekova-gw-03(config-router)# network eth2
msk-antoychubekova-gw-03(config-router)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 21: Настройка RIP в msk-antoychubekova-gw-03

5.24 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# router rip
msk-antoychubekova-gw-04(config-router)# version 2
msk-antoychubekova-gw-04(config-router)# network eth0
msk-antoychubekova-gw-04(config-router)# network eth1
msk-antoychubekova-gw-04(config-router)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 22: Настройка IPv6 в msk-antoychubekova-gw-04

5.25 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Убедимся, что маршрутизация по RIP настроена.

msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/2] via 10.0.1.2, eth1, weight 1, 00:04:19
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:01:19
R>* 10.0.11.0/24 [120/3] via 10.0.1.2, eth1, weight 1, 00:02:45
msk-antoychubekova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network        Next Hop        Metric From        Tag Time
C(i) 10.0.1.0/24      0.0.0.0          1 self              0
R(n) 10.0.2.0/24      10.0.1.2          2 10.0.1.2            0 02:56
R(n) 10.0.3.0/24      10.0.4.1          2 10.0.4.1            0 02:55
C(i) 10.0.4.0/24      0.0.0.0          1 self              0
C(i) 10.0.10.0/24     0.0.0.0          1 self              0
R(n) 10.0.11.0/24     10.0.1.2          3 10.0.1.2            0 02:56
msk-antoychubekova-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 0 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface        Send Recv  Key-chain
    eth0              2      2
    eth1              2      2
```

5.26 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.3.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:04:24
R>* 10.0.4.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:05:56
R>* 10.0.10.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:05:56
R>* 10.0.11.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:04:20
msk-antoychubekova-gw-02# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network        Next Hop        Metric From        Tag Time
C(i) 10.0.1.0/24      0.0.0.0          1 self              0
C(i) 10.0.2.0/24      0.0.0.0          1 self              0
R(n) 10.0.3.0/24      10.0.2.2         2 10.0.2.2           0 02:32
R(n) 10.0.4.0/24      10.0.1.1         2 10.0.1.1           0 02:31
R(n) 10.0.10.0/24     10.0.1.1         2 10.0.1.1           0 02:31
R(n) 10.0.11.0/24     10.0.2.2         2 10.0.2.2           0 02:32
msk-antoychubekova-gw-02# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 13 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface        Send Recv  Key-chain
    eth0              2      2
    eth1              2      2
  Routing for Networks:
    eth0,
```

5.27 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - FBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.2.1, eth1, weight 1, 00:05:52
R>* 10.0.4.0/24 [120/2] via 10.0.3.2, eth2, weight 1, 00:04:28
R>* 10.0.10.0/24 [120/3] via 10.0.2.1, eth1, weight 1, 00:05:52
msk-antoychubekova-gw-03# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network          Next Hop        Metric From      Tag Time
R(n) 10.0.1.0/24      10.0.2.1         2 10.0.2.1       0 02:37
C(i) 10.0.2.0/24      0.0.0.0          1 self           0
C(i) 10.0.3.0/24      0.0.0.0          1 self           0
R(n) 10.0.4.0/24      10.0.3.2         2 10.0.3.2       0 02:29
R(n) 10.0.10.0/24     10.0.2.1         3 10.0.2.1       0 02:37
C(i) 10.0.11.0/24     0.0.0.0          1 self           0
msk-antoychubekova-gw-03# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 19 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2     2
    eth1           2     2
    eth2           2     2
  Routing for Networks:
    eth0
```

5.28 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:05:21
R>* 10.0.2.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:05:24
R>* 10.0.10.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:05:21
R>* 10.0.11.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:05:24
msk-antoychubekova-gw-04# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
(n) - normal, (s) - static, (d) - default, (r) - redistribute,
(i) - interface

      Network          Next Hop          Metric From          Tag Time
R(n) 10.0.1.0/24       10.0.4.2           2 10.0.4.2           0 02:39
R(n) 10.0.2.0/24       10.0.3.1           2 10.0.3.1           0 02:31
C(i) 10.0.3.0/24       0.0.0.0            1 self              0
C(i) 10.0.4.0/24       0.0.0.0            1 self              0
R(n) 10.0.10.0/24      10.0.4.2           2 10.0.4.2           0 02:39
R(n) 10.0.11.0/24      10.0.3.1           2 10.0.3.1           0 02:31
msk-antoychubekova-gw-04# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 33 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2    2
    eth1           2    2
  Routing for Networks:
```

5.29 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим пути прохождения пакетов. С PC1 пропингуем PC2 и определим путь следования пакетов. Мы видим, что у нас все успешно пингуется.

```
PC1> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=48.637 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=43.257 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=12.921 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=21.304 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=16.118 ms

PC1> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    23.710 ms   3.291 ms   2.873 ms
 2  10.0.1.2     24.448 ms   9.283 ms   5.918 ms
 3  10.0.2.2     23.805 ms  13.906 ms  11.431 ms
 4  10.0.11.10   23.134 ms  19.548 ms  17.666 ms

PC1> █
```

Рисунок 27: Проверка доступа P1 к P2

5.30 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим метрики протокола RIP. В таблице видны как подключённые сети (C), так и изученные по RIP (R) маршруты. Указаны сети 10.0.1.0/24, 10.0.4.0/24 и 10.0.10.0/24 как напрямую подключённые, метрика — 1. Сети 10.0.2.0/24, 10.0.3.0/24 и 10.0.11.0/24 получены по RIP от соседних маршрутизаторов через следующий хоп 10.0.1.2 с метрикой 2–3. В колонке Time показано время с последнего обновления. Это подтверждает, что динамическая маршрутизация RIP настроена корректно и маршрутизатор успешно обменивается маршрутами с соседями.

5.31 Настройка динамической маршрутизации в сетях IPv4 и IPv6

```
msk-antoychubekova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface

   Network        Next Hop        Metric From        Tag Time
C(i) 10.0.1.0/24   0.0.0.0          1 self             0
R(n) 10.0.2.0/24   10.0.1.2         2 10.0.1.2         0 02:35
R(n) 10.0.3.0/24   10.0.4.1         2 10.0.4.1         0 02:47
C(i) 10.0.4.0/24   0.0.0.0          1 self             0
C(i) 10.0.10.0/24  0.0.0.0          1 self             0
R(n) 10.0.11.0/24  10.0.1.2         3 10.0.1.2         0 02:35
msk-antoychubekova-gw-01#
```

Рисунок 28: Метрики протокола RIP

5.32 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Предположим, что пакет проходит через маршрутизатор msk-antoychubekova-gw-02. Отключим на маршрутизаторе msk-antoychubekova-gw-02 интерфейс.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# shutdown
msk-antoychubekova-gw-02(config-if)#
```

Рисунок 29: Отключение интерфейса

5.33 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим метрики протокола RIP. Мы видим, что в предыдущем выводе сети 10.0.2.0/24, 10.0.3.0/24 и 10.0.11.0/24 приходили на gw-01 через 10.0.1.2 (сосед gw-02). В новом выводе эти же маршруты теперь получаются через 10.0.4.1 (сосед gw-04). Метрики (2–3) при этом сохранились, обновилось только время получения маршрутов (таймеры).

```
msk-antoychubekova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface

  Network      Next Hop      Metric From      Tag Time
C(i) 10.0.1.0/24 0.0.0.0        1 self           0
R(n) 10.0.2.0/24 10.0.4.1       3 10.0.4.1       0 02:48
R(n) 10.0.3.0/24 10.0.4.1       2 10.0.4.1       0 02:48
C(i) 10.0.4.0/24 0.0.0.0        1 self           0
C(i) 10.0.10.0/24 0.0.0.0        1 self           0
R(n) 10.0.11.0/24 10.0.4.1       3 10.0.4.1       0 02:48
msk-antoychubekova-gw-01#
```

Рисунок 30: Метрики протокола RIP

5.34 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуйте PC2 и определим путь следования пакетов. Примерное время, за которое восстановится маршрут.

```
PC1> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=63.760 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=13.854 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=11.323 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=15.572 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=19.291 ms

PC1> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1  15.223 ms  24.298 ms  1.886 ms
 2  10.0.4.1   22.870 ms  7.705 ms  13.982 ms
 3  10.0.3.1   28.961 ms  9.473 ms  8.994 ms
 4  10.0.11.10 13.034 ms 12.154 ms 13.540 ms

PC1> █
```

Рисунок 31: PC1 пинг PC2

5.35 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Включим на маршрутизаторе msk-antoychubekova-gw-02 интерфейс.

```
msk-antoychubekova-gw-02(config)# interface eth0  
msk-antoychubekova-gw-02(config-if)# no shutdown  
msk-antoychubekova-gw-02(config-if)#
```

Рисунок 32: Включение интерфейса

5.36 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуйте PC2 и определим путь следования пакетов.

```
PC1> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=18.299 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=16.899 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=17.727 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=13.804 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=19.993 ms

PC1> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1   8.341 ms  3.638 ms  2.918 ms
 2  10.0.4.1    7.570 ms  5.531 ms  9.214 ms
 3  10.0.3.1    20.871 ms 17.276 ms 24.707 ms
 4  10.0.11.10  23.075 ms 54.960 ms 17.125 ms

PC1> █
```

Рисунок 33: PC1 пинг PC2

5.37 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Посмотрим захваченный трафик.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.10.10 (Request)
2	0.995397	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.10.10 (Request)
3	1.995474	Private_66:68:00	Broadcast	ARP	64	Gratuitous ARP for 10.0.10.10 (Request)
4	191.000923	::	ff02::1:6	ICMPv6	130	Multicast Listener Report Message v2
5	191.187155	::	ff02::1:6	ICMPv6	130	Multicast Listener Report Message v2
6	191.198894	::	ff02::1:1:ffb2:0	ICMPv6	86	Neighbor Solicitation for fe80::e7f:f6ff:feb2:0
7	192.278880	Fe80::e7f:f6ff:feb2_	ff02::1:6	ICMPv6	150	Multicast Listener Report Message v2
8	192.355894	Fe80::e7f:f6ff:feb2_	ff02::1:6	ICMPv6	90	Multicast Listener Report Message v2
9	192.712133	Fe80::e7f:f6ff:feb2_	ff02::1:6	ICMPv6	150	Multicast Listener Report Message v2
10	193.587294	Fe80::e7f:f6ff:feb2_	ff02::1:6	ICMPv6	90	Multicast Listener Report Message v2
11	2366.541620	2001::10::a	ff02::1:2	ICMPv6	62	Router Solicitation
12	2645.406046	Fe80::e7f:f6ff:feb2_	ff02::1:6	ICMPv6	170	Multicast Listener Report Message v2
13	2645.825217	::	ff02::1:1:ff00:1	ICMPv6	86	Neighbor Solicitation for 2001::10::1
14	2646.116471	Fe80::e7f:f6ff:feb2_	ff02::1:6	ICMPv6	170	Multicast Listener Report Message v2
15	2673.982476	Fe80::e7f:f6ff:feb2_	ff02::1:1	ICMPv6	110	Router Advertisement from 0c:f7:f6:b2:00:00
16	2674.810545	Fe80::e7f:f6ff:feb2_	ff02::1:1	ICMPv6	110	Router Advertisement from 0c:f7:f6:b2:00:00
17	2674.967407	Fe80::e7f:f6ff:feb2_	ff02::1:1	ICMPv6	110	Router Advertisement from 0c:f7:f6:b2:00:00
18	2675.968275	Fe80::e7f:f6ff:feb2_	ff02::1:1	ICMPv6	110	Router Advertisement from 0c:f7:f6:b2:00:00
19	2676.971775	Fe80::e7f:f6ff:feb2_	ff02::1:1	ICMPv6	110	Router Advertisement from 0c:f7:f6:b2:00:00
20	3280.495135	Fe80::e7f:f6ff:feb2_	ff02::1:1	ICMPv6	110	Router Advertisement from 0c:f7:f6:b2:00:00
21	3682.261295	Fe80::e7f:f6ff:feb2_	ff02::1:1	ICMPv6	110	Router Advertisement from 0c:f7:f6:b2:00:00
22	4024.897798	10.0.10.1	224.0.0.9	RIPv2	66	Request
23	4024.112550	10.0.10.1	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
24	4024.322803	10.0.10.1	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
25	4027.783029	10.0.10.1	224.0.0.9	RIPv2	66	Response
26	4028.757873	10.0.10.1	224.0.0.9	RIPv2	66	Request
27	4028.779888	10.0.10.1	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
28	4028.992017	10.0.10.1	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
29	4034.155529	10.0.10.1	224.0.0.9	RIPv2	66	Response
30	4035.156495	10.0.10.1	224.0.0.9	RIPv2	66	Request
31	4035.181419	10.0.10.1	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
32	4035.711005	10.0.10.1	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
33	4035.855273	10.0.10.1	224.0.0.9	RIPv2	86	Response
34	4064.883314	10.0.10.1	224.0.0.9	RIPv2	86	Response
35	4098.899058	10.0.10.1	224.0.0.9	RIPv2	86	Response
36	4118.155122	10.0.10.1	224.0.0.9	RIPv2	66	Response
37	4133.910095	10.0.10.1	224.0.0.9	RIPv2	106	Response
38	4160.919147	10.0.10.1	224.0.0.9	RIPv2	106	Response
39	4188.923961	10.0.10.1	224.0.0.9	RIPv2	106	Response
40	4207.920900	10.0.10.1	224.0.0.9	RIPv2	66	Response
41	4212.928263	10.0.10.1	224.0.0.9	RIPv2	146	Response
42	4240.934431	10.0.10.1	224.0.0.9	RIPv2	146	Response
43	4264.943098	10.0.10.1	224.0.0.9	RIPv2	146	Response
44	4297.950891	10.0.10.1	224.0.0.9	RIPv2	146	Response
45	4298.350343	10.0.10.1	224.0.0.9	RIPv2	86	Response
46	4320.953247	10.0.10.1	224.0.0.9	RIPv2	146	Response
47	4346.958625	10.0.10.1	224.0.0.9	RIPv2	146	Response
48	4377.977895	10.0.10.1	224.0.0.9	RIPv2	146	Response
49	4414.991285	10.0.10.1	224.0.0.9	RIPv2	146	Response
50	4450.997952	10.0.10.1	224.0.0.9	RIPv2	146	Response

Рисунок 35: Анализ трафика

5.39 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Протокол маршрутизации: OSPFv2

No.	Time	Source	Destination	Protocol	Length	Info
139	00:01:35:4022	10.0.0.1	224.0.0.9	RIPv2	146	Response
140	00:01:36:2237	10.0.0.1	224.0.0.9	RIPv2	146	Response
141	00:01:36:3051	10.0.0.1	224.0.0.9	RIPv2	146	Response
142	00:01:37:0361	10.0.0.1	224.0.0.9	RIPv2	146	Response
143	00:01:37:1124	10.0.0.1	224.0.0.9	RIPv2	146	Response
144	00:01:38:0361	10.0.0.1	224.0.0.9	RIPv2	146	Response
145	00:01:39:0180	10.0.0.1	224.0.0.9	RIPv2	146	Response
146	00:01:39:0561	10.0.0.1	224.0.0.9	RIPv2	146	Response
147	00:01:40:0001	10.0.0.1	224.0.0.9	RIPv2	146	Response
148	00:01:40:0001	10.0.0.1	224.0.0.9	RIPv2	146	Response
149	00:01:41:0115	Private, 10.0.0.0	Broadcast	ARP	64	Who has 10.0.0.10? I'll tell 10.0.0.10
150	00:01:42:0011	60:79:fb:02:00:00	Private, 10.0.0.0	NDP	60	10.0.0.10 is at 60:79:fb:02:00:00
151	00:01:42:0447	10.0.0.10	10.0.1.1	ICMP	60	Echo (ping) request: id=6046, seq=1256, ttl=64 (reply in 152)
152	00:01:42:0449	10.0.1.1	10.0.1.10	ICMP	60	kuha (ping) reply: id=6046, seq=1256, ttl=64 (request in 151)
153	00:01:43:0004	10.0.0.10	10.0.1.1	ICMP	60	Echo (ping) request: id=6046, seq=1212, ttl=64 (reply in 154)
154	00:01:43:0700	10.0.1.1	10.0.1.10	ICMP	60	kuha (ping) reply: id=6046, seq=1212, ttl=64 (request in 153)
155	00:01:43:0701	10.0.0.10	10.0.1.1	ICMP	60	Echo (ping) request: id=6046, seq=1760, ttl=64 (reply in 155)
156	00:01:43:0801	10.0.1.1	10.0.1.10	ICMP	60	kuha (ping) reply: id=6046, seq=1760, ttl=64 (request in 155)
157	00:01:44:0320	10.0.0.10	10.0.1.1	ICMP	60	Echo (ping) request: id=6046, seq=1304, ttl=64 (reply in 158)
158	00:01:44:0576	10.0.1.1	10.0.1.10	ICMP	60	kuha (ping) reply: id=6046, seq=1304, ttl=64 (request in 157)
159	00:01:44:0576	10.0.0.10	10.0.1.1	ICMP	60	Echo (ping) request: id=6046, seq=1208, ttl=64 (reply in 159)
160	00:01:44:0576	10.0.1.1	10.0.1.10	ICMP	60	kuha (ping) reply: id=6046, seq=1208, ttl=64 (request in 159)
161	00:01:45:0001	60:79:fb:02:00:00	Private, 10.0.0.0	NDP	60	Who has 10.0.0.10? I'll tell 10.0.0.10
162	00:01:45:0348	Private, 10.0.0.0	60:79:fb:02:00:00	ARP	60	10.0.0.10 is at 60:79:fb:02:00:00
163	00:01:45:0377	10.0.0.10	10.0.1.1	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
164	00:01:45:0377	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
165	00:01:45:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
166	00:01:45:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
167	00:01:46:0218	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
168	00:01:46:0218	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
169	00:01:46:0424	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
170	00:01:46:0424	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
171	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
172	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
173	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
174	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
175	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
176	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
177	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
178	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
179	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
180	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
181	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
182	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
183	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
184	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
185	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
186	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
187	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
188	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
189	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
190	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
191	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
192	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
193	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
194	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
195	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
196	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
197	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
198	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)
199	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	74	[TCP Retransmission] 4577 → 4578 [WIN] Seq=1400 Len=105-1408 Tval=176181238 Tsc=0 sds=
200	00:01:46:0576	10.0.0.10	10.0.1.1	ICMP	100	Time-to-live exceeded (Time to live exceeded in transit)

Frame 195: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface, 44 B

Рисунок 36: Анализ трафика

Рисунок 37: Анализ трафика

◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ 🔍 ↺

5.41 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
232	6416.524059	10.0.10.1	224.0.0.9	IGMPv2	60	IGMP Request
238	6416.582256	10.0.10.10	10.0.11.10	ICMP	60	Echo (ping) request
239	6416.595778	10.0.11.10	10.0.10.10	ICMP	60	Echo (ping) reply
240	6417.482337	10.0.10.10	10.0.11.10	ICMP	60	Echo (ping) request
241	6417.437071	10.0.11.10	10.0.10.10	ICMP	60	Echo (ping) reply
242	6418.420531	10.0.10.10	10.0.11.10	ICMP	60	Echo (ping) request
243	6418.470876	10.0.11.10	10.0.10.10	ICMP	60	Echo (ping) reply
244	6419.477089	10.0.10.10	10.0.11.10	ICMP	60	Echo (ping) request
245	6419.469623	10.0.11.10	10.0.10.10	ICMP	60	Echo (ping) reply
246	6419.454859	10.0.10.10	10.0.11.10	ICMP	60	Echo (ping) request
247	6419.457262	10.0.11.10	10.0.10.10	ICMP	60	Echo (ping) reply
248	6422.430467	80.77.76.12:80:80	Private: 80:80:80	ARP	60	80 who has 10.0.10.10? Tell 10.0.10.1
249	6422.447731	Private: 60:60:60	80:77:76.12:80:80	ARP	60	10.0.10.10 is at 60:77:76.12:80:80
250	6422.471252	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
251	6422.490923	10.0.10.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
252	6422.491040	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
253	6422.495140	10.0.10.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
254	6422.496132	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
255	6422.498124	10.0.10.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
256	6422.498454	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
257	6422.498761	10.0.0.0.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
258	6422.498856	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
259	6422.499337	10.0.0.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
260	6422.500480	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
261	6422.502832	10.0.0.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
262	6422.503180	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
263	6422.505752	10.0.0.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
264	6422.506888	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
265	6422.508265	10.0.0.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
266	6422.509231	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
267	6422.509324	10.0.0.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
268	6422.509870	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
269	6422.510104	10.0.10.10	10.0.11.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
270	6422.510599	10.0.0.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
271	6422.510639	10.0.11.10	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
272	6422.510923	10.0.0.1	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
273	6422.510950	10.0.11.10	10.0.10.10	ICMP	60	Time-to-live exceeded (Time to live exceeded in transit)
274	6448.570524	10.0.10.1	224.0.0.9	IGMPv2	140	Response
275	6452.577090	10.0.10.1	224.0.0.9	IGMPv2	140	Response
276	6507.584295	10.0.10.1	224.0.0.9	IGMPv2	140	Response
277	6508.584993	10.0.10.1	224.0.0.9	IGMPv2	140	Response
278	6509.684040	10.0.10.1	224.0.0.9	IGMPv2	140	Response
279	6503.629973	10.0.10.1	224.0.0.9	IGMPv2	140	Response
280	6453.647740	10.0.10.1	224.0.0.9	IGMPv2	140	Response
281	6539.642087	10.0.10.1	224.0.0.9	IGMPv2	140	Response
282	6463.580241	10.0.10.1	224.0.0.9	IGMPv2	140	Response
283	6509.684051	10.0.10.1	224.0.0.9	IGMPv2	140	Response
284	6723.912182	10.0.10.1	224.0.0.9	IGMPv2	140	Response
285	6747.917572	10.0.10.1	224.0.0.9	IGMPv2	140	Response

Рисунок 38: Анализ трафика

5.42 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Private_66:6d:01	Broadcast	ARP	64	Gratuitous ARP for 10.0.11.10 (Request)
2	0.997585	Private_66:6d:01	Broadcast	ARP	64	Gratuitous ARP for 10.0.11.10 (Request)
3	1.999346	Private_66:6d:01	Broadcast	ARP	64	Gratuitous ARP for 10.0.11.10 (Request)
4	494.587839	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
5	494.588079	::	ff02::ff2a:0	ICMPv6	86	Neighbor Solicitation for fe80::e5b:38ff:fe2a::
6	495.369402	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
7	496.818712	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	158	Multicast Listener Report Message v2
8	496.838625	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
9	496.588054	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	158	Multicast Listener Report Message v2
10	496.628531	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
11	1364.173988	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
12	1364.457547	::	ff02::1:ff2a:0	ICMPv6	86	Neighbor Solicitation for fe80::e5b:38ff:fe2a::
13	1365.157305	::	ff02::16	ICMPv6	138	Multicast Listener Report Message v2
14	1365.477239	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	158	Multicast Listener Report Message v2
15	1365.585519	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
16	1366.835317	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	158	Multicast Listener Report Message v2
17	1366.437221	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	90	Multicast Listener Report Message v2
18	2351.217051	2001::1:1	ff02::2	ICMPv6	62	Router Solicitation
19	3828.282115	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	170	Multicast Listener Report Message v2
20	3828.738958	::	ff02::1:ff00:1	ICMPv6	86	Neighbor Solicitation for 2001::1:1
21	3829.855363	fe80::e5b:38ff:fe2a::	ff02::16	ICMPv6	170	Multicast Listener Report Message v2
22	3849.835901	fe80::e5b:38ff:fe2a::	ff02::1	ICMPv6	118	Router Advertisement from fc::5b:38:2a:00:00
23	3849.855566	fe80::e5b:38ff:fe2a::	ff02::1	ICMPv6	118	Router Advertisement from fc::5b:38:2a:00:00
24	3850.819027	fe80::e5b:38ff:fe2a::	ff02::1	ICMPv6	118	Router Advertisement from fc::5b:38:2a:00:00
25	3851.819529	fe80::e5b:38ff:fe2a::	ff02::1	ICMPv6	118	Router Advertisement from fc::5b:38:2a:00:00
26	3852.825228	fe80::e5b:38ff:fe2a::	ff02::1	ICMPv6	118	Router Advertisement from fc::5b:38:2a:00:00
27	3855.574285	fe80::e5b:38ff:fe2a::	ff02::1	ICMPv6	118	Router Advertisement from fc::5b:38:2a:00:00
28	4115.481711	10.0.11.1	224.0.0.9	RIPv2	66	Request
29	4115.893780	10.0.11.1	224.0.0.22	IGMPv3	68	Membership Report / Join group 224.0.0.9 for any sources
30	4115.215941	10.0.11.1	224.0.0.22	IGMPv3	68	Membership Report / Join group 224.0.0.9 for any sources
31	4117.389612	10.0.11.1	224.0.0.9	RIPv2	66	Response
32	4118.574945	10.0.11.1	224.0.0.9	RIPv2	66	Request
33	4118.415715	10.0.11.1	224.0.0.22	IGMPv3	68	Membership Report / Join group 224.0.0.9 for any sources
34	4119.123476	10.0.11.1	224.0.0.22	IGMPv3	68	Membership Report / Join group 224.0.0.9 for any sources
35	4120.582818	10.0.11.1	224.0.0.9	RIPv2	66	Request
36	4120.522336	10.0.11.1	224.0.0.22	IGMPv3	68	Membership Report / Join group 224.0.0.9 for any sources
37	4121.163158	10.0.11.1	224.0.0.22	IGMPv3	68	Membership Report / Join group 224.0.0.9 for any sources
38	4121.394656	10.0.11.1	224.0.0.9	RIPv2	126	Response
39	4125.694562	10.0.11.1	224.0.0.9	RIPv2	146	Response
40	4150.789185	10.0.11.1	224.0.0.9	RIPv2	146	Response
41	4180.717369	10.0.11.1	224.0.0.9	RIPv2	146	Response
42	4281.628042	10.0.11.1	224.0.0.9	RIPv2	66	Response
43	4288.721983	10.0.11.1	224.0.0.9	RIPv2	146	Response
44	4232.729971	10.0.11.1	224.0.0.9	RIPv2	146	Response
45	4258.985865	fe80::e5b:38ff:fe2a::	ff02::1	ICMPv6	118	Router Advertisement from fc::5b:38:2a:00:00
46	4269.737185	10.0.11.1	224.0.0.9	RIPv2	146	Response
47	4299.744399	10.0.11.1	224.0.0.9	RIPv2	146	Response

> Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface --, id 0
 > Ethernet II, Src: Private_66:6d:01 (00:50:79:6d:6d:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 > Address Resolution Protocol (request/gratuitous ARP)

Рисунок 39: Анализ трафика

5.43 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
43	4288.725985	10.0.0.11	224.0.0.9	RIPv2	144	Response
44	4292.725971	10.0.0.11	224.0.0.9	RIPv2	144	Response
45	4298.905863	6000::c50:10ff:fa2a	ff02::1	ICMPv6	118	Router Advertisement from 6c10:10:1a:00:00
46	4298.777148	10.0.0.11	224.0.0.9	RIPv2	144	Response
47	4299.744390	10.0.0.11	224.0.0.9	RIPv2	144	Response
48	4304.740402	10.0.0.11	224.0.0.9	RIPv2	144	Response
49	4304.751201	10.0.0.11	224.0.0.9	RIPv2	144	Response
50	4308.748227	10.0.0.11	224.0.0.9	RIPv2	144	Response
51	4315.740484	10.0.0.11	224.0.0.9	RIPv2	144	Response
52	4444.775709	10.0.0.11	224.0.0.9	RIPv2	144	Response
53	4476.787190	10.0.0.11	224.0.0.9	RIPv2	144	Response
54	4505.812780	10.0.0.11	224.0.0.9	RIPv2	144	Response
55	4531.898035	10.0.0.11	224.0.0.9	RIPv2	144	Response
56	4561.897525	10.0.0.11	224.0.0.9	RIPv2	144	Response
57	4574.888794	6c10:10:1a:00:00	ff02::1	ICMPv6	118	Router Advertisement from 6c10:10:1a:00:00
58	4575.112287	Private, 10.0.0.0/8	6c10:10:1a:00:00	ARP	60	Who has 10.0.11.18? 10.0.11.18
59	4574.321158	10.0.0.10	10.0.11.18	ICMP	96	Echo (ping) request 10.0.11.18, seq=1256, ttl=64 (reply in 60)
60	4574.125959	10.0.0.11	10.0.10.18	ICMP	96	Echo (ping) reply 10.0.11.18, seq=1256, ttl=64 (request in 58)
61	4575.300912	10.0.0.10	10.0.11.18	ICMP	96	Echo (ping) request 10.0.11.18, seq=1712, ttl=64 (reply in 61)
62	4575.300884	10.0.0.11	10.0.10.18	ICMP	96	Echo (ping) reply 10.0.11.18, seq=1712, ttl=64 (request in 61)
63	4575.300285	10.0.0.10	10.0.11.18	ICMP	96	Echo (ping) request 10.0.11.18, seq=3768, ttl=64 (reply in 64)
64	4575.303751	10.0.0.11	10.0.10.18	ICMP	96	Echo (ping) reply 10.0.11.18, seq=3768, ttl=64 (request in 63)
65	4577.410397	10.0.0.10	10.0.11.18	ICMP	96	Echo (ping) request 10.0.11.18, seq=4768, ttl=64 (reply in 65)
66	4577.410886	10.0.0.11	10.0.10.18	ICMP	96	Echo (ping) reply 10.0.11.18, seq=4768, ttl=64 (request in 65)
67	4578.432139	10.0.0.10	10.0.11.18	ICMP	96	Echo (ping) request 10.0.11.18, seq=5768, ttl=64 (reply in 67)
68	4578.434042	10.0.0.11	10.0.10.18	ICMP	96	Echo (ping) reply 10.0.11.18, seq=5768, ttl=64 (request in 67)
69	4588.814145	10.0.0.11	224.0.0.9	RIPv2	144	Response
70	4588.905862	10.0.0.10	10.0.11.18	ICMP	74	ICMP Echo (ping) 10.0.11.18, seq=6768, ttl=64 (request in 70)
71	4588.910778	10.0.0.11	10.0.10.18	ICMP	74	ICMP Echo (ping) 10.0.11.18, seq=6768, ttl=64 (request in 71)
72	4588.911084	10.0.0.10	10.0.11.18	ICMP	74	ICMP Echo (ping) 10.0.11.18, seq=6768, ttl=64 (request in 72)
73	4588.911778	10.0.0.11	10.0.10.18	ICMP	74	ICMP Echo (ping) 10.0.11.18, seq=6768, ttl=64 (request in 73)
74	4588.913115	10.0.0.10	10.0.11.18	ICMP	74	ICMP Echo (ping) 10.0.11.18, seq=6768, ttl=64 (request in 74)
75	4588.914929	10.0.0.11	10.0.10.18	ICMP	74	ICMP Echo (ping) 10.0.11.18, seq=6768, ttl=64 (request in 75)
76	4587.922217	10.0.0.11	224.0.0.9	RIPv2	144	Response
77	4588.906973	10.0.0.11	224.0.0.9	RIPv2	144	Response
78	4588.909864	10.0.0.11	224.0.0.9	RIPv2	144	Response
79	4796.903872	10.0.0.11	224.0.0.9	RIPv2	144	Response
80	4796.917186	10.0.0.11	224.0.0.9	RIPv2	144	Response
81	4792.979620	10.0.0.11	224.0.0.9	RIPv2	144	Response
82	4798.989514	10.0.0.11	224.0.0.9	RIPv2	144	Response
83	4827.992080	10.0.0.11	224.0.0.9	RIPv2	144	Response
84	4868.876747	6000::c50:10ff:fa2a	ff02::1	ICMPv6	118	Router Advertisement from 6c10:10:1a:00:00
85	4862.884780	10.0.0.11	224.0.0.9	RIPv2	144	Response
86	4885.818868	10.0.0.11	224.0.0.9	RIPv2	144	Response
87	4913.839422	10.0.0.11	224.0.0.9	RIPv2	144	Response
88	4943.846285	10.0.0.11	224.0.0.9	RIPv2	144	Response
89	4977.898036	10.0.0.11	224.0.0.9	RIPv2	144	Response

Frame 1: 68 bytes on wire (542 bits), 68 bytes captured (542 bits) on interface 0, 16 B
 Ethernet II, Src: Private, 08:00:00:00:00:00, Dst: Broadcast, ff:ff:ff:ff:ff:ff

Рисунок 40: Анализ трафика

5.44 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
180	5547.188952	18.0.11.1	224.0.0.9	IGMPv2	146	Response
186	5570.126866	18.0.11.1	224.0.0.9	IGMPv2	146	Response
185	5485.284554	18.0.11.1	224.0.0.9	IGMPv2	146	Response
180	5430.218557	18.0.11.1	224.0.0.9	IGMPv2	146	Response
187	5452.224739	18.0.11.1	224.0.0.9	IGMPv2	146	Response
188	5462.292055	f80b:ef5b:3eff:fc2b::f802:1	224.0.0.9	IGMPv6	118	Router Advertisement from R10:10:2a:00:00
189	5466.223993	18.0.11.1	224.0.0.9	IGMPv2	146	Response
118	5523.228851	18.0.11.1	224.0.0.9	IGMPv2	146	Response
111	5547.222558	18.0.11.1	224.0.0.9	IGMPv2	146	Response
112	5577.228388	18.0.11.1	224.0.0.9	IGMPv2	146	Response
113	5686.243288	18.0.11.1	224.0.0.9	IGMPv2	146	Response
114	5620.248375	18.0.11.1	224.0.0.9	IGMPv2	146	Response
115	5659.252751	18.0.11.1	224.0.0.9	IGMPv2	146	Response
110	5655.309948	18.0.11.1	224.0.0.9	IGMPv2	146	Response
117	5737.322541	18.0.11.1	224.0.0.9	IGMPv2	146	Response
118	5757.328898	18.0.11.1	224.0.0.9	IGMPv2	146	Response
119	5781.325461	18.0.11.1	224.0.0.9	IGMPv2	146	Response
120	5888.327458	18.0.11.1	224.0.0.9	IGMPv2	146	Response
121	5875.342244	18.0.11.1	224.0.0.9	IGMPv2	146	Response
122	5860.346571	18.0.11.1	224.0.0.9	IGMPv2	146	Response
123	5981.393822	18.0.11.1	224.0.0.9	IGMPv2	146	Response
124	5958.392528	18.0.11.1	224.0.0.9	IGMPv2	146	Response
125	5968.394315	18.0.11.1	224.0.0.9	IGMPv2	146	Response
126	6081.797554	18.0.18.18	18.0.11.18	TCP	60	80 kshu (ping) request 18-0-0-0-0-0, seq=1280, ttl=64 (reply is 129)
127	6081.772295	Private 04:00:00	18.0.11.18	ARP	64	who has 18.0.11.1? Tell 18.0.11.18
128	6081.763228	04:00:00:00:00:00	Private 04:00:00	ARP	60	18.0.11.1 is at 04:00:00:00:00:00
129	6081.769028	18.0.18.18	18.0.18.18	TCP	60	80 kshu (ping) reply 18-0-0-0-0-0, seq=1280, ttl=64 (request is 126)
136	6082.802538	18.0.18.18	18.0.11.18	TCP	60	80 kshu (ping) request 18-0-0-0-0-0, seq=1532, ttl=64 (reply is 131)
135	6080.808884	18.0.18.18	18.0.18.18	TCP	60	80 kshu (ping) reply 18-0-0-0-0-0, seq=1532, ttl=64 (request is 130)
132	6080.795228	18.0.11.1	224.0.0.9	IGMPv2	146	Response
133	6080.821731	18.0.18.18	18.0.11.18	TCP	60	80 kshu (ping) request 18-0-0-0-0-0, seq=1788, ttl=64 (reply is 136)
134	6080.828181	18.0.18.18	18.0.18.18	TCP	60	80 kshu (ping) reply 18-0-0-0-0-0, seq=1788, ttl=64 (request is 133)
135	6080.833557	18.0.18.18	18.0.11.18	TCP	60	80 kshu (ping) request 18-0-0-0-0-0, seq=1824, ttl=64 (reply is 138)
134	6080.834882	18.0.18.18	18.0.18.18	TCP	60	80 kshu (ping) reply 18-0-0-0-0-0, seq=1824, ttl=64 (request is 135)
137	6080.853886	18.0.18.18	18.0.11.18	TCP	60	80 kshu (ping) request 18-0-0-0-0-0, seq=11386, ttl=64 (reply is 143)
138	6080.853857	18.0.18.18	18.0.18.18	TCP	60	80 kshu (ping) reply 18-0-0-0-0-0, seq=11386, ttl=64 (request is 137)
140	6080.792288	04:00:00:00:00:00	Private 04:00:00	ARP	64	who has 18.0.11.18? Tell 18.0.11.1
140	6080.793554	Private 04:00:00	04:00:00:00:00:00	ARP	60	18.0.11.18 is at 04:00:00:00:00:00
142	6082.874634	18.0.18.18	18.0.11.18	TCP	74	45777 -> 45778 [SYN] Seq=45766 Len=0 MSS=65535 Win=0 Len=0 Len=0 Len=0 Len=0
143	6082.874634	18.0.18.18	18.0.18.18	TCP	74	45777 -> 45778 [ACK] Seq=45766 Len=0 Len=0 Len=0 Len=0
143	6082.880817	18.0.11.18	18.0.11.18	TCP	74	TCP Retransmission! 45777 -> 45778 [SYN] Seq=45766 Len=0 MSS=65535 Win=0 Len=0 Len=0 Len=0
145	6082.180483	18.0.18.18	18.0.11.18	TCP	74	TCP Retransmission! 45777 -> 45778 [SYN] Seq=45766 Len=0 MSS=65535 Win=0 Len=0 Len=0 Len=0
146	6082.874634	18.0.11.18	18.0.18.18	TCP	74	TCP Retransmission! TCP Port number reused! 45778 -> 45777 [FIN, ACK] Seq=45807336 Ack=45766 Len=0
147	6080.808884	18.0.11.1	224.0.0.9	IGMPv2	146	Response
148	6084.439857	f80b:ef5b:3eff:fc2b::f802:1	224.0.0.9	IGMPv6	118	Router Advertisement from R10:10:2a:00:00
149	6084.457223	18.0.11.1	224.0.0.9	IGMPv2	146	Response

> Frame 11: 60 bytes on wire (121 bits), 60 bytes captured (122 bits) on interface 0, 0 s
 > Ethernet II, Src: Private 04:00:00 (08-00-27-00-00-00), Dst: Broadcast ff:ff:ff:ff:ff:ff

Рисунок 41: Анализ трафика

5.45 Настройка динамической маршрутизации в сетях IPv4 и IPv6

254	8217.530054	39.8.11.1	224.0.0.9	R2P2	148	Response
255	8218.000028	39.8.11.1	224.0.0.9	R2P2	68	Response
256	8219.330032	39.8.11.1	224.0.0.9	R2P2	148	Response
257	8220.000032	39.8.11.1	224.0.0.9	R2P2	148	Response
258	8386.673878	39.8.11.1	224.0.0.9	R2P2	148	Response
259	8313.423451	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1256, ttl=63 (reply in 182))
260	8313.423664	Priority_66-60-00	Extended	AB0	64	who has 39.8.0.11.1? Ttl 18.8.11.18
261	8313.420845	60:150:38:1a:00:00	Priority_66-60-00	AB0	64	39.8.11.1 is at 60:150:38:1a:00:00
262	8313.431881	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1256, ttl=64 (request in 190))
263	8314.487788	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1312, ttl=63 (reply in 182))
264	8314.490504	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1312, ttl=64 (request in 182))
265	8315.470128	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1768, ttl=63 (reply in 182))
266	8315.470471	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1768, ttl=64 (request in 182))
267	8316.491566	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1824, ttl=63 (reply in 182))
268	8316.490842	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1824, ttl=64 (request in 182))
269	8317.512288	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1128, ttl=63 (reply in 178))
270	8317.512588	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1128, ttl=64 (request in 182))
271	8318.340006	60:150:38:1a:00:00	Priority_66-60-00	AB0	64	who has 39.8.0.11.1? Ttl 18.8.11.18
272	8318.470538	Priority_66-60-00	60:150:38:1a:00:00	AB0	64	39.8.0.11.1 is at 60:150:38:1a:00:00
273	8320.673423	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
274	8320.673423	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
275	8320.688312	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
276	8320.688312	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
277	8320.182128	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
278	8320.182128	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
279	8321.673414	39.8.11.1	224.0.0.9	R2P2	148	Response
280	8320.670721	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1256, ttl=63 (reply in 182))
281	8320.671761	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1256, ttl=64 (request in 190))
282	8386.673878	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1312, ttl=63 (reply in 182))
283	8386.673878	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1312, ttl=64 (request in 182))
284	8381.903348	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1768, ttl=63 (reply in 182))
285	8381.903337	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1768, ttl=64 (request in 182))
286	8382.911478	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1824, ttl=63 (reply in 182))
287	8382.912321	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1824, ttl=64 (request in 182))
288	8383.340006	39.8.10.18	39.8.11.18	T2P0	98	Echo (ping) request (len=64, seq=1128, ttl=63 (reply in 178))
289	8383.911477	39.8.11.18	39.8.10.18	T2P0	98	Echo (ping) reply (len=64, seq=1128, ttl=64 (request in 182))
290	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
291	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
292	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
293	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
294	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
295	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
296	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
297	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
298	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
299	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600
300	8383.911477	39.8.10.18	39.8.11.18	T2P0	74	AB0 > 6400 [SN] Seq=1000 Len=8051600 Ttl=11516000 Seq=1000 Len=8051600

Рисунок 42: Анализ трафика

5.46 Настройка динамической маршрутизации в сетях IPv4 и IPv6

На маршрутизаторах настроим RIPng для сетей IPv6.
msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# configure terminal
msk-antoychubekova-gw-01(config)# router ripng
msk-antoychubekova-gw-01(config-router)# network eth0
msk-antoychubekova-gw-01(config-router)# network eth1
msk-antoychubekova-gw-01(config-router)# network eth2
msk-antoychubekova-gw-01(config-router)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 43: Настройка RIPng в msk-antoychubekova-gw-01

5.47 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# router ripng
msk-antoychubekova-gw-02(config-router)# network eth0
msk-antoychubekova-gw-02(config-router)# network eth1
msk-antoychubekova-gw-02(config-router)# network eth2
msk-antoychubekova-gw-02(config-router)# no network eth2
msk-antoychubekova-gw-02(config-router)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 44: Настройка RIPng в msk-antoychubekova-gw-02

5.48 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# router ripng
msk-antoychubekova-gw-03(config-router)# network eth0
msk-antoychubekova-gw-03(config-router)# network eth1
msk-antoychubekova-gw-03(config-router)# network eth2
msk-antoychubekova-gw-03(config-router)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 45: Настройка RIPng в msk-antoychubekova-gw-03

5.49 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# rouret ripng
% Unknown command: rouret ripng
msk-antoychubekova-gw-04(config)# router ripng
msk-antoychubekova-gw-04(config-router)# network eth0
msk-antoychubekova-gw-04(config-router)# network eth1
msk-antoychubekova-gw-04(config-router)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 46: Настройка RIPng в msk-antoychubekova-gw-04

5.50 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим пути прохождения пакетов. С PC1 пропингуем PC2 и определим путь следования пакетов.

```
PC1> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=39.715 ms
2001:11::a icmp6_seq=2 ttl=58 time=16.511 ms
2001:11::a icmp6_seq=3 ttl=58 time=14.775 ms
2001:11::a icmp6_seq=4 ttl=58 time=19.016 ms
2001:11::a icmp6_seq=5 ttl=58 time=10.283 ms

PC1> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1  14.120 ms  2.307 ms  4.046 ms
 2 2001:1::2   25.093 ms  6.457 ms  7.442 ms
 3 2001:2::2   18.343 ms  11.762 ms 11.149 ms
 4 2001:11::a  14.890 ms  18.829 ms 20.439 ms

PC1> █
```

Рисунок 47: Пингование PC1 к PC2

5.51 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим метрики протокола RIPng. Мы видим, что маршрутизатор имеет несколько подключённых (C) IPv6-сетей (2001:1::/64, 2001:4::/64, 2001:11::/64), а также обученные по RIPng (R) маршруты к удалённым подсетям (2001:2::/64, 2001:3::/64, 2001:6::/64, 2001:7::/64). Для удалённых сетей указаны следующие параметры: — Next Hop — link-local адрес соседнего маршрутизатора; — Via — интерфейс получения маршрута (eth1, eth2); — Metric — стоимость маршрута; — Time — время с момента последнего обновления.

5.52 Настройка динамической маршрутизации в сетях IPv4 и IPv6

```
msk-antoychubekova-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

  Network      Next Hop          Via      Metric Tag Time
C(i) 2001:1::/64      ::                      self       1   0
R(n) 2001:2::/64      fe80::ebd:8dff:fee7:0    eth1       2   0 02:40
R(n) 2001:3::/64      fe80::e36:caff:fea9:1    eth2       2   0 02:32
C(i) 2001:4::/64      ::                      self       1   0
C(i) 2001:10::/64     ::                      self       1   0
R(n) 2001:11::/64     fe80::ebd:8dff:fee7:0    eth1       3   0 02:40
msk-antoychubekova-gw-01#
```

Рисунок 48: Метрики протокола RIPng

5.53 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Предположим, что пакет проходит через маршрутизатор msk-antoychubekova-gw-02. Отключим на маршрутизаторе msk-antoychubekova-gw-02 интерфейс.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# shutdown
msk-antoychubekova-gw-02(config-if)#
```

Рисунок 49: Отключение интерфейса

5.54 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим метрики протокола RIPvng. Мы видим, что по сравнению с предыдущим выводом изменились маршруты, полученные по RIPvng и их параметры:

Для сетей 2001:2::/64 и 2001:3::/64 теперь next-hop идёт через другой соседний маршрутизатор fe80::e36:caff:fea9:1 и интерфейс eth2 (раньше был другой link-local или интерфейс).

Изменились метрики маршрутов (стали 2 и 3), что говорит о том, что путь до них теперь рассчитывается иначе (возможно, изменился топология/доступность соседей).

Время (Time) обновилось — маршруты продолжают регулярно приниматься.

5.55 Настройка динамической маршрутизации в сетях IPv4 и IPv6

```
mek-antoychubekova-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

  Network      Next Hop          Via      Metric Tag Time
C(i) 2001:1::/64
      ::                self      1      0
R(n) 2001:2::/64
      fe80::e36:caff:fea9:1 eth2      3      0 02:55
R(n) 2001:3::/64
      fe80::e36:caff:fea9:1 eth2      2      0 02:55
C(i) 2001:4::/64
      ::                self      1      0
C(i) 2001:10::/64
      ::                self      1      0
R(n) 2001:11::/64
      fe80::ebd:8dff:fee7:0 eth1      3      0 02:55
mek-antoychubekova-gw-01#
```

Рисунок 50: Метрики протокола RIPng

5.56 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуем PC2 и определим путь следования пакетов. Мы видим, что пакеты теперь идут по gw-04. Примерное время, за которое восстанавливается маршрут равно 1 минуте.

```
PC1> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=135.692 ms
2001:11::a icmp6_seq=2 ttl=58 time=14.402 ms
2001:11::a icmp6_seq=3 ttl=58 time=29.057 ms
2001:11::a icmp6_seq=4 ttl=58 time=34.305 ms
2001:11::a icmp6_seq=5 ttl=58 time=16.725 ms

PC1> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1  13.939 ms  3.432 ms  3.391 ms
 2 2001:1::2   16.328 ms  12.132 ms  10.553 ms
 3 2001:2::2   66.320 ms  65.786 ms  21.236 ms
 4 2001:11::a  29.501 ms  71.723 ms  21.625 ms

PC1> 
```

Рисунок 51: Пингование PC1 к PC2

5.57 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Посмотрим захваченный трафик.

No.	Time	Source	Destination	Protocol	Length	Info
350	7591.087961	fe80::e7f:feff:feb2::	ff02::9	RIPng	86	Command Response, Version 1
351	7618.415177	fe80::e7f:feff:feb2::	ff02::9	RIPng	186	Command Response, Version 1
352	7614.186049	10.0.0.0	224.0.0.0	RIPv2	146	Response
353	7624.031440	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) request id=bc5ea, seq=1, hop limit=64 (reply in 356)
354	7624.063559	fe80::e7f:feff:feb2::	ff02::1::ff	ICMPv6	86	Neighbor Solicitation for 2001::1::a from bc:7f:fe:32:00:00
355	7624.954137	2001::1::a	fe80::e7f:feff:feb2::	ICMPv6	86	Neighbor Advertisement 2001::1::a (sol, ovr) is at 00:50:79:00:00:00
356	7624.968403	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) reply id=bc5ea, seq=1, hop limit=58 (request in 355)
357	7625.972734	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) request id=bc5ea, seq=2, hop limit=64 (reply in 358)
358	7625.984224	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) reply id=bc5ea, seq=2, hop limit=58 (request in 357)
359	7626.998803	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) request id=bc5ea, seq=3, hop limit=64 (reply in 360)
360	7627.004669	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) reply id=bc5ea, seq=3, hop limit=58 (request in 359)
361	7628.008549	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) request id=bc5ea, seq=4, hop limit=64 (reply in 362)
362	7628.024100	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) reply id=bc5ea, seq=4, hop limit=58 (request in 361)
363	7629.028157	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) request id=bc5ea, seq=5, hop limit=64 (reply in 364)
364	7629.037637	2001::1::a	2001::1::a	ICMPv6	118	Echo (ping) reply id=bc5ea, seq=5, hop limit=58 (request in 363)
365	7641.071570	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
366	7641.087422	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
367	7641.936421	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
368	7641.948131	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
369	7641.999325	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
370	7641.994204	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
371	7641.995246	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
372	7642.019001	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
373	7642.020504	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
374	7642.026005	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
375	7642.027090	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
376	7642.033900	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
377	7642.040591	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
378	7642.053410	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
379	7642.054296	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
380	7642.064254	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
381	7642.065000	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
382	7642.073106	2001::1::a	2001::1::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
383	7642.080445	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
384	7642.094201	2001::1::a	2001::1::a	ICMPv6	174	Destination unreachable (Port unreachable)
385	7642.095560	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
386	7642.112649	2001::1::a	2001::1::a	ICMPv6	174	Destination unreachable (Port unreachable)
387	7642.115070	2001::1::a	2001::1::a	UDP	126	64049 → 64050 Len=64
388	7642.116049	2001::1::a	2001::1::a	ICMPv6	174	Destination unreachable (Port unreachable)
389	7645.119011	10.0.0.0	224.0.0.0	RIPv2	146	Response
390	7652.426702	fe80::e7f:feff:feb2::	ff02::9	RIPng	186	Command Response, Version 1
391	7670.437809	fe80::e7f:feff:feb2::	ff02::9	RIPng	186	Command Response, Version 1
392	7676.298076	10.0.0.0	224.0.0.0	RIPv2	146	Response
393	7676.299201	10.0.0.0	224.0.0.0	RIPv2	146	Response
394	7705.441711	fe80::e7f:feff:feb2::	ff02::9	RIPng	186	Command Response, Version 1
395	7732.210428	10.0.0.0	224.0.0.0	RIPv2	146	Response
396	7738.401403	fe80::e7f:feff:feb2::	ff02::9	RIPng	186	Command Response, Version 1
397	7757.220453	10.0.0.0	224.0.0.0	RIPv2	146	Response
398	7757.462751	fe80::e7f:feff:feb2::	ff02::9	RIPng	186	Command Response, Version 1
399	7792.239600	10.0.0.0	224.0.0.0	RIPv2	146	Response

Рисунок 52: Анализ трафика

5.58 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
413	7937.895588	10.0.0.0.9	224.0.0.9	RIPv2	60	Response
414	7953.518090	fe80::e7ff:feff:feb2::	ff02::9	RIPng	106	Command Response, Version 1
415	7971.296584	10.0.0.0.1	224.0.0.9	RIPv2	146	Response
416	7974.780814	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b23ec, seq=1, hop limit=64 (reply in 752)
417	7976.780207	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b23ec, seq=1, hop limit=64 (reply in 753)
418	7976.780907	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b23ec, seq=3, hop limit=64 (reply in 754)
419	7980.780534	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b23ec, seq=4, hop limit=64 (reply in 755)
420	7982.780435	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b23ec, seq=5, hop limit=64 (reply in 756)
421	7986.355149	2001:10::a	2001:11::a	UDP	126	8224 → 8225 Len=64
422	7986.371543	2001:10::1	2001:10::1	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
423	7986.371085	2001:10::a	2001:11::a	UDP	126	8224 → 8225 Len=64
424	7986.361952	2001:10::1	2001:10::1	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
425	7986.351443	2001:10::a	2001:11::a	UDP	126	8224 → 8225 Len=64
426	7986.352374	2001:10::1	2001:10::1	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
427	7986.355864	2001:10::a	2001:11::a	UDP	126	8224 → 8225 Len=64
428	7987.387771	2001:10::a	2001:11::a	UDP	126	8224 → 8225 Len=64
429	7986.386297	2001:10::a	2001:11::a	UDP	126	8224 → 8225 Len=64
430	7986.389763	2001:10::a	2001:11::a	UDP	126	8224 → 8225 Len=64
431	7986.518063	2001:10::1	2001:10::1	ICMPv6	174	Destination Unreachable (Address unreachable)
432	7986.518783	2001:10::1	2001:10::1	ICMPv6	174	Destination Unreachable (Address unreachable)
433	7989.517092	2001:10::1	2001:10::1	ICMPv6	174	Destination Unreachable (Address unreachable)
434	7989.518322	2001:10::1	2001:10::1	ICMPv6	174	Destination Unreachable (Address unreachable)
435	7991.412362	fe80::e7ff:feff:feb2::	2001:10::1	ICMPv6	86	Neighbor Solicitation for 2001:10::a from bc:7f:f6:b2:00:00
436	7992.451442	fe80::e7ff:feff:feb2::	2001:10::1	ICMPv6	86	Neighbor Solicitation for 2001:10::a from bc:7f:f6:b2:00:00
437	7993.491813	fe80::e7ff:feff:feb2::	2001:10::1	ICMPv6	86	Neighbor Solicitation for 2001:10::a from bc:7f:f6:b2:00:00
438	7997.518423	fe80::e7ff:feff:feb2::	ff02::9	RIPng	106	Command Response, Version 1
439	7986.474595	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b3cae, seq=1, hop limit=64 (no response found)
440	7999.476797	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b3cae, seq=2, hop limit=64 (no response found)
441	8001.477418	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b3cae, seq=3, hop limit=64 (no response found)
442	8001.573177	2001:10::1	ff02::1:ff00::a	ICMPv6	86	Neighbor Solicitation for 2001:10::a from bc:7f:f6:b2:00:00
443	8001.573853	2001:10::1	2001:10::1	ICMPv6	86	Neighbor advertisement 2001:10::a (sol, over) is at 00:50:79:66:68:00
444	8006.518063	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
445	8001.581749	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
446	8001.581761	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
447	8002.582798	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b3cae, seq=4, hop limit=64 (no response found)
448	8001.589351	10.0.0.0.9	224.0.0.9	RIPv2	146	Response
449	8005.589623	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b3cae, seq=5, hop limit=64 (no response found)
450	8005.732208	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
451	8006.732699	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
452	8024.518063	fe80::e7ff:feff:feb2::	ff02::9	RIPng	106	Command Response, Version 1
453	8026.580757	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b58ec, seq=1, hop limit=64 (no response found)
454	8030.580902	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b58ec, seq=2, hop limit=64 (no response found)
455	8033.581749	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
456	8031.571754	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
457	8032.582134	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b58ec, seq=3, hop limit=64 (no response found)
458	8033.584895	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b58ec, seq=4, hop limit=64 (no response found)
459	8034.585200	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=8b58ec, seq=5, hop limit=64 (no response found)
460	8035.571148	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
461	8035.572529	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)
462	8035.572586	2001:10::1	2001:10::1	ICMPv6	166	Destination Unreachable (Address unreachable)

Рисунок 53: Анализ трафика

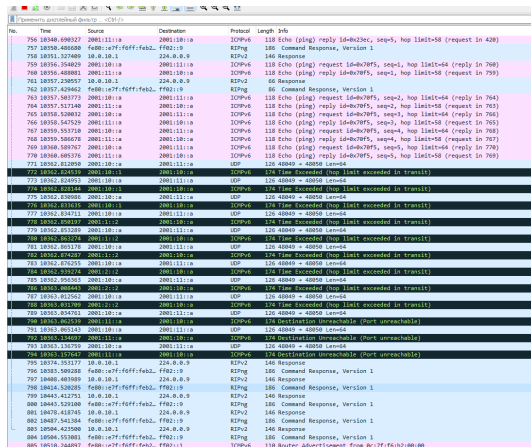
5.59 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
632	9048.644280	10.0.10.1	224.0.0.9	RIPv2	146	Response
633	9071.050207	10.0.10.1	224.0.0.9	RIPv2	146	Response
634	9076.940034	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
635	9106.055708	10.0.10.1	224.0.0.9	RIPv2	146	Response
636	9118.953685	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
637	9136.065793	10.0.10.1	224.0.0.9	RIPv2	146	Response
638	9144.959001	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
639	9165.964908	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
640	9173.075858	10.0.10.1	224.0.0.9	RIPv2	146	Response
641	9194.971795	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
642	9206.082946	10.0.10.1	224.0.0.9	RIPv2	146	Response
643	9228.094691	10.0.10.1	224.0.0.9	RIPv2	146	Response
644	9234.977218	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
645	9258.089793	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
646	9262.784342	10.0.10.1	224.0.0.9	RIPv2	146	Response
647	9279.954997	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
648	9285.712521	10.0.10.1	224.0.0.9	RIPv2	146	Response
649	9304.140571	fe80::e7f:fofff:feb2::ff02::11	ff02::11	ICMPv6	110	Router Advertisement from 0c::7f:fb:b2::00:00
650	9312.720371	10.0.10.1	224.0.0.9	RIPv2	146	Response
651	9316.001284	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
652	9342.466205	2001::1::1a	2001::1::1a	ICMPv6	110	Echo (ping) request id=0x7af1, seq=1, hop limit=64 (no response found!)
653	9342.724054	10.0.10.1	224.0.0.9	RIPv2	146	Response
654	9343.470310	2001::1::1a	2001::1::1a	ICMPv6	110	Echo (ping) request id=0x7af1, seq=2, hop limit=64 (no response found!)
655	9345.470952	2001::1::1a	2001::1::1a	ICMPv6	110	Echo (ping) request id=0x7af1, seq=3, hop limit=64 (no response found!)
656	9345.575804	2001::1::11	ff02::1:ff00::a	ICMPv6	86	Neighbor Solicitation for 2001::1::1a from 0c::7f:fb:b2::00:00
657	9345.576688	2001::1::1a	2001::1::11	ICMPv6	88	Neighbor Advertisement 2001::1::1a (sol, adv) is at 00:50:79:66:68:00
658	9346.574004	2001::1::11	2001::1::1a	ICMPv6	166	Destination Unreachable (Address unreachable)
659	9345.587053	2001::1::11	2001::1::1a	ICMPv6	166	Destination Unreachable (Address unreachable)
660	9345.587058	2001::1::11	2001::1::1a	ICMPv6	166	Destination Unreachable (Address unreachable)
661	9346.590924	2001::1::1a	2001::1::11	ICMPv6	110	Echo (ping) request id=0x7af1, seq=4, hop limit=64 (no response found!)
662	9347.592013	2001::1::1a	2001::1::11	ICMPv6	110	Echo (ping) request id=0x7af1, seq=5, hop limit=64 (no response found!)
663	9348.023047	2001::1::1a	2001::1::11	UDP	120	11930 → 11937 Len=64
664	9348.025736	2001::1::11	2001::1::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
665	9348.027089	2001::1::1a	2001::1::11	UDP	120	11930 → 11937 Len=64
666	9348.027091	2001::1::1a	2001::1::11	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
667	9348.029730	2001::1::1a	2001::1::11	UDP	120	11930 → 11937 Len=64
668	9348.031216	2001::1::11	2001::1::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
669	9348.033625	2001::1::1a	2001::1::11	UDP	120	11930 → 11937 Len=64
670	9348.072121	2001::1::11	2001::1::1a	ICMPv6	166	Destination Unreachable (Address unreachable)
671	9348.750047	2001::1::11	2001::1::1a	ICMPv6	166	Destination Unreachable (Address unreachable)
672	9349.730138	2001::1::11	2001::1::1a	ICMPv6	174	Destination Unreachable (Address unreachable)
673	9308.054494	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
674	9377.7120013	10.0.10.1	224.0.0.9	RIPv2	146	Response
675	9398.120379	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
676	9405.735764	10.0.10.1	224.0.0.9	RIPv2	146	Response
677	9430.753469	10.0.10.1	224.0.0.9	RIPv2	146	Response
678	9432.127786	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
679	9457.772107	10.0.10.1	224.0.0.9	RIPv2	146	Response
680	9476.140511	fe80::e7f:fofff:feb2::ff02::19	ff02::19	RIPng	186	Command Response, Version 1
681	9493.217083	2001::1::1a	2001::1::11	ICMPv6	110	Echo (ping) request id=0x11f2, seq=1, hop limit=64 (no response found!)

Frame 95: 96 bytes on wire (768 bits) (66 bytes captured (528 bits) on interface 0 - 14.0)

Рисунок 54: Анализ трафика

5.60 Настройка динамической маршрутизации в сетях IPv4 и IPv6



No.	Time	Source	Destination	Protocol	Length	Info
756	10340.698327	2001:111::a	2001:101::a	ICMPv6	118	Echo (ping) reply id=0x23ec, seq=5, hop limit=58 (request in 428)
757	10350.406080	fe80::e7f:f6ff:feb2::ff02::9	ff02::9	RIPng	186	Command Response, Version 1
758	10351.327400	10.0.0.1	224.0.0.9	RIPv2	146	Response
759	10356.354029	2001:111::a	2001:101::a	ICMPv6	118	Echo (ping) request id=0x70f5, seq=1, hop limit=64 (reply in 760)
760	10356.488081	2001:111::a	2001:101::a	ICMPv6	118	Echo (ping) reply id=0x70f5, seq=1, hop limit=58 (request in 759)
761	10357.290557	10.0.0.1	224.0.0.9	RIPv2	66	Response
762	10357.429662	fe80::e7f:f6ff:feb2::ff02::9	ff02::9	RIPng	86	Command Response, Version 1
763	10357.503773	2001:101::a	2001:111::a	ICMPv6	118	Echo (ping) request id=0x70f5, seq=2, hop limit=64 (reply in 764)
764	10357.517140	2001:111::a	2001:101::a	ICMPv6	118	Echo (ping) reply id=0x70f5, seq=2, hop limit=58 (request in 763)
765	10358.520032	2001:101::a	2001:111::a	ICMPv6	118	Echo (ping) request id=0x70f5, seq=3, hop limit=64 (reply in 766)
766	10358.547529	2001:111::a	2001:101::a	ICMPv6	118	Echo (ping) reply id=0x70f5, seq=3, hop limit=58 (request in 765)
767	10359.555726	2001:101::a	2001:111::a	ICMPv6	118	Echo (ping) request id=0x70f5, seq=4, hop limit=64 (reply in 768)
768	10359.566670	2001:111::a	2001:101::a	ICMPv6	118	Echo (ping) reply id=0x70f5, seq=4, hop limit=58 (request in 767)
769	10360.509767	2001:101::a	2001:111::a	ICMPv6	118	Echo (ping) request id=0x70f5, seq=5, hop limit=64 (reply in 770)
770	10360.605376	2001:111::a	2001:101::a	ICMPv6	118	Echo (ping) reply id=0x70f5, seq=5, hop limit=58 (request in 769)
771	10362.612968	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
772	10362.624536	2001:101::a	2001:111::a	UDP	174	Time Exceeded (hop limit exceeded in transit)
773	10362.624953	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
774	10362.628144	2001:101::a	2001:101::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
775	10362.630905	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
776	10362.631581	2001:101::a	2001:101::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
777	10362.634711	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
778	10362.638197	2001:111::a	2001:101::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
779	10362.651289	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
780	10362.653291	2001:111::a	2001:101::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
781	10362.655179	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
782	10362.674057	2001:111::a	2001:101::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
783	10362.676255	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
784	10362.692674	2001:111::a	2001:101::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
785	10362.695615	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
786	10363.000641	2001:121::a	2001:101::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
787	10363.012562	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
788	10363.031709	2001:121::a	2001:101::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
789	10363.034761	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
790	10363.051251	2001:111::a	2001:101::a	ICMPv6	174	Destination Unreachable (Port unreachable)
791	10363.055143	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
792	10363.134097	2001:111::a	2001:101::a	ICMPv6	174	Destination Unreachable (Port unreachable)
793	10363.136759	2001:101::a	2001:111::a	UDP	126	40040 → 40050 Len=64
794	10363.151547	2001:101::a	2001:111::a	ICMPv6	174	Destination Unreachable (Port unreachable)
795	10374.353117	10.0.0.1	224.0.0.9	RIPv2	146	Response
796	10385.509280	fe80::e7f:f6ff:feb2::ff02::9	ff02::9	RIPng	186	Command Response, Version 1
797	10400.403080	10.0.0.1	224.0.0.9	RIPv2	146	Response
798	10414.526285	fe80::e7f:f6ff:feb2::ff02::9	ff02::9	RIPng	186	Command Response, Version 1
799	10443.412791	10.0.0.1	224.0.0.9	RIPv2	146	Response
800	10443.529100	fe80::e7f:f6ff:feb2::ff02::9	ff02::9	RIPng	186	Command Response, Version 1
801	10478.418745	10.0.0.1	224.0.0.9	RIPv2	146	Response
802	10487.541384	fe80::e7f:f6ff:feb2::ff02::9	ff02::9	RIPng	186	Command Response, Version 1
803	10504.421500	10.0.0.1	224.0.0.9	RIPv2	146	Response
804	10504.553081	fe80::e7f:f6ff:feb2::ff02::9	ff02::9	RIPng	186	Command Response, Version 1
805	10518.244897	fe80::e7f:f6ff:feb2::ff02::11	ff02::11	ICMPv6	118	Router Advertisement from bc:f7:f6b2:80::80

Рисунок 55: Анализ трафика

5.61 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Применить дисплейный фильтр: «C1»

No.	Time	Source	Destination	Protocol	Length	Info
244	7457.756078	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
245	7444.344031	10.0.11.1	224.0.0.9	RIPv2	140	Response
246	7467.744258	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
247	7474.351268	10.0.11.1	224.0.0.9	RIPv2	140	Response
248	7481.874484	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	96	Command Response, Version 1
249	7487.751458	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
250	7499.368890	10.0.11.1	224.0.0.9	RIPv2	140	Response
251	7523.766842	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
252	7527.366089	10.0.11.1	224.0.0.9	RIPv2	140	Response
253	7538.431441	Fe80::e5b:30ff:fe2a::f902:1:ff00::a	ff02::1:ff00::a	ICMPv6	86	Neighbor Solicitation for 2001::1::a from dc:5b:30:2a:00:00
254	7538.433889	2001::1::a	fe80::e5b:30ff:fe2a::	ICMPv6	86	Neighbor Advertisement 2001::1::a (sol, ovr) is at 00:50:79:06:08:01
255	7538.435373	2001::10::a	2001::1::a	ICMPv6	118	Echo (ping) request id=0xc5ea, seq=1, hop limit=61 (reply in 256)
256	7538.435938	2001::12::a	2001::10::a	ICMPv6	118	Echo (ping) reply id=0xc5ea, seq=1, hop limit=61 (request in 255)
257	7539.467506	2001::10::a	2001::11::a	ICMPv6	118	Echo (ping) request id=0xc5ea, seq=2, hop limit=61 (reply in 258)
258	7539.468956	2001::11::a	2001::10::a	ICMPv6	118	Echo (ping) reply id=0xc5ea, seq=2, hop limit=61 (request in 257)
259	7540.405643	2001::10::a	2001::11::a	ICMPv6	118	Echo (ping) request id=0xc5ea, seq=3, hop limit=61 (reply in 260)
260	7540.406080	2001::11::a	2001::10::a	ICMPv6	118	Echo (ping) reply id=0xc5ea, seq=3, hop limit=61 (request in 259)
261	7541.505822	2001::10::a	2001::11::a	ICMPv6	118	Echo (ping) request id=0xc5ea, seq=4, hop limit=61 (reply in 262)
262	7541.507817	2001::11::a	2001::10::a	ICMPv6	118	Echo (ping) reply id=0xc5ea, seq=4, hop limit=61 (request in 261)
263	7542.510759	2001::10::a	2001::11::a	ICMPv6	118	Echo (ping) request id=0xc5ea, seq=5, hop limit=61 (reply in 264)
264	7542.520558	2001::11::a	2001::10::a	ICMPv6	118	Echo (ping) reply id=0xc5ea, seq=5, hop limit=61 (request in 263)
265	7553.772081	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
266	7555.373546	2001::10::a	2001::11::a	UDP	126	64040 → 64050 len=64
267	7555.374573	2001::11::a	2001::10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
268	7555.587814	2001::10::a	2001::11::a	UDP	126	64040 → 64050 len=64
269	7555.589997	2001::11::a	2001::10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
270	7555.634577	2001::10::a	2001::11::a	UDP	126	64040 → 64050 len=64
271	7555.635604	2001::11::a	2001::10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
272	7562.373380	10.0.11.1	224.0.0.9	RIPv2	140	Response
273	7561.777677	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
274	7560.381466	10.0.11.1	224.0.0.9	RIPv2	140	Response
275	7621.784619	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
276	7631.384457	10.0.11.1	224.0.0.9	RIPv2	140	Response
277	7681.387142	10.0.11.1	224.0.0.9	RIPv2	140	Response
278	7681.390467	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
279	7681.491851	10.0.11.1	224.0.0.9	RIPv2	96	Response
280	7684.790668	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
281	7684.515613	10.0.11.1	224.0.0.9	RIPv2	96	Response
282	7671.282023	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
283	7691.393339	10.0.11.1	224.0.0.9	RIPv2	140	Response
284	7709.799761	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
285	7725.397018	10.0.11.1	224.0.0.9	RIPv2	140	Response
286	7734.803577	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
287	7751.408024	10.0.11.1	224.0.0.9	RIPv2	140	Response
288	7779.812922	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1
289	7795.408779	10.0.11.1	224.0.0.9	RIPv2	140	Response
290	7813.810411	Fe80::e5b:30ff:fe2a::f902:19	ff02::1:9	RIPng	108	Command Response, Version 1

Frame 1: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface "... id 0

Рисунок 56: Анализ трафика

5.62 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
450	18253.000010	fc00::e5b:38ff:fe2a::	ff02::9	RIPng	80	Command Response, Version 1
460	18254.115634	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b23ec, seq=1, hop limit=61 (reply in 461)
461	18254.122569	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b23ec, seq=1, hop limit=61 (request in 460)
462	18254.125314	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b23ec, seq=2, hop limit=61 (reply in 464)
463	18254.125422	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b23ec, seq=3, hop limit=61 (reply in 465)
464	18254.126128	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b23ec, seq=2, hop limit=61 (request in 462)
465	18254.126148	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b23ec, seq=3, hop limit=61 (request in 463)
466	18254.130670	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b23ec, seq=4, hop limit=61 (reply in 468)
467	18254.130930	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b23ec, seq=5, hop limit=61 (reply in 469)
468	18254.131020	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b23ec, seq=4, hop limit=61 (request in 466)
469	18254.131205	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b23ec, seq=5, hop limit=61 (request in 467)
470	18255.105100	fc00::e5b:38ff:fe2a::	2001::11:1a	ICMPv6	86	Neighbor Solicitation for 2001::11:1a from 8c:5b:38:2a:00:00
471	18260.170881	fc00::e5b:38ff:fe2a::	2001::11:1a	ICMPv6	86	Neighbor Solicitation for 2001::11:1a from 8c:5b:38:2a:00:00
472	18261.242908	fc00::e5b:38ff:fe2a::	2001::11:1a	ICMPv6	86	Neighbor Solicitation for 2001::11:1a from 8c:5b:38:2a:00:00
473	18266.030401	fc00::e5b:38ff:fe2a::	ff02::1:ff02::1	ICMPv6	86	Neighbor Solicitation for 2001::11:1a from 8c:5b:38:2a:00:00
474	18269.932719	2001::11:1a	fc00::e5b:38ff:fe2a::	ICMPv6	86	Neighbor Advertisement 2001::11:1a (sol, adv) is at 00:50:79:0a:68:81
475	18269.952758	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b2705, seq=1, hop limit=61 (reply in 476)
476	18269.954524	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b2705, seq=1, hop limit=61 (request in 475)
477	18270.996783	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b2705, seq=2, hop limit=61 (reply in 478)
478	18270.998132	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b2705, seq=2, hop limit=61 (request in 477)
479	18272.827722	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b2705, seq=3, hop limit=61 (reply in 480)
480	18272.828689	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b2705, seq=3, hop limit=61 (request in 479)
481	18273.855235	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b2705, seq=4, hop limit=61 (reply in 482)
482	18273.860376	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b2705, seq=4, hop limit=61 (request in 481)
483	18274.885292	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) request 1d4b2705, seq=5, hop limit=61 (reply in 484)
484	18274.886491	2001::11:1a	2001::11:1a	ICMPv6	118	Echo (ping) reply 1d4b2705, seq=5, hop limit=61 (request in 483)
485	18276.532490	2001::11:1a	2001::11:1a	UDP	120	40800 → 40808 Len=64
486	18276.532520	2001::11:1a	2001::11:1a	ICMPv6	174	Destination unreachable (Port unreachable)
487	18276.580967	2001::11:1a	2001::11:1a	UDP	120	40800 → 40808 Len=64
488	18276.580920	2001::11:1a	2001::11:1a	ICMPv6	174	Destination unreachable (Port unreachable)
489	18276.635241	2001::11:1a	2001::11:1a	ICMPv6	174	Destination unreachable (Port unreachable)
490	18276.636427	2001::11:1a	2001::11:1a	ICMPv6	174	Destination unreachable (Port unreachable)
491	18283.194540	fc00::e5b:38ff:fe2a::	ff02::1	ICMPv6	118	Router Advertisement from 8c:5b:38:2a:00:00
492	18286.380951	10.0.0.1.1	224.0.0.9	RIPv2	146	Response
493	18289.068135	fc00::e5b:38ff:fe2a::	ff02::9	RIPng	186	Command Response, Version 1
494	18314.144306	fc00::e5b:38ff:fe2a::	ff02::9	RIPng	186	Command Response, Version 1
495	18315.480908	10.0.0.1.1	224.0.0.9	RIPv2	146	Response
496	18340.153179	fc00::e5b:38ff:fe2a::	ff02::9	RIPng	186	Command Response, Version 1
497	18344.406915	10.0.0.1.1	224.0.0.9	RIPv2	146	Response
498	18377.413151	10.0.0.1.1	224.0.0.9	RIPv2	146	Response
499	18386.167952	fc00::e5b:38ff:fe2a::	ff02::9	RIPng	186	Command Response, Version 1
500	18480.425393	10.0.0.1.1	224.0.0.9	RIPv2	146	Response
501	18412.187824	fc00::e5b:38ff:fe2a::	ff02::9	RIPng	186	Command Response, Version 1
502	18437.444260	10.0.0.1.1	224.0.0.9	RIPv2	146	Response
503	18443.195638	fc00::e5b:38ff:fe2a::	ff02::9	RIPng	186	Command Response, Version 1
504	18449.462790	10.0.0.1.1	224.0.0.9	RIPv2	146	Response
505	18475.262080	fc00::e5b:38ff:fe2a::	ff02::9	RIPng	186	Command Response, Version 1

Frame 11: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface ..., 1d 8
 Ethernet II, Src: Private_8c:5b:38:2a, Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 Address Resolution Protocol (request/unsolicited ARP)

Рисунок 57: Анализ трафика

5.63 Настройка динамической маршрутизации в сетях IPv4 и IPv6

OSPF — протокол динамической маршрутизации на основе состояния каналов. Поддерживает IPv4 (OSPFv2) и IPv6 (OSPFv3), работает на оборудовании разных производителей, поддерживает бесклассовую маршрутизацию, балансировку нагрузки и суммирование маршрутов. Метрика — стоимость, зависящая от скорости интерфейса, административное расстояние — 110.

Маршрутизаторы обмениваются Hello-пакетами (224.0.0.5 / FF02::5) для автоматического обнаружения соседей и формирования отношений соседства. OSPF использует три таблицы: соседей, топологии и маршрутизации. Передача обновлений ведётся по принципу SPF-алгоритма, который рассчитывает кратчайшие пути.

Сеть разделяется на области (area) для уменьшения таблиц и ускорения сходимости. Главная — область 0. Маршрутизаторы с несколькими областями — ABR, маршрутизаторы, соединяющие разные протоколы — ASBR. Для IPv6 используется OSPFv3, применяющий групповые адреса FF02::5 и FF02::6.

5.64 Настройка динамической маршрутизации в сетях IPv4 и IPv6

На маршрутизаторах настроим OSPFv2 для сетей IPv4.
msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# configure terminal
msk-antoychubekova-gw-01(config)# router ospf
msk-antoychubekova-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-antoychubekova-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-antoychubekova-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-antoychubekova-gw-01(config-router)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 58: Настройка OSPFv2 в msk-antoychubekova-gw-01 для IPv4

5.65 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# router ospf
msk-antoychubekova-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-antoychubekova-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-antoychubekova-gw-02(config-router)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 59: Настройка OSPFv2 в msk-antoychubekova-gw-02 для IPv4

5.66 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# router ospf
msk-antoychubekova-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-antoychubekova-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-antoychubekova-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-antoychubekova-gw-03(config-router)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 60: Настройка OSPFv2 в msk-antoychubekova-gw-03 для IPv4

5.67 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# router ospf
msk-antoychubekova-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-antoychubekova-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-antoychubekova-gw-04(config-router)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 61: Настройка OSPFv2 в msk-antoychubekova-gw-04 для IPv4

5.68 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуем PC2 и определим путь следования пакетов.

```
PC1> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=97.933 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=26.350 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=19.303 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=29.997 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=14.548 ms

PC1> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1   16.848 ms  3.237 ms  3.588 ms
 2  10.0.4.1    20.421 ms  40.928 ms 12.591 ms
 3  10.0.2.2    37.480 ms  30.227 ms 19.924 ms
 4  10.0.11.10  19.092 ms  17.531 ms 29.635 ms

PC1> 
```

Рисунок 62: Пингование PC1 к PC2

5.69 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим таблицу маршрутизации протокола OSPFv2.

```
msk-antoychubekova-gw-01# show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.0.2.1	1	Full/Backup	33.017s	10.0.1.2	eth1:10.0.1.1
10.0.4.1	1	Full/Backup	39.139s	10.0.4.1	eth2:10.0.4.2

```
msk-antoychubekova-gw-01# show ip ospf route
```

```
===== OSPF network routing table =====
```

N	10.0.1.0/24	[100] area: 0.0.0.0
		directly attached to eth1
N	10.0.2.0/24	[200] area: 0.0.0.0
		via 10.0.1.2, eth1
N	10.0.3.0/24	[200] area: 0.0.0.0
		via 10.0.4.1, eth2
N	10.0.4.0/24	[100] area: 0.0.0.0
		directly attached to eth2
N	10.0.10.0/24	[100] area: 0.0.0.0
		directly attached to eth0
N	10.0.11.0/24	[300] area: 0.0.0.0
		via 10.0.1.2, eth1
		via 10.0.4.1, eth2

```
===== OSPF router routing table =====
```

```
===== OSPF external routing table =====
```

```
msk-antoychubekova-gw-01#
```

Рисунок 63: Таблица маршрутизации протокола OSPFv2

5.70 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Предположим, что пакет проходит через маршрутизатор msk-antoychubekova-gw-02. Отключим на маршрутизаторе msk-antoychubekova-gw-02 интерфейс.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02 (config)# interface eth0
msk-antoychubekova-gw-02 (config-if)# shutdown
msk-antoychubekova-gw-02 (config-if)#
```

Рисунок 64: Отключение интерфейса

5.71 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим таблицу маршрутизации протокола OSPFv2.

```
msk-antoychubekova-gw-01# show ip ospf neighbor

Neighbor ID      Pri State          Dead Time Address      Interface
  RXmtL RqstL DBsmL
10.0.4.1         1 Full/Backup    34.456s 10.0.4.1      eth2:10.0.4.2
    0      0      0

msk-antoychubekova-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24      [100] area: 0.0.0.0
    directly attached to eth1
N   10.0.2.0/24      [300] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.3.0/24      [200] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.4.0/24      [100] area: 0.0.0.0
    directly attached to eth2
N   10.0.10.0/24     [100] area: 0.0.0.0
    directly attached to eth0
N   10.0.11.0/24     [300] area: 0.0.0.0
    via 10.0.4.1, eth2

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-antoychubekova-gw-01#
```

Рисунок 65: Таблица маршрутизации протокола OSPFv2

5.72 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуйте PC2 и определим путь следования пакетов. Примерное время, за которое восстановится маршрут равняется 1 минуте.

```
PC1> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=22.471 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=16.943 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=24.385 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=27.916 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=11.665 ms

PC1> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    9.792 ms  4.100 ms  2.861 ms
 2  10.0.4.1     11.503 ms  6.535 ms  5.965 ms
 3  10.0.3.1     22.658 ms  9.824 ms 12.700 ms
 4  10.0.11.10   16.471 ms 12.831 ms 11.932 ms

PC1> █
```

Рисунок 66: Пингование PC1 к PC2

5.73 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Включим на маршрутизаторе msk-antoychubekova-gw-02 интерфейс.

```
msk-antoychubekova-gw-02(config)# interface eth0  
msk-antoychubekova-gw-02(config-if)# no shutdown  
msk-antoychubekova-gw-02(config-if)#
```

Рисунок 67: Включение интерфейса

5.74 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуйте PC2 и определим путь следования пакетов.

```
PC1> ping 10.0.11.10
84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=45.692 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=18.135 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=16.573 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=17.874 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=18.955 ms

PC1> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    7.368 ms  2.363 ms  4.924 ms
 2  10.0.1.2     14.062 ms  6.223 ms  4.896 ms
 3  10.0.3.1     15.856 ms  9.181 ms  5.697 ms
 4  10.0.11.10   16.981 ms 12.160 ms 14.056 ms

PC1> █
```

Рисунок 68: Пингование PC1 к PC2

Просмотрим захваченный трафик.

No	Time	Source	Destination	Protocol	Length	Info
888	112308.572808	10.0.0.1	224.0.0.5	IGMP	40	Host Hello Packet
889	112308.797878	fe80::c7::ffff::fe02::	ff02::c	IGMP	108	Common Response, Version 3
890	112308.802039	10.0.0.1	224.0.0.5	IGMP	78	Host Hello Packet
900	112310.616388	10.0.0.1	224.0.0.9	IGMP	40	Host Hello Packet
901	112310.620069	10.0.0.1	224.0.0.5	IGMP	78	Host Hello Packet
901	112308.840005	10.0.0.1	224.0.0.5	IGMP	78	Host Hello Packet
902	112311.433191	10.0.0.1	224.0.0.9	IGMP	40	Host Hello Packet
902	112312.423315	ff02::c7::ffff::fe02::	ff02::c	IGMP	108	Common Response, Version 3
903	112312.423315	ff02::c7::ffff::fe02::	ff02::c	IGMP	108	Common Response, Version 3
904	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
905	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
906	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
907	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
908	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
909	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
910	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
911	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
912	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
913	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
914	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
915	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
916	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
917	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
918	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
919	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
920	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
921	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
922	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
923	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
924	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
925	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
926	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
927	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
928	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
929	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
930	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
931	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
932	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
933	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
934	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
935	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
936	112312.721318	10.0.11.10	10.0.0.30	10DP	96	800.0.0.0.1 is at 8c17ff7f32:00:00:00
937	112308.620909	10.0.0.1	224.0.0.5	IGMP	78	Host Hello Packet

Рисунок 69: Анализ трафика

5.76 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Type	Source	Destination	Protocol	Length	Info
1000	11000.000100	10.0.10.1	224.0.0.1	IGMP	70	IGMPv2 Packet
1001	11000.000100	10.0.10.1	224.0.0.1	IGMP	70	IGMPv2 Packet
1002	11000.002107	10.0.10.1	224.0.0.9	RIPv2	140	Response
1003	11000.002107	10.0.10.1	224.0.0.1	IGMP	70	IGMPv2 Packet
1004	11000.000100	10.0.10.1	224.0.0.1	RIPng	80	Command Response, Version 1
1005	11011.000100	10.0.10.1	224.0.0.1	RIPng	160	Command Response, Version 1
1006	11011.000100	10.0.10.1	224.0.0.1	IGMP	70	IGMPv2 Packet
1007	11010.000100	Private: 10.0.0.0	Broadcast	ARP	60	Who has 10.0.10.1? Tell 10.0.10.10
1008	11010.000100	10.0.10.10.1	10.0.10.1	ARP	60	10.0.10.1 is at 0:17:f0:02:00:00
1009	11010.002107	10.0.10.10	10.0.11.10	ICMP	80	Echo (ping) request 10.0.10.10, seq=1234, ttl=64 (reply is 10018)
1010	11010.002107	10.0.11.10	10.0.10.10	ICMP	80	Echo (ping) reply 10.0.10.10, seq=1234, ttl=64 (request is 1009)
1011	11010.002107	10.0.10.10	10.0.11.10	ICMP	80	Echo (ping) request 10.0.10.10, seq=1234, ttl=64 (reply is 10011)
1012	11010.002107	10.0.11.10	10.0.10.10	ICMP	80	Echo (ping) reply 10.0.10.10, seq=1234, ttl=64 (request is 1011)
1013	11010.002107	10.0.10.10	10.0.11.10	ICMP	80	Echo (ping) request 10.0.10.10, seq=1234, ttl=64 (reply is 10014)
1014	11010.002107	10.0.11.10	10.0.10.10	ICMP	80	Echo (ping) reply 10.0.10.10, seq=1234, ttl=64 (request is 1013)
1015	11010.002107	10.0.10.10	10.0.11.10	ICMP	80	Echo (ping) request 10.0.10.10, seq=1234, ttl=64 (reply is 10016)
1016	11010.002107	10.0.11.10	10.0.10.10	ICMP	80	Echo (ping) reply 10.0.10.10, seq=1234, ttl=64 (request is 1015)
1017	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1018	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1019	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1020	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1021	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1022	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1023	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1024	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1025	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1026	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1027	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1028	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1029	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1030	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1031	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1032	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1033	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1034	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1035	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1036	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1037	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1038	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1039	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1040	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1041	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1042	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1043	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1044	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1045	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1046	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1047	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1048	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1049	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1050	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1051	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1052	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1053	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1054	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1055	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1056	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1057	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1058	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1059	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1060	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1061	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1062	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1063	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1064	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1065	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1066	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1067	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1068	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1069	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1070	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1071	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1072	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1073	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1074	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1075	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1076	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1077	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1078	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1079	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1080	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1081	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1082	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1083	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1084	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1085	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1086	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1087	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1088	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1089	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1090	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1091	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1092	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1093	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1094	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1095	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1096	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1097	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1098	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1099	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet
1100	11010.002107	10.0.10.10	224.0.0.1	IGMP	70	IGMPv2 Packet

Рисунок 70: Анализ трафика

5.77 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
577	11:10:10.00748	Fd00::a50::30FF::FE2A	Fd02::1	ICMPv6	280	Command Response, Version 1
578	11:10:10.137011	10.0.0.10	10.0.11.10	ICMP	90	Echo (ping) request 10-0-0-10, seq=1/254, ttl=64 (reply to 581)
579	11:10:10.140000	Private, 10.0.0.0	Broadcast	ARP	64	who has 10.0.11.1? tell 10.0.11.10
580	11:10:10.140000	10.0.10.10	10.0.10.10	ARP	60	who has 10.0.11.1? tell 10.0.10.10
581	11:10:10.139040	10.0.11.10	10.0.10.10	ICMP	90	Echo (ping) reply 10-0-0-10, seq=1/254, ttl=64 (request to 578)
582	11:10:10.211040	10.0.11.1	224.0.0.5	OSPF	70	Hello Packet
583	11:10:10.211040	10.0.11.1	10.0.11.10	ICMP	90	Echo (ping) request 10-0-0-10, seq=2/512, ttl=64 (reply to 584)
584	11:10:10.212759	10.0.11.10	10.0.11.10	ICMP	90	Echo (ping) reply 10-0-0-10, seq=2/512, ttl=64 (request to 583)
585	11:10:10.210000	10.0.10.10	10.0.11.10	ICMP	90	Echo (ping) request 10-0-0-10, seq=3/768, ttl=64 (reply to 586)
586	11:10:10.250721	10.0.11.10	10.0.11.10	ICMP	90	Echo (ping) reply 10-0-0-10, seq=3/768, ttl=64 (request to 585)
587	11:10:10.282767	10.0.10.10	10.0.11.10	ICMP	90	Echo (ping) request 10-0-0-10, seq=4/1024, ttl=64 (reply to 588)
588	11:10:10.284770	10.0.11.10	10.0.11.10	ICMP	90	Echo (ping) reply 10-0-0-10, seq=4/1024, ttl=64 (request to 587)
589	11:10:10.710517	10.0.11.1	224.0.0.5	OSPFv2	140	Response
590	11:10:10.100000	10.0.11.10	10.0.11.10	ICMP	90	Echo (ping) request 10-0-0-10, seq=5/1280, ttl=64 (reply to 591)
591	11:10:10.100000	10.0.11.10	10.0.11.10	ICMP	90	Echo (ping) reply 10-0-0-10, seq=5/1280, ttl=64 (request to 590)
592	11:10:10.200000	Private, 10.0.0.0	Broadcast	ARP	60	who has 10.0.11.1? tell 10.0.11.1
593	11:10:10.200000	Private, 10.0.0.0	Broadcast	ARP	60	who has 10.0.11.1? tell 10.0.11.1
594	11:10:10.211040	10.0.11.1	224.0.0.5	OSPF	70	Hello Packet
595	11:10:10.211040	Fd00::a50::30FF::FE2A	Fd02::1	ICMPv6	280	Command Response, Version 1
596	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
597	11:10:10.211040	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
598	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
599	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
600	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
601	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
602	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
603	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
604	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
605	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
606	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
607	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
608	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
609	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
610	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
611	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
612	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
613	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
614	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
615	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
616	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
617	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
618	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
619	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
620	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
621	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
622	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0
623	11:10:10.100000	10.0.11.10	10.0.11.10	TCP	74	4024 -> 4024 [WIN] Seq=1000000000 Win=1000000000 Len=0

Рисунок 71: Анализ трафика

Рисунок 72: Анализ трафика

[illegible]

5.79 Настройка динамической маршрутизации в сетях IPv4 и IPv6

На маршрутизаторах настроим OSPFv3 для сетей IPv6.

msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# configure terminal
msk-antoychubekova-gw-01(config)# router ospf6
msk-antoychubekova-gw-01(config-ospf6)# ospf6 router-id 1.1.1.1
msk-antoychubekova-gw-01(config-ospf6)# exit
msk-antoychubekova-gw-01(config)# interface eth0
msk-antoychubekova-gw-01(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth1
msk-antoychubekova-gw-01(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth2
msk-antoychubekova-gw-01(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 73: Настройка OSPFv3 в msk-antoychubekova-gw-01 для IPv6

5.80 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# router ospf6
msk-antoychubekova-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-antoychubekova-gw-02(config-ospf6)# exit
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth1
msk-antoychubekova-gw-02(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 74: Настройка OSPFv3 в msk-antoychubekova-gw-02 для IPv6

5.81 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# router ospf6
msk-antoychubekova-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-antoychubekova-gw-03(config-ospf6)# exit
msk-antoychubekova-gw-03(config)# interface wth0
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth0
msk-antoychubekova-gw-03(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth1
msk-antoychubekova-gw-03(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth2
msk-antoychubekova-gw-03(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 75: Настройка OSPFv3 в msk-antoychubekova-gw-03 для IPv6

5.82 Настройка динамической маршрутизации в сетях IPv4 и IPv6

msk-antoychubekova-gw-04.

```
[OK]
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# router ospf6
msk-antoychubekova-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-antoychubekova-gw-04(config-ospf6)# exit
msk-antoychubekova-gw-04(config)# interface eth0
msk-antoychubekova-gw-04(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# interface eth1
msk-antoychubekova-gw-04(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 76: Настройка OSPFv3 в msk-antoychubekova-gw-04 для IPv6

5.83 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуем PC2 и определим путь следования пакетов.

```
PC1> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=31.585 ms
2001:11::a icmp6_seq=2 ttl=58 time=20.081 ms
2001:11::a icmp6_seq=3 ttl=58 time=15.333 ms
2001:11::a icmp6_seq=4 ttl=58 time=9.701 ms
2001:11::a icmp6_seq=5 ttl=58 time=13.517 ms

PC1> trace 2001:11::a

Trace to 2001:11::a, 64 hops max
 1 2001:10::1  19.523 ms  8.181 ms  7.899 ms
 2 2001:4::1   15.734 ms  7.406 ms  10.357 ms
 3 2001:3::1   32.033 ms  28.507 ms  14.146 ms
 4 2001:11::a  24.200 ms  25.733 ms  18.170 ms

PC1> █
```

Рисунок 77: Пингование PC1 к PC2

5.84 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим таблицу маршрутизации протокола OSPFv3.

```
msk-antoychubekova-gw-01# show ipv6 ospf6 neighbor
Neighbor ID      Pri   DeadTime   State/IfState      Duration I/F[State]
2.2.2.2          1     00:00:37   Full/BDR           00:07:36 eth1[DR]
4.4.4.4          1     00:00:38   Full/BDR           00:02:23 eth2[DR]
msk-antoychubekova-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64      ::                eth1 00:07:48
*N IA 2001:2::/64      fe80::abd:8dff:fee7:0 eth1 00:05:12
*N IA 2001:3::/64      fe80::e36:caff:fea9:1 eth2 00:02:35
*N IA 2001:4::/64      ::                eth2 00:02:35
*N IA 2001:10::/64     ::                eth0 00:10:06
*N IA 2001:11::/64     fe80::abd:8dff:fee7:0 eth1 00:02:35
                        fe80::e36:caff:fea9:1 eth2
msk-antoychubekova-gw-01#
```

Рисунок 78: Таблица маршрутизации протокола OSPFv3

5.85 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Предположим, что пакет проходит через маршрутизатор msk-antoychubekova-gw-02. Отключим на маршрутизаторе msk-antoychubekova-gw-02 интерфейс.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# stutdown
% Unknown command: stutdown
msk-antoychubekova-gw-02(config-if)# shutdown
msk-antoychubekova-gw-02(config-if)# 
1e80::e36:caff:ea9:1 eth2 00
```

Рисунок 79: Отключение интерфейса

5.86 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Проверим таблицу маршрутизации протокола OSPFv3.

```
msk-antoychubekova-gw-01# show ipv6 ospf6 neighbor
Neighbor ID Pri DeadTime State/IfState Duration I/F[State]
4.4.4.4 1 00:00:37 Full/BDR 00:05:24 eth2[DR]
msk-antoychubekova-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64 :: eth1 00:00:17
*N IA 2001:2::/64 fe80::e36:caff:fea9:1 eth2 00:01:48
*N IA 2001:3::/64 fe80::e36:caff:fea9:1 eth2 00:05:27
*N IA 2001:4::/64 :: eth2 00:05:27
*N IA 2001:10::/64 :: eth0 00:01:17
*N IA 2001:11::/64 fe80::e36:caff:fea9:1 eth2 00:01:48
msk-antoychubekova-gw-01#
```

Рисунок 80: Таблица маршрутизации протокола OSPFv3

5.87 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуем PC2 и определим путь следования пакетов. Примерное время, за которое восстановится маршрут равно минуте.

```
PC1> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=19.553 ms
2001:11::a icmp6_seq=2 ttl=58 time=10.304 ms
2001:11::a icmp6_seq=3 ttl=58 time=10.474 ms
2001:11::a icmp6_seq=4 ttl=58 time=18.785 ms
2001:11::a icmp6_seq=5 ttl=58 time=12.991 ms

PC1> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    4.048 ms  2.652 ms  7.011 ms
 2 2001:4::1     5.448 ms  3.983 ms  6.943 ms
 3 2001:3::1     11.739 ms 8.437 ms  9.533 ms
 4 2001:11::a    14.923 ms 27.194 ms 19.533 ms

PC1> 
```

Рисунок 81: Пингование PC1 к PC2

5.88 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Включим на маршрутизаторе msk-antoychubekova-gw-02 интерфейс.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)#
```

Рисунок 82: Включение интерфейса

5.89 Настройка динамической маршрутизации в сетях IPv4 и IPv6

С PC1 пропингуем PC2 и определим путь следования пакетов.

```
PC1> ping 2001:11::a
2001:11::a icmp6_seq=1 ttl=58 time=21.314 ms
2001:11::a icmp6_seq=2 ttl=58 time=15.131 ms
2001:11::a icmp6_seq=3 ttl=58 time=11.889 ms
2001:11::a icmp6_seq=4 ttl=58 time=13.315 ms
2001:11::a icmp6_seq=5 ttl=58 time=12.649 ms

PC1> trace 2001:11::a
trace to 2001:11::a, 64 hops max
 1 2001:10::1  6.797 ms  2.155 ms  1.762 ms
 2 2001:4::1   10.266 ms  6.243 ms  13.909 ms
 3 2001:3::1   18.115 ms  12.964 ms  7.897 ms
 4 2001:11::a  23.980 ms  13.775 ms  16.764 ms

PC1> 
```

Рисунок 83: Пингование PC1 к PC2

5.90 Настройка динамической маршрутизации в сетях IPv4 и IPv6

Посмотрим захваченный трафик.

No.	Time	Source	Destination	Protocol	Length	Info
1385	12589.209871	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1386	12590.213428	10.0.10.1	224.0.0.9	RIPv2	146	Response
1387	12591.254219	fe80::e7ff::f0ff::feb2::	ff02::15	OSPF	90	Hello Packet
1388	12599.271508	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1389	12595.264519	fe80::e7ff::f0ff::feb2::	ff02::15	OSPF	90	Hello Packet
1310	12599.279800	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1311	12610.159821	2001:10::1a	2001:11::1a	ICMPv6	118	Echo (ping) request id=0x3efe, seq=1, hop limit=64 (reply in 1312)
1312	12610.225190	2001:11::1a	2001:10::1a	ICMPv6	118	Echo (ping) reply id=0x3efe, seq=1, hop limit=58 (request in 1311)
1313	12611.229936	2001:10::1a	2001:11::1a	ICMPv6	118	Echo (ping) request id=0x3efe, seq=2, hop limit=64 (reply in 1314)
1314	12611.248761	2001:11::1a	2001:10::1a	ICMPv6	118	Echo (ping) reply id=0x3efe, seq=2, hop limit=58 (request in 1313)
1315	12612.251355	2001:10::1a	2001:11::1a	ICMPv6	118	Echo (ping) request id=0x3efe, seq=3, hop limit=64 (reply in 1316)
1316	12612.269539	2001:11::1a	2001:10::1a	ICMPv6	118	Echo (ping) reply id=0x3efe, seq=3, hop limit=58 (request in 1315)
1317	12613.287273	2001:10::1a	2001:11::1a	ICMPv6	118	Echo (ping) request id=0x3efe, seq=4, hop limit=64 (reply in 1318)
1318	12613.276168	2001:11::1a	2001:10::1a	ICMPv6	118	Echo (ping) reply id=0x3efe, seq=4, hop limit=58 (request in 1317)
1319	12614.220802	10.0.10.1	224.0.0.9	RIPv2	146	Response
1320	12614.277899	2001:10::1a	2001:11::1a	ICMPv6	118	Echo (ping) request id=0x3efe, seq=5, hop limit=64 (reply in 1321)
1321	12614.290365	2001:11::1a	2001:10::1a	ICMPv6	118	Echo (ping) reply id=0x3efe, seq=5, hop limit=58 (request in 1320)
1322	12615.205922	fe80::e7ff::f0ff::feb2::	2001:10::1a	ICMPv6	86	Neighbor Solicitation for 2001:10::1a from 0c::7f::6b2:00:00
1323	12615.279380	fe80::e7ff::f0ff::feb2::	ff02::15	OSPF	90	Hello Packet
1324	12616.306376	fe80::e7ff::f0ff::feb2::	2001:10::1a	ICMPv6	86	Neighbor Solicitation for 2001:10::1a from 0c::7f::6b2:00:00
1325	12617.346280	fe80::e7ff::f0ff::feb2::	2001:10::1a	ICMPv6	86	Neighbor Solicitation for 2001:10::1a from 0c::7f::6b2:00:00
1326	12619.279475	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1327	12621.273575	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1328	12621.287995	2001:10::1	ff02::1:ff00::a	ICMPv6	86	Neighbor Solicitation for 2001:10::1a from 0c::7f::6b2:00:00
1329	12621.288421	2001:10::1a	2001:10::1	ICMPv6	86	Neighbor Advertisement 2001:10::1a (sol, mcr) is at 00:50:79:64:62:00
1330	12621.292424	2001:10::1a	2001:11::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1331	12621.296161	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1332	12622.305239	2001:10::1	2001:10::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1333	12622.309579	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1334	12622.312456	2001:10::1a	2001:11::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1335	12623.314430	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1336	12623.320150	2001:4::1	2001:10::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1337	12623.331450	2001:10::1	2001:11::1a	UDP	126	32001 = 32002 Len=64
1338	12623.332424	2001:4::1	2001:10::1	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1339	12623.340303	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1340	12623.347875	2001:4::1	2001:10::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1341	12623.352456	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1342	12623.352421	2001:20::1	2001:10::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1343	12623.359511	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1344	12623.412600	2001:3::1	2001:10::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1345	12623.412722	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1346	12623.427706	2001:3::1	2001:10::1a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1347	12623.428815	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1348	12623.451301	2001:11::1a	2001:10::1a	ICMPv6	174	Destination Unreachable (Port unreachable)
1349	12623.459150	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1350	12623.460029	2001:11::1a	2001:10::1a	ICMPv6	174	Destination Unreachable (Port unreachable)
1351	12623.461933	2001:10::1a	2001:11::1a	UDP	126	32001 = 32002 Len=64
1352	12623.462706	2001:11::1a	2001:10::1a	ICMPv6	174	Destination Unreachable (Port unreachable)
1353	12624.004210	fe80::e7ff::f0ff::feb2::	ff02::15	RIPv2	126	Command Response, Version 1
1354	12625.279193	fe80::e7ff::f0ff::feb2::	ff02::15	OSPF	90	Hello Packet
1355	12625.280063	fe80::e7ff::f0ff::feb2::	ff02::15	OSPF	90	Hello Packet

Рисунок 84: Анализ трафика

5.91 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
1425	12899.356389	10.0.0.1	224.0.0.5	OSPF	70	Hello Packet
1426	12899.745101	fe80::e7f:f6ff:feb2::ff02::5	224.0.0.5	RIPng	100	Command Response, Version 1
1427	12899.361908	fe80::e7f:f6ff:feb2::ff02::5	224.0.0.5	OSPF	90	Hello Packet
1428	12899.357281	10.0.0.1	224.0.0.5	OSPF	70	Hello Packet
1429	12905.364049	fe80::e7f:f6ff:feb2::ff02::5	224.0.0.5	OSPF	90	Hello Packet
1430	12909.363580	10.0.0.1	224.0.0.5	OSPF	70	Hello Packet
1431	12913.399829	10.0.0.1	224.0.0.9	RIPv2	246	Response
1432	12914.224143	10.0.0.1	224.0.0.9	RIPv2	66	Response
1433	12915.369180	fe80::e7f:f6ff:feb2::ff02::5	224.0.0.5	OSPF	90	Hello Packet
1434	12919.363801	10.0.0.1	224.0.0.5	OSPF	70	Hello Packet
1435	12919.632622	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) request id=0x73ff, seq=1, hop limit=64 (reply in 1436)
1436	12919.644661	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x73ff, seq=1, hop limit=64 (request in 1435)
1437	12920.668288	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) request id=0x73ff, seq=2, hop limit=64 (reply in 1438)
1438	12920.678382	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x73ff, seq=2, hop limit=64 (request in 1437)
1439	12921.679795	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) request id=0x73ff, seq=3, hop limit=64 (reply in 1440)
1440	12921.693527	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x73ff, seq=3, hop limit=64 (request in 1439)
1441	12922.695454	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) request id=0x73ff, seq=4, hop limit=64 (reply in 1442)
1442	12922.707741	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x73ff, seq=4, hop limit=64 (request in 1441)
1443	12923.711006	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) request id=0x73ff, seq=5, hop limit=64 (reply in 1444)
1444	12923.726308	2001:10::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0x73ff, seq=5, hop limit=64 (request in 1443)
1445	12924.705823	fe80::e7f:f6ff:feb2::ff02::1	2001:10::a	ICMPv6	86	Neighbor Solicitation for 2001:10::a from 8c:7f:f6:b2:00:00
1446	12924.741497	fe80::e7f:f6ff:feb2::ff02::1	2001:10::a	ICMPv6	110	Router Advertisement from 8c:7f:f6:b2:00:00
1447	12925.370592	fe80::e7f:f6ff:feb2::ff02::5	224.0.0.5	OSPF	90	Hello Packet
1448	12925.746712	fe80::e7f:f6ff:feb2::ff02::5	2001:10::a	ICMPv6	86	Neighbor Solicitation for 2001:10::a from 8c:7f:f6:b2:00:00
1449	12926.396556	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1450	12926.403947	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1451	12926.405508	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1452	12926.406063	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1453	12926.412154	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1454	12926.412629	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1455	12926.413776	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1456	12926.421063	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1457	12926.425162	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1458	12926.432104	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1459	12926.434473	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1460	12926.440159	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1461	12926.442269	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1462	12926.451197	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1463	12926.454742	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1464	12926.467301	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1465	12926.469547	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1466	12926.472761	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1467	12926.478934	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1468	12926.494938	2001:10::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
1469	12926.497450	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1470	12926.501595	2001:10::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
1471	12926.512194	2001:10::a	2001:10::a	UDP	128	38337 → 38338 Len=64
1472	12926.522090	2001:10::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
1473	12926.786791	fe80::e7f:f6ff:feb2::ff02::1	2001:10::a	ICMPv6	86	Neighbor Solicitation for 2001:10::a from 8c:7f:f6:b2:00:00
1474	12927.971893	10.0.0.1	224.0.0.9	RIPv2	66	Response

Рисунок 85: Анализ трафика

5.92 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
1482	12949.371921	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1483	12955.377568	fe80::e7f:feff:feb2::ff02::5	ff02::5	OSPF	90	Hello Packet
1484	12959.374771	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1485	12963.123241	fe80::e7f:feff:feb2::ff02::9	ff02::9	RIPng	86	Command Response, Version 1
1486	12965.379981	fe80::e7f:feff:feb2::ff02::5	ff02::5	OSPF	90	Hello Packet
1487	12967.667379	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xa3ff, seq=1, hop limit=64 (reply in 1490)
1488	12967.682474	fe80::e7f:feff:feb2::ff02::1:fff00::a	ff02::1:fff00::a	ICMPv6	86	Neighbor Solicitation for 2001:10::a from 0c::7f:fe:b2:00:00
1489	12967.683435	2001:10::a	fe80::e7f:feff:feb2::ff02::1:fff00::a	ICMPv6	86	Neighbor Advertisement 2001:10::a (sol, cur) is at 00:50:79:66:68:00
1490	12967.688895	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xa3ff, seq=1, hop limit=58 (request in 1487)
1491	12968.688837	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xa3ff, seq=2, hop limit=64 (reply in 1492)
1492	12968.697646	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xa3ff, seq=2, hop limit=58 (request in 1491)
1493	12969.375225	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1494	12969.700348	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xa3ff, seq=3, hop limit=64 (reply in 1495)
1495	12969.709925	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xa3ff, seq=3, hop limit=58 (request in 1494)
1496	12970.713862	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xa3ff, seq=4, hop limit=64 (reply in 1497)
1497	12970.731719	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xa3ff, seq=4, hop limit=58 (request in 1496)
1498	12971.736194	2001:10::a	2001:11::a	ICMPv6	118	Echo (ping) request id=0xa3ff, seq=5, hop limit=64 (reply in 1499)
1499	12971.748109	2001:11::a	2001:10::a	ICMPv6	118	Echo (ping) reply id=0xa3ff, seq=5, hop limit=58 (request in 1498)
1500	12971.756116	fe80::e7f:feff:feb2::ff02::9	ff02::9	RIPng	186	Command Response, Version 1
1501	12973.822845	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1502	12973.825837	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1503	12973.828775	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1504	12973.830665	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1505	12973.832429	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1506	12973.837624	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1507	12973.839713	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1508	12973.842475	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1509	12973.845412	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1510	12973.848675	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1511	12973.851358	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1512	12973.854248	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1513	12973.856889	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1514	12973.860794	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1515	12973.872800	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1516	12973.875383	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1517	12973.881883	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1518	12973.884921	2001:10::a	2001:10::a	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1519	12973.892843	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1520	12973.895928	2001:10::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
1521	12973.899803	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1522	12973.904749	2001:10::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
1523	12973.910340	2001:10::a	2001:11::a	UDP	126	44240 → 44241 Len=64
1524	12973.914174	2001:11::a	2001:10::a	ICMPv6	174	Destination Unreachable (Port unreachable)
1525	12973.915748	10.0.10.1	224.0.0.9	RIPv2	146	Response
1526	12975.381972	fe80::e7f:feff:feb2::ff02::5	ff02::5	OSPF	90	Hello Packet
1527	12979.376459	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1528	12985.384713	fe80::e7f:feff:feb2::ff02::5	ff02::5	OSPF	90	Hello Packet
1529	12989.377272	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
1530	12991.382273	fe80::e7f:feff:feb2::ff02::9	ff02::9	RIPng	186	Command Response, Version 1
1531	12995.389226	fe80::e7f:feff:feb2::ff02::5	ff02::5	OSPF	90	Hello Packet

Frame 15: 86 bytes on wire (688 bits) 86 bytes captured (688 bits) on interface 0 - 14.8

Рисунок 86: Анализ трафика

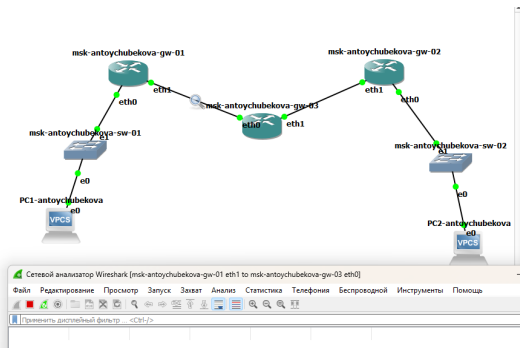
5.93 Настройка динамической маршрутизации в сетях IPv4 и IPv6

No.	Time	Source	Destination	Protocol	Length	Info
1554	13070.755889	fe80::c7f:f6ff:feb2::	ff02::1:9	RIPng	186	Command Response, Version 1
1555	13075.410627	fe80::c7f:f6ff:feb2::	ff02::1:5	OSPF	90	Hello Packet
1556	13079.420540	10.0.10.1	224.0.0.5	OSPF	70	Hello Packet
1557	13085.432094	fe80::c7f:f6ff:feb2::	ff02::1:5	OSPF	90	Hello Packet
1558	13089.415239	10.0.10.1	224.0.0.5	OSPF	70	Hello Packet
1559	13089.789851	fe80::c7f:f6ff:feb2::	ff02::1:9	RIPng	186	Command Response, Version 1
1560	13093.347028	10.0.10.1	224.0.0.9	RIPv2	146	Response
1561	13094.221592	2001:10:10::	2001:11:10::	ICMPv6	118	Echo (ping) request id=0x2200, seq=1, hop limit=64 (reply in 1562)
1562	13094.240948	2001:11:10::	2001:10:10::	ICMPv6	118	Echo (ping) reply id=0x2200, seq=1, hop limit=50 (request in 1561)
1563	13095.243028	2001:10:10::	2001:11:10::	ICMPv6	118	Echo (ping) request id=0x2200, seq=2, hop limit=64 (reply in 1564)
1564	13095.250823	2001:11:10::	2001:10:10::	ICMPv6	118	Echo (ping) reply id=0x2200, seq=2, hop limit=50 (request in 1563)
1565	13095.426088	fe80::c7f:f6ff:feb2::	ff02::1:5	OSPF	90	Hello Packet
1566	13096.260817	2001:10:10::	2001:11:10::	ICMPv6	118	Echo (ping) request id=0x2200, seq=3, hop limit=64 (reply in 1567)
1567	13096.271588	2001:11:10::	2001:10:10::	ICMPv6	118	Echo (ping) reply id=0x2200, seq=3, hop limit=50 (request in 1566)
1568	13097.274590	2001:10:10::	2001:11:10::	ICMPv6	118	Echo (ping) request id=0x2200, seq=4, hop limit=64 (reply in 1569)
1569	13097.286961	2001:11:10::	2001:10:10::	ICMPv6	118	Echo (ping) reply id=0x2200, seq=4, hop limit=50 (request in 1568)
1570	13098.208099	2001:10:10::	2001:11:10::	ICMPv6	118	Echo (ping) request id=0x2200, seq=5, hop limit=64 (reply in 1571)
1571	13098.301518	2001:11:10::	2001:10:10::	ICMPv6	118	Echo (ping) reply id=0x2200, seq=5, hop limit=50 (request in 1570)
1572	13099.289038	fe80::c7f:f6ff:feb2::	2001:10:10::	ICMPv6	86	Neighbor Solicitation for 2001:10:10:: from 8c:7f:f6:b2:00:00
1573	13099.417493	10.0.10.1	224.0.0.5	OSPF	70	Hello Packet
1574	13099.511196	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1575	13099.516823	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1576	13099.519584	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1577	13099.521674	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1578	13099.541091	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1579	13099.543053	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1580	13099.545105	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1581	13099.545474	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1582	13099.555592	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1583	13099.561422	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1584	13099.564282	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1585	13099.572279	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1586	13099.579211	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1587	13099.586297	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1588	13099.597946	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1589	13099.607077	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1590	13099.611143	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1591	13099.617940	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1592	13099.619280	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1593	13099.624167	2001:10:10::	2001:10:10::	ICMPv6	174	Time Exceeded (hop limit exceeded in transit)
1594	13099.644636	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1595	13099.657446	2001:10:10::	2001:10:10::	ICMPv6	174	Destination Unreachable (Port unreachable)
1596	13099.658418	2001:10:10::	2001:11:10::	UDP	126	50016 → 50017 len=64
1597	13100.000000	2001:10:10::	2001:10:10::	ICMPv6	174	Destination Unreachable (Host unreachable)
1598	13100.003149	fe80::c7f:f6ff:feb2::	2001:10:10::	ICMPv6	86	Neighbor Solicitation for 2001:10:10:: from 8c:7f:f6:b2:00:00
1599	13101.347017	fe80::c7f:f6ff:feb2::	2001:10:10::	ICMPv6	86	Neighbor Solicitation for 2001:10:10:: from 8c:7f:f6:b2:00:00
1600	13105.437066	fe80::c7f:f6ff:feb2::	ff02::1:5	OSPF	90	Hello Packet
1601	13109.420641	10.0.10.1	224.0.0.5	OSPF	70	Hello Packet
1602	13115.439586	fe80::c7f:f6ff:feb2::	ff02::1:5	OSPF	90	Hello Packet
1603	13119.424584	10.0.10.1	224.0.0.5	OSPF	70	Hello Packet

Рисунок 87: Анализ трафика

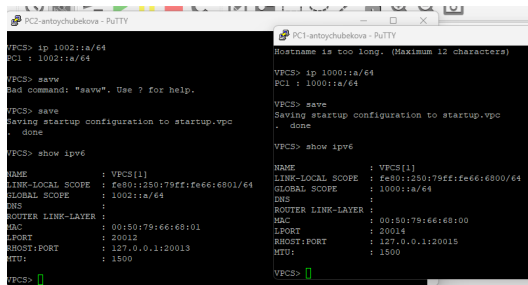
5.94 Построение туннеля IPv6–IPv4

Создадим новый проект в GNS3. В рабочем пространстве разместим и соединим устройства в соответствии с топологией, приведенной в лабораторной работе. Используем маршрутизаторы VyOS. Изменим отображаемые названия устройств. На соединении между первым и третьим маршрутизаторами подключим анализатор трафика.



5.95 Построение туннеля IPv6–IPv4

Присвойте адреса оконечным устройствам PC1 и PC2



The image shows two side-by-side PuTTY terminal windows. The left window is titled 'PC2-antoychubekova - PuTTY' and the right window is titled 'PC1-antoychubekova - PuTTY'. Both windows show a VPCS prompt. In the PC2 window, the user has entered 'ip 1002::a/64', 'PC1 : 1002::a/64', 'savv' (which results in an error 'Bad command: "savv". Use ? for help.'), 'save' (which results in 'Saving startup configuration to startup.vpc' and 'done'), and 'show ipv6'. The output of 'show ipv6' shows the configuration for VPCS[1] with a link-local scope of fe80::250:79ff:fe66:6801/64, a global scope of 1002::a/64, and a router link-layer of 00:50:79:66:68:01. The right window shows the user entering 'ip 1000::a/64', 'PC1 : 1000::a/64', 'save' (resulting in 'Saving startup configuration to startup.vpc' and 'done'), and 'show ipv6'. The output of 'show ipv6' shows the configuration for VPCS[1] with a link-local scope of fe80::250:79ff:fe66:6800/64, a global scope of 1000::a/64, and a router link-layer of 00:50:79:66:68:00.

```
PC2-antoychubekova - PuTTY
VPCS> ip 1002::a/64
PC1 : 1002::a/64

VPCS> savv
Bad command: "savv". Use ? for help.

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPORT               : 20012
RHOST:PORT          : 127.0.0.1:20013
MTU                 : 1500

VPCS>

PC1-antoychubekova - PuTTY
Hostname is too long. (Maximum 12 characters)

VPCS> ip 1000::a/64
PC1 : 1000::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1000::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPORT               : 20014
RHOST:PORT          : 127.0.0.1:20015
MTU                 : 1500

VPCS>
```

Рисунок 89: Присвоение адресов

5.96 Построение туннеля IPv6–IPv4

Устройство	Интерфейс	Адрес	Шлюз по умолчанию
R1	eth0	1000::1/64	–
R1	eth1	10.0.0.1/8	–
R1	Tunnel0	1001::1/64	–
R2	eth0	1002::1/64	–
R2	eth1	20.0.0.2/8	–
R2	Tunnel0	1001::2/64	–
R3	eth0	10.0.0.2/8	–
R3	eth1	20.0.0.1/8	–
PC1	–	1000::a/64	1000::1
PC2	–	1002::a/64	1002::1

Рисунок 90: Таблица адресации

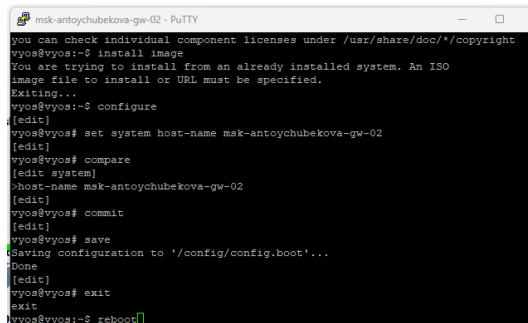
5.97 Построение туннеля IPv6–IPv4

Установим систему на маршрутизаторы VyOS. На маршрутизаторах перейдем в режим конфигурирования, изменим имя каждого устройства.

```
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-antoychubekova-gw-01
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-antoychubekova-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
```

Рисунок 91: Изменение имени устройства

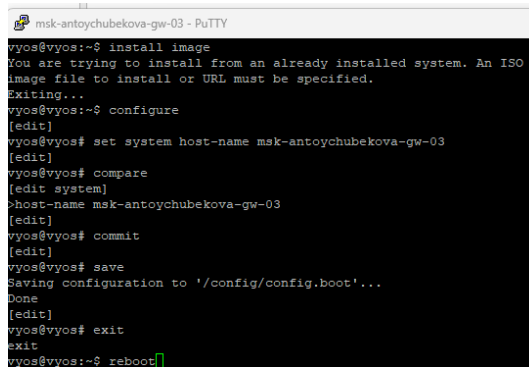
5.98 Построение туннеля IPv6–IPv4



```
msh-antoychubekova-gw-02 - PuTTY
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-antoychubekova-gw-02
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-antoychubekova-gw-02
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
```

Рисунок 92: Изменение имени устройства

5.99 Построение туннеля IPv6–IPv4



```
msh-antoychubekova-gw-03 - PuTTY
vyos@vyos:~$ install image
You are trying to install from an already installed system. An ISO
image file to install or URL must be specified.
Exiting...
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-antoychubekova-gw-03
[edit]
vyos@vyos# compare
[edit system]
>host-name msk-antoychubekova-gw-03
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos# exit
exit
vyos@vyos:~$ reboot
```

Рисунок 93: Изменение имени устройства

5.100 Построение туннеля IPv6–IPv4

Настроим адреса на интерфейсах маршрутизаторов.
msk-antoychubekova-gw-01.

```
vyos@msk-antoychubekova-gw-01:~$ configure
[edit]
vyos@msk-antoychubekova-gw-01# set interfaces ethernet eth0 address 1000::1/64
[edit]
vyos@msk-antoychubekova-gw-01# set interfaces ethernet eth1 address 10.0.0.1/8
[edit]
^C
[edit]
000::/64-antoychubekova-gw-01# set service router-advert interface eth0 prefix 1
[edit]
vyos@msk-antoychubekova-gw-01# commit
[edit]
vyos@msk-antoychubekova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-01#
```

Рисунок 94: Настройка адресов

5.101 Построение туннеля IPv6–IPv4

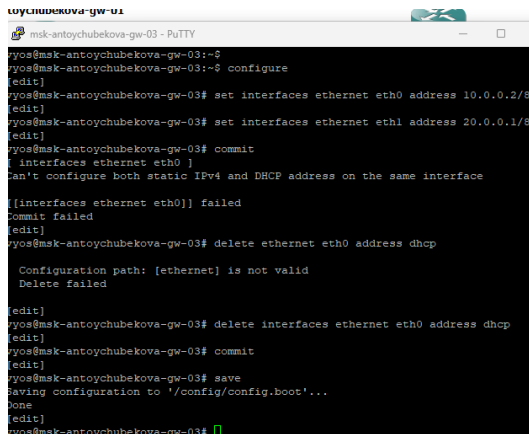
msk-antoychubekova-gw-02.

```
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@msk-antoychubekova-gw-02:~$ configure
[edit]
vyos@msk-antoychubekova-gw-02# set interfaces ethernet eth0 address 1002::1/64
[edit]
vyos@msk-antoychubekova-gw-02# set interfaces ethernet eth1 address 20.0.0.2/8
[edit]
vyos@msk-antoychubekova-gw-02# set service router-advert interface eth0 prefix 1
002::/64
[edit]
vyos@msk-antoychubekova-gw-02# commit
[edit]
vyos@msk-antoychubekova-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-02#
```

Рисунок 95: Настройка адресов

5.102 Построение туннеля IPv6–IPv4

msk-antoychubekova-gw-03.



```
msk-antoychubekova-gw-03 - PuTTY
msk-antoychubekova-gw-03:~$
msk-antoychubekova-gw-03:~$ configure
[edit]
msk-antoychubekova-gw-03# set interfaces ethernet eth0 address 10.0.0.2/8
[edit]
msk-antoychubekova-gw-03# set interfaces ethernet eth1 address 20.0.0.1/8
[edit]
msk-antoychubekova-gw-03# commit
[ interfaces ethernet eth0 ]
Can't configure both static IPv4 and DHCP address on the same interface

[[interfaces ethernet eth0]] failed
Commit failed
[edit]
msk-antoychubekova-gw-03# delete ethernet eth0 address dhcp

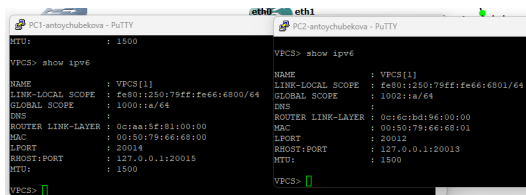
Configuration path: [ethernet] is not valid
Delete failed

[edit]
msk-antoychubekova-gw-03# delete interfaces ethernet eth0 address dhcp
[edit]
msk-antoychubekova-gw-03# commit
[edit]
msk-antoychubekova-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
msk-antoychubekova-gw-03#
```

Рисунок 96: Настройка адресов

5.103 Построение туннеля IPv6–IPv4

Убедимся, что на PC появились адреса ближайших к ним маршрутизаторов.



The image shows two PuTTY terminal windows side-by-side. The left window is titled 'PC1-antoychubekova - PuTTY' and shows the output of the 'show ipv6' command on a VPCS interface. The right window is titled 'PC2-antoychubekova - PuTTY' and shows the output of the 'show ipv6' command on a VPCS interface. Both windows show the same configuration details for the VPCS[1] interface.

```
PC1-antoychubekova - PuTTY
VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 1002::a/64
DNS                  :
ROUTER LINK-LAYER    : 0c:aa:5f:81:00:00
MAC                  : 00:50:79:66:68:00
LPORT                : 20014
RHOST:PORT           : 127.0.0.1:20015
MTU                  : 1500
VPCS>
```

```
PC2-antoychubekova - PuTTY
VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 1002::a/64
DNS                  :
ROUTER LINK-LAYER    : 0c:6c:bd:96:00:00
MAC                  : 00:50:79:66:68:01
LPORT                : 20012
RHOST:PORT           : 127.0.0.1:20013
MTU                  : 1500
VPCS>
```

Рисунок 97: Адреса ближних маршрутизаторов

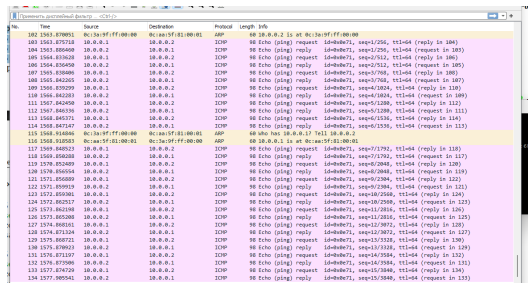
5.104 Построение туннеля IPv6–IPv4

Проверим маршруты с маршрутизатора R1. Мы видим, что пингуется только первый адрес.

```
vyos@msk-antoychubekova-gw-01:~$  
vyos@msk-antoychubekova-gw-01:~$ ping 10.0.0.2  
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.  
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=59.0 ms  
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=5.85 ms  
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=10.9 ms  
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=4.49 ms  
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=6.10 ms  
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=3.75 ms  
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=3.06 ms  
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=7.09 ms  
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=6.86 ms  
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=4.70 ms  
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=4.77 ms  
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=6.57 ms  
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=3.84 ms  
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=4.27 ms  
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=35.2 ms  
^C  
--- 10.0.0.2 ping statistics ---  
15 packets transmitted, 15 received, 0% packet loss, time 61ms  
rtt min/avg/max/mdev = 3.063/11.102/59.039/14.901 ms  
vyos@msk-antoychubekova-gw-01:~$ ping 20.0.0.1  
connect: Network is unreachable  
vyos@msk-antoychubekova-gw-01:~$ ping 20.0.0.2  
connect: Network is unreachable  
vyos@msk-antoychubekova-gw-01:~$
```

5.105 Построение туннеля IPv6–IPv4

Просмотрим анализ трафика.



No.	Time	Source	Destination	Protocol	Length	Info
102	1563.870951	0c:3a:5f:ff:ff:00:00	0c:3a:5f:81:00:01	ARP	60	10.0.0.2 is at 0c:3a:5f:ff:ff:00:00
103	1563.875718	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=1/256, ttl=64 (reply in 104)
104	1563.880409	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=1/256, ttl=64 (request in 103)
105	1564.813629	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=2/512, ttl=64 (reply in 106)
106	1564.820450	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=2/512, ttl=64 (request in 105)
107	1565.830406	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=3/768, ttl=64 (reply in 108)
108	1565.842205	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=3/768, ttl=64 (request in 107)
109	1566.850299	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=4/1024, ttl=64 (reply in 110)
110	1566.862283	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=4/1024, ttl=64 (request in 109)
111	1567.862450	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=5/1280, ttl=64 (reply in 112)
112	1567.866139	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=5/1280, ttl=64 (request in 111)
113	1568.845371	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=6/1536, ttl=64 (reply in 114)
114	1568.847147	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=6/1536, ttl=64 (request in 113)
115	1568.914886	0c:3a:5f:ff:ff:00:00	0c:3a:5f:81:00:01	ARP	60	Who has 10.0.0.1? Tell 10.0.0.2
116	1568.915553	0c:3a:5f:81:00:01	0c:3a:5f:ff:ff:00:00	ARP	60	10.0.0.1 is at 0c:3a:5f:81:00:01
117	1569.848523	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=7/1792, ttl=64 (reply in 118)
118	1569.850288	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=7/1792, ttl=64 (request in 117)
119	1570.852409	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=8/2048, ttl=64 (reply in 120)
120	1570.856554	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=8/2048, ttl=64 (request in 119)
121	1571.856809	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=9/2304, ttl=64 (reply in 122)
122	1571.859919	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=9/2304, ttl=64 (request in 121)
123	1572.859361	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=10/2560, ttl=64 (reply in 124)
124	1572.862517	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=10/2560, ttl=64 (request in 123)
125	1573.862198	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=11/2816, ttl=64 (reply in 126)
126	1573.865208	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=11/2816, ttl=64 (request in 125)
127	1574.868181	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=12/3072, ttl=64 (reply in 128)
128	1574.873324	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=12/3072, ttl=64 (request in 127)
129	1575.880711	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=13/3328, ttl=64 (reply in 129)
130	1575.879923	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=13/3328, ttl=64 (request in 129)
131	1576.871197	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=14/3584, ttl=64 (reply in 132)
132	1576.873586	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=14/3584, ttl=64 (request in 131)
133	1577.874729	10.0.0.1	10.0.0.2	ICMP	60	Echo (ping) request id=0x0e71, seq=15/3840, ttl=64 (reply in 134)
134	1577.880541	10.0.0.2	10.0.0.1	ICMP	60	Echo (ping) reply id=0x0e71, seq=15/3840, ttl=64 (request in 133)

Рисунок 99: Анализ трафика

Рисунок 100: Анализ трафика

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	::	ff02::1	IGMPv6	138	Multicast Listener Report Message v2
2	0.005120	::	ff02::1:ff01:1	IGMPv6	86	Neighbor Solicitation for fe80::aaa:5fff::fe81:1
3	0.308250	::	ff02::16	IGMPv6	138	Multicast Listener Report Message v2
4	1.087927	fe80::aaa:5fff::fe81::	ff02::16	IGMPv6	150	Multicast Listener Report Message v2
5	1.103071	fe80::aaa:5fff::fe81::	ff02::16	IGMPv6	90	Multicast Listener Report Message v2
6	1.320009	fe80::aaa:5fff::fe81::	ff02::16	IGMPv6	150	Multicast Listener Report Message v2
7	1.937037	fe80::aaa:5fff::fe81::	ff02::16	IGMPv6	90	Multicast Listener Report Message v2
8	0.100866	::	ff02::16	IGMPv6	138	Multicast Listener Report Message v2
9	11.335417	::	ff02::1	IGMPv6	138	Multicast Listener Report Message v2
10	11.335420	::	ff02::1:ffff:0	IGMPv6	86	Neighbor Solicitation for fe80::a3a:5fff::feff:0
11	12.455133	fe80::a3a:5fff::feff::	ff02::16	IGMPv6	150	Multicast Listener Report Message v2
12	12.460579	fe80::a3a:5fff::feff::	ff02::16	IGMPv6	90	Multicast Listener Report Message v2
13	13.344845	fe80::a3a:5fff::feff::	ff02::16	IGMPv6	150	Multicast Listener Report Message v2
14	13.417853	fe80::a3a:5fff::feff::	ff02::16	IGMPv6	90	Multicast Listener Report Message v2
15	18.857283	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x95a5ed23
16	30.668845	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
17	32.310662	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
18	33.511523	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
19	95.075011	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x34660b14
20	70.067905	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
21	71.131776	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
22	72.170727	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
23	131.284572	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x9e05dc2e
24	131.284518	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x9e05dc2e
25	116.542000	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
26	117.570946	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
27	117.635680	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x9e05dc2e
28	118.506321	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
29	124.463860	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x9e05dc2e
30	142.640832	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
31	144.333689	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1
32	145.105733	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x9e05dc2e
33	146.137943	0c::a5:f81:00:01	Broadcast	ARP	60	Who has 10.0.2.2? Tell 10.0.0.1

5.107 Построение туннеля IPv6–IPv4

Настроим маршрутизацию IPv4.
msk-antoychubekova-gw-01.

```
vyos@msk-antoychubekova-gw-01:~$ configure
[edit]
vyos@msk-antoychubekova-gw-01# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-antoychubekova-gw-01# commit
[edit]
vyos@msk-antoychubekova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-01#
```

Рисунок 101: Настройка маршрутизации IPv4

5.108 Построение туннеля IPv6–IPv4

msk-antoychubekova-gw-02.

```
vyos@msk-antoychubekova-gw-02:~$ configure
[edit]
vyos@msk-antoychubekova-gw-02# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-antoychubekova-gw-02# commit
[edit]
vyos@msk-antoychubekova-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-02# █
```

Рисунок 102: Настройка маршрутизации IPv4

5.109 Построение туннеля IPv6–IPv4

msk-antoychubekova-gw-03.

```
[edit]
vyos@msk-antoychubekova-gw-03# set protocols rip network 10.0.0.0/8
[edit]
vyos@msk-antoychubekova-gw-03# set protocols rip network 20.0.0.0/8
[edit]
vyos@msk-antoychubekova-gw-03# commit
[edit]
vyos@msk-antoychubekova-gw-03# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-03# █
```

Рисунок 103: Настройка маршрутизации IPv4

5.110 Построение туннеля IPv6–IPv4

Проверим маршруты. Мы видим, что все адреса пингуются.

```
vyos@msk-antoychubekova-gw-01# ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=26.8 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=5.21 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=3.86 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=5.17 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=3.86 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=4.88 ms
^C
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 19ms
rtt min/avg/max/mdev = 3.856/8.300/26.829/8.305 ms
[edit]
vyos@msk-antoychubekova-gw-01# ping 20.0.0.1
PING 20.0.0.1 (20.0.0.1) 56(84) bytes of data.
64 bytes from 20.0.0.1: icmp_seq=1 ttl=64 time=4.90 ms
64 bytes from 20.0.0.1: icmp_seq=2 ttl=64 time=4.01 ms
64 bytes from 20.0.0.1: icmp_seq=3 ttl=64 time=5.91 ms
64 bytes from 20.0.0.1: icmp_seq=4 ttl=64 time=6.63 ms
64 bytes from 20.0.0.1: icmp_seq=5 ttl=64 time=10.2 ms
^C
--- 20.0.0.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 15ms
rtt min/avg/max/mdev = 4.008/6.325/10.178/2.121 ms
[edit]
vyos@msk-antoychubekova-gw-01# ping 20.0.0.2
PING 20.0.0.2 (20.0.0.2) 56(84) bytes of data.
64 bytes from 20.0.0.2: icmp_seq=1 ttl=63 time=85.6 ms
64 bytes from 20.0.0.2: icmp_seq=2 ttl=63 time=6.57 ms
64 bytes from 20.0.0.2: icmp_seq=3 ttl=63 time=8.67 ms
64 bytes from 20.0.0.2: icmp_seq=4 ttl=63 time=13.4 ms
```

5.111 Построение туннеля IPv6–IPv4

Просмотрим анализ трафика.

No.	Time	Source	Destination	Protocol	Length	Info
228	15781.686556	10.0.0.1	224.0.0.9	RIPv2	66	Request
229	15781.975672	10.0.0.1	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
230	16344.176717	10.0.0.2	224.0.0.9	RIPv2	66	Request
231	16344.182091	10.0.0.2	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
232	16344.952336	10.0.0.2	224.0.0.22	IGMPv3	60	Membership Report / Join group 224.0.0.9 for any sources
233	16377.457156	10.0.0.2	224.0.0.9	RIPv2	66	Response
234	16485.484508	10.0.0.2	224.0.0.9	RIPv2	66	Response
235	16429.494535	10.0.0.2	224.0.0.9	RIPv2	66	Response
236	16464.598720	10.0.0.2	224.0.0.9	RIPv2	66	Response
237	16496.512863	10.0.0.2	224.0.0.9	RIPv2	66	Response
238	16526.526517	10.0.0.2	224.0.0.9	RIPv2	66	Response
239	16558.535579	10.0.0.2	224.0.0.9	RIPv2	66	Response
240	16588.543846	10.0.0.2	224.0.0.9	RIPv2	66	Response
241	16613.553773	10.0.0.2	224.0.0.9	RIPv2	66	Response
242	16643.572295	10.0.0.2	224.0.0.9	RIPv2	66	Response
243	16673.582289	10.0.0.2	224.0.0.9	RIPv2	66	Response
244	16684.685532	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request 10.0.0.1 → 10.0.0.2 (reply in 245)
245	16684.580818	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply 10.0.0.2 → 10.0.0.1 (request in 244)
246	16685.479816	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request 10.0.0.1 → 10.0.0.2 (reply in 247)
247	16685.483200	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply 10.0.0.2 → 10.0.0.1 (request in 246)
248	16686.488494	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request 10.0.0.1 → 10.0.0.2 (reply in 249)
249	16686.482791	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply 10.0.0.2 → 10.0.0.1 (request in 248)
250	16687.484150	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request 10.0.0.1 → 10.0.0.2 (reply in 251)
251	16687.487442	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply 10.0.0.2 → 10.0.0.1 (request in 250)
252	16688.486138	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request 10.0.0.1 → 10.0.0.2 (reply in 253)
253	16688.483205	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply 10.0.0.2 → 10.0.0.1 (request in 252)
254	16689.488649	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request 10.0.0.1 → 10.0.0.2 (reply in 255)
255	16689.492039	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply 10.0.0.2 → 10.0.0.1 (request in 254)
256	16689.670279	0c:aa:5f:ff:00:00	0c:aa:5f:ff:00:00	ARP	60	Who has 10.0.0.1? Tell 10.0.0.2
257	16689.674606	0c:aa:5f:ff:00:00	0c:aa:5f:ff:00:00	ARP	60	10.0.0.1 is at 0c:aa:5f:ff:00:00
258	16689.798297	0c:aa:5f:ff:00:00	0c:aa:5f:ff:00:00	ARP	60	Who has 10.0.0.2? Tell 10.0.0.1
259	16689.793673	0c:aa:5f:ff:00:00	0c:aa:5f:ff:00:00	ARP	60	10.0.0.2 is at 0c:aa:5f:ff:00:00
260	16699.767608	10.0.0.1	224.0.0.1	ICMP	98	Echo (ping) request 10.0.0.1 → 224.0.0.1 (reply in 261)

Рисунок 105: Анализ трафика

5.112 Построение туннеля IPv6–IPv4

No.	Time	Source	Destination	Protocol	Length	Info
256	16689.678279	0c:3a:9f:ff:00:00	0c:3a:9f:ff:00:00	ARP	60	00 Who has 10.0.0.1? Tell 10.0.0.2
257	16689.674606	0c:3a:9f:ff:00:00	0c:3a:9f:ff:00:00	ARP	60	00 10.0.0.1 is at 0c:3a:9f:ff:00:00
258	16689.798297	0c:3a:9f:ff:00:00	0c:3a:9f:ff:00:00	ARP	60	00 Who has 10.0.0.2? Tell 10.0.0.1
259	16689.793675	0c:3a:9f:ff:00:00	0c:3a:9f:ff:00:00	ARP	60	00 10.0.0.2 is at 0c:3a:9f:ff:00:00
260	16689.767668	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) request id=0x28c8, seq=1/256, ttl=64 (reply in 261)
261	16689.770713	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c8, seq=1/256, ttl=64 (request in 260)
262	16700.595584	10.0.0.2	224.0.0.9	RIPv2	66	Response
263	16700.769954	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) request id=0x28c8, seq=2/512, ttl=64 (reply in 264)
264	16700.772125	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c8, seq=2/512, ttl=64 (request in 263)
265	16701.772138	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) request id=0x28c8, seq=3/768, ttl=64 (reply in 266)
266	16701.776338	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c8, seq=3/768, ttl=64 (request in 265)
267	16702.774288	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) request id=0x28c8, seq=4/1024, ttl=64 (reply in 268)
268	16702.777040	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c8, seq=4/1024, ttl=64 (request in 267)
269	16703.777352	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) request id=0x28c8, seq=5/1280, ttl=64 (reply in 270)
270	16703.780035	10.0.0.1	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c8, seq=5/1280, ttl=64 (request in 269)
271	16713.405028	10.0.0.1	10.0.0.2	ICMP	98	08 Echo (ping) request id=0x28c9, seq=1/256, ttl=64 (reply in 272)
272	16713.407051	10.0.0.2	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c9, seq=1/256, ttl=63 (request in 271)
273	16714.406401	10.0.0.1	10.0.0.2	ICMP	98	08 Echo (ping) request id=0x28c9, seq=2/512, ttl=64 (reply in 274)
274	16714.411323	10.0.0.2	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c9, seq=2/512, ttl=63 (request in 273)
275	16715.409042	10.0.0.1	10.0.0.2	ICMP	98	08 Echo (ping) request id=0x28c9, seq=3/768, ttl=64 (reply in 276)
276	16715.415964	10.0.0.2	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c9, seq=3/768, ttl=63 (request in 275)
277	16716.415376	10.0.0.1	10.0.0.2	ICMP	98	08 Echo (ping) request id=0x28c9, seq=4/1024, ttl=64 (reply in 278)
278	16716.424869	10.0.0.2	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c9, seq=4/1024, ttl=63 (request in 277)
279	16717.418910	10.0.0.1	10.0.0.2	ICMP	98	08 Echo (ping) request id=0x28c9, seq=5/1280, ttl=64 (reply in 280)
280	16717.428078	10.0.0.2	10.0.0.1	ICMP	98	08 Echo (ping) reply id=0x28c9, seq=5/1280, ttl=63 (request in 279)
281	16718.837804	0c:3a:9f:ff:00:00	0c:3a:9f:ff:00:00	ARP	60	00 Who has 10.0.0.1? Tell 10.0.0.2
282	16718.837400	0c:3a:9f:ff:00:00	0c:3a:9f:ff:00:00	ARP	60	00 10.0.0.1 is at 0c:3a:9f:ff:00:00
283	16724.603159	10.0.0.2	224.0.0.9	RIPv2	66	Response
284	16754.615252	10.0.0.2	224.0.0.9	RIPv2	66	Response
285	16778.629562	10.0.0.2	224.0.0.9	RIPv2	66	Response
286	16805.634901	10.0.0.2	224.0.0.9	RIPv2	66	Response
287	16833.656589	10.0.0.2	224.0.0.9	RIPv2	66	Response
288	16861.669085	10.0.0.2	224.0.0.9	RIPv2	66	Response

Рисунок 106: Анализ трафика

5.113 Построение туннеля IPv6–IPv4

Создадим туннель IPv6 через сеть IPv4.
msk-antoychubekova-gw-01.

```
[edit]
vyos@msk-antoychubekova-gw-01# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-antoychubekova-gw-01# set interfaces tunnel tun0 source-address 10.0.0.
1
[edit]
vyos@msk-antoychubekova-gw-01# set interfaces tunnel tun0 remote 20.0.0.2
[edit]
vyos@msk-antoychubekova-gw-01# set interfaces tunnel tun0 address 1001::1/64
[edit]
vyos@msk-antoychubekova-gw-01# commit
[edit]
vyos@msk-antoychubekova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-01#
```

Рисунок 107: Создание туннеля IPv6 через сеть IPv4

5.114 Построение туннеля IPv6–IPv4

msk-antoychubekova-gw-02.

```
vyos@msk-antoychubekova-gw-02# set interfaces tunnel tun0 source-address 20.0.0.2
[edit]
vyos@msk-antoychubekova-gw-02# set interfaces tunnel tun0 encapsulation sit
[edit]
vyos@msk-antoychubekova-gw-02# set interfaces tunnel tun0 remote 10.0.0.1
[edit]
vyos@msk-antoychubekova-gw-02# set interfaces tunnel tun0 address 1001::2/64
[edit]
vyos@msk-antoychubekova-gw-02# commit
[edit]
vyos@msk-antoychubekova-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-02#
```

Рисунок 108: Создание туннеля IPv6 через сеть IPv4

5.115 Построение туннеля IPv6–IPv4

Настроим статическую маршрутизацию IPv6.
msk-antoychubekova-gw-01.

```
vyos@msk-antoychubekova-gw-01# set protocols static route6 1002::0/64 next-hop 1
001::2
[edit]
vyos@msk-antoychubekova-gw-01# commit
[edit]
vyos@msk-antoychubekova-gw-01# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-01#
```

Рисунок 109: Настройка статичной маршрутизации IPv6

5.116 Построение туннеля IPv6–IPv4

msk-antoychubekova-gw-02.

```
[edit]
vyos@msk-antoychubekova-gw-02# set protocols static route6 1000::0/64 next-hop 1
001::1
[edit]
vyos@msk-antoychubekova-gw-02# commit
[edit]
vyos@msk-antoychubekova-gw-02# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@msk-antoychubekova-gw-02#
```

Рисунок 110: Настройка статичной маршрутизации IPv6

5.117 Построение туннеля IPv6–IPv4

Проверим доступность конечных устройств. Мы видим, что все корректно обрабатывается.

```
VPCS> ping 1002::a

1002::a icmp6_seq=1 ttl=60 time=209.655 ms
1002::a icmp6_seq=2 ttl=60 time=15.693 ms
1002::a icmp6_seq=3 ttl=60 time=17.691 ms
1002::a icmp6_seq=4 ttl=60 time=23.744 ms
1002::a icmp6_seq=5 ttl=60 time=23.878 ms

VPCS> trace 1002::a

trace to 1002::a, 64 hops max
 1 1000::1  22.954 ms  3.641 ms  3.809 ms
 2 1001::2  32.166 ms  14.858 ms  16.861 ms
 3 1002::a  28.959 ms  25.756 ms  12.470 ms

VPCS> 
```

Рисунок 111: Отправка запроса с PC1 к PC2

5.118 Построение туннеля IPv6–IPv4

```
VPCS> ping 1000::a

1000::a icmp6_seq=1 ttl=60 time=22.529 ms
1000::a icmp6_seq=2 ttl=60 time=15.069 ms
1000::a icmp6_seq=3 ttl=60 time=26.041 ms
1000::a icmp6_seq=4 ttl=60 time=56.882 ms
1000::a icmp6_seq=5 ttl=60 time=60.004 ms

VPCS> trace 1000::a

trace to 1000::a, 64 hops max
 1 1002::1  15.421 ms  12.209 ms  4.178 ms
 2 1001::1  25.571 ms  16.710 ms  32.226 ms
 3 1000::a  34.680 ms  18.754 ms  18.203 ms

VPCS> 
```

Рисунок 112: Отправка запроса с PC1 к PC2

5.119 Построение туннеля IPv6–IPv4

Просмотрим анализ трафика.

Time	Source	Destination	Protocol	Length	Info
303	17319.794735	10.0.0.2	224.0.0.9	RIPv2	66 Response
304	17344.008122	10.0.0.2	224.0.0.9	RIPv2	66 Response
305	17376.055083	10.0.0.2	224.0.0.9	RIPv2	66 Response
306	17406.508005	10.0.0.2	224.0.0.9	RIPv2	66 Response
307	17437.902095	10.0.0.2	224.0.0.9	RIPv2	66 Response
308	17470.946817	10.0.0.2	224.0.0.9	RIPv2	66 Response
309	17493.987082	10.0.0.2	224.0.0.9	RIPv2	66 Response
310	17522.833323	10.0.0.2	224.0.0.9	RIPv2	66 Response
311	17548.040659	10.0.0.2	224.0.0.9	RIPv2	66 Response
312	17579.859675	10.0.0.2	224.0.0.9	RIPv2	66 Response
313	17610.065681	10.0.0.2	224.0.0.9	RIPv2	66 Response
314	17638.075354	10.0.0.2	224.0.0.9	RIPv2	66 Response
315	17662.081342	10.0.0.2	224.0.0.9	RIPv2	66 Response
316	17694.087995	10.0.0.2	224.0.0.9	RIPv2	66 Response
317	17726.096282	10.0.0.2	224.0.0.9	RIPv2	66 Response
318	17755.171544	10.0.0.2	224.0.0.9	RIPv2	66 Response
319	17778.242383	10.0.0.2	224.0.0.9	RIPv2	66 Response
320	17807.250189	10.0.0.2	224.0.0.9	RIPv2	66 Response
321	17824.364484	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0xc3ba, seq=1, hop limit=63 (reply in 322)
322	17824.486063	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0xc3ba, seq=1, hop limit=61 (request in 321)
323	17825.540118	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0xc3ba, seq=2, hop limit=63 (reply in 324)
324	17825.547983	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0xc3ba, seq=2, hop limit=61 (request in 323)
325	17826.556084	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0xc3ba, seq=3, hop limit=63 (reply in 326)
326	17826.566796	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0xc3ba, seq=3, hop limit=61 (request in 325)
327	17827.574459	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0xc3ba, seq=4, hop limit=63 (reply in 328)
328	17827.591188	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0xc3ba, seq=4, hop limit=61 (request in 327)
329	17828.603112	1000::a	1002::a	ICMPv6	138 Echo (ping) request id=0xc3ba, seq=5, hop limit=63 (reply in 330)
330	17828.615145	1002::a	1000::a	ICMPv6	138 Echo (ping) reply id=0xc3ba, seq=5, hop limit=61 (request in 329)
331	17829.498597	0c1a:5f:f1:00:01	0c1a:5f:f1:00:00	ARP	60 who has 10.0.0.2? Tell 10.0.0.1
332	17829.504247	0c1a:5f:f1:00:00	0c1a:5f:f1:00:01	ARP	60 10.0.0.2 is at 0c1a:5f:f1:00:01
333	17829.505967	0c1a:5f:f1:00:00	0c1a:5f:f1:00:01	ARP	60 who has 10.0.0.2? Tell 10.0.0.2
334	17829.609358	0c1a:5f:f1:00:01	0c1a:5f:f1:00:00	ARP	60 10.0.0.1 is at 0c1a:5f:f1:00:00
335	17836.118700	1000::a	1002::a	UDP	146 52097 → 52098 Len=64
336	17836.143341	1001::2	1000::a	ICMPv6	104 Time Exceeded (hop limit exceeded in transit)
337	17836.153173	1000::a	1001::2	ICMPv6	146 52097 → 52098 Len=64
338	17836.160204	1001::2	1000::a	ICMPv6	104 Time Exceeded (hop limit exceeded in transit)
339	17836.168223	1000::a	1001::2	ICMPv6	146 52097 → 52098 Len=64
340	17836.176320	1001::2	1000::a	ICMPv6	104 Time Exceeded (hop limit exceeded in transit)
341	17836.184822	1000::a	1001::2	ICMPv6	146 52097 → 52098 Len=64
342	17836.200602	1002::a	1000::a	ICMPv6	104 Destination Unreachable (Port unreachable)
343	17836.217211	1000::a	1002::a	ICMPv6	146 52097 → 52098 Len=64
344	17836.237113	1002::a	1000::a	ICMPv6	104 Destination Unreachable (Port unreachable)
345	17836.242449	1000::a	1002::a	ICMPv6	146 52097 → 52098 Len=64
346	17836.250166	10.0.0.2	224.0.0.9	RIPv2	104 Destination Unreachable (Port unreachable)
347	17837.202530	10.0.0.2	224.0.0.9	RIPv2	66 Response
348	17867.278245	10.0.0.2	224.0.0.9	RIPv2	66 Response

Рисунок 113: Анализ трафика

5.120 Построение туннеля IPv6–IPv4

349	17867.489961	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xeeba, seq=1, hop limit=63 (reply in 350)
350	17867.419170	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xeeba, seq=1, hop limit=61 (request in 349)
351	17868.436549	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xeeba, seq=2, hop limit=63 (reply in 352)
352	17868.441348	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xeeba, seq=2, hop limit=61 (request in 351)
353	17869.456814	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xeeba, seq=3, hop limit=63 (reply in 354)
354	17869.466181	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xeeba, seq=3, hop limit=61 (request in 353)
355	17870.491001	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xeeba, seq=4, hop limit=63 (reply in 356)
356	17870.511620	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xeeba, seq=4, hop limit=61 (request in 355)
357	17871.551418	1002::a	1000::a	ICMPv6	138 Echo (ping) request id=0xeeba, seq=5, hop limit=63 (reply in 358)
358	17871.585488	1000::a	1002::a	ICMPv6	138 Echo (ping) reply id=0xeeba, seq=5, hop limit=61 (request in 357)
359	17872.503417	0c:aa:5f:01:00:01	0c:3a:9f:ff:00:00	ARP	60 who has 10.0.0.2? Tell 10.0.0.1
360	17872.506806	0c:3a:9f:ff:00:00	0c:aa:5f:01:00:01	ARP	60 10.0.0.2 is at 0c:3a:9f:ff:00:00
361	17872.873774	0c:3a:9f:ff:00:00	0c:aa:5f:01:00:01	ARP	60 who has 10.0.0.2? Tell 10.0.0.2
362	17872.879330	0c:aa:5f:01:00:01	0c:3a:9f:ff:00:00	ARP	60 10.0.0.1 is at 0c:aa:5f:01:00:01
363	17879.809555	1002::a	1000::a	UDP	146 52097 → 52098 Len=64
364	17879.809701	1000::a	1002::a	ICMPv6	194 Time exceeded (hop limit exceeded in transit)
365	17879.120551	1002::a	1000::a	UDP	146 52097 → 52098 Len=64
366	17879.123870	1001::1	1002::a	ICMPv6	194 Time exceeded (hop limit exceeded in transit)
367	17879.152070	1002::a	1000::a	UDP	146 52097 → 52098 Len=64
368	17879.155402	1001::1	1002::a	ICMPv6	194 Time exceeded (hop limit exceeded in transit)
369	17879.175628	1002::a	1000::a	UDP	146 52097 → 52098 Len=64
370	17879.197267	1000::a	1002::a	ICMPv6	194 Destination Unreachable (Port unreachable)
371	17879.209371	1002::a	1000::a	UDP	146 52097 → 52098 Len=64
372	17879.214113	1000::a	1002::a	ICMPv6	194 Destination Unreachable (Port unreachable)
373	17879.230553	1002::a	1000::a	UDP	146 52097 → 52098 Len=64
374	17879.237130	1000::a	1002::a	ICMPv6	194 Destination Unreachable (Port unreachable)
375	17900.282060	10.0.0.2	224.0.0.9	RIPv2	66 Response
376	17927.288732	10.0.0.2	224.0.0.9	RIPv2	66 Response
377	17953.303144	10.0.0.2	224.0.0.9	RIPv2	66 Response
378	17981.322921	10.0.0.2	224.0.0.9	RIPv2	66 Response
379	18006.332704	10.0.0.2	224.0.0.9	RIPv2	66 Response

▼ Frame 337: 146 bytes on wire (1168 bits), 146 bytes captured (1168 bits) on interface -, id 0

Рисунок 114: Анализ трафика

5.121 Построение туннеля IPv6–IPv4

Пакеты, проходящие по туннелю, имеют одновременно поля IPv4 и IPv6.

The screenshot displays the Wireshark interface with a packet capture list on the left and a detailed view of a selected packet (packet 332) on the right. The packet list shows various network events, including Ethernet II, Internet Protocol Version 4, and Internet Protocol Version 6 packets. The packet details pane on the right provides a hierarchical view of the selected packet's structure. It starts with an Ethernet II header, followed by an Internet Protocol Version 4 header (source: 10.0.0.1, destination: 10.0.0.2), and then an Internet Protocol Version 6 header (source: ::1, destination: ::2). The payload is an ICMP Echo (ping) request.

Рисунок 115: Анализ трафика

Рисунок 116: Анализ трафика



5.123 Задание для самостоятельного выполнения

У нас задана топология сети, адреса сегментов сети, адреса, назначаемые интерфейсам устройств

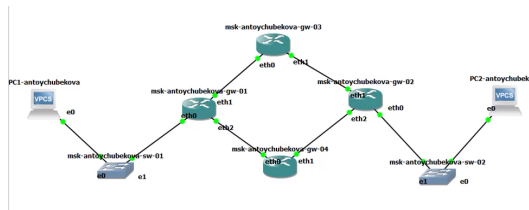


Рисунок 117: Топология сети

5.124 Задание для самостоятельного выполнения

Устройства	Сеть IPv4	Сеть IPv6
PC1 – gw-01	10.10.1.96/27	2001:db8:1:1::/64
PC2 – gw-02	10.10.1.64/28	2001:db8:1:6::/64
gw-01 – gw-03	10.10.1.4/30	2001:db8:1:2::/64
gw-02 – gw-04	10.10.1.8/30	2001:db8:1:3::/64
gw-03 – gw-02	10.10.1.16/30	2001:db8:1:4::/64
gw-04 – gw-02	10.10.1.32/30	2001:db8:1:5::/64

Рисунок 118: Адреса сегментов сети

5.125 Задание для самостоятельного выполнения

Устройство	Интерфейс	Адрес IP/префикс	Шлюз по умолчанию	Следующее устройство
gw-01	eth0	10.10.1.97/27	n/a	PC1
gw-01	eth0	2001:db8:1:1::1/64	n/a	PC1
gw-01	eth1	10.10.1.5/30	n/a	gw-03
gw-01	eth1	2001:db8:1:2::1/64	n/a	gw-03
gw-01	eth2	10.10.1.9/30	n/a	gw-04
gw-01	eth2	2001:db8:1:3::1/64	n/a	gw-04
gw-02	eth0	10.10.1.65/28	n/a	PC2
gw-02	eth0	2001:db8:1:6::1/64	n/a	PC2
gw-02	eth1	10.10.1.18/30	n/a	gw-03
gw-02	eth1	2001:db8:1:4::1/64	n/a	gw-03
gw-02	eth2	10.10.1.33/30	n/a	gw-04
gw-02	eth2	2001:db8:1:5::1/64	n/a	gw-04
gw-03	eth0	10.10.1.6/30	n/a	gw-01
gw-03	eth0	2001:db8:1:2::2/64	n/a	gw-01
gw-03	eth1	10.10.1.17/30	n/a	gw-02
gw-03	eth1	2001:db8:1:4::2/64	n/a	gw-02
gw-04	eth0	10.10.1.10/30	n/a	gw-01
gw-04	eth0	2001:db8:1:3::2/64	n/a	gw-01
gw-04	eth1	10.10.1.34/30	n/a	gw-02
gw-04	eth1	2001:db8:1:5::2/64	n/a	gw-02
PC1	NIC	10.10.1.98/27	10.10.1.97	gw-01
PC1	NIC	2001:db8:1:1::2/64	n/a	gw-01
PC2	NIC	10.10.1.66/28	10.10.1.65	gw-02
PC2	NIC	2001:db8:1:6::2/64	n/a	gw-02

5.126 Задание для самостоятельного выполнения

Присвоим IPv4 и IPv6-адресов оконечному устройству PC1 в соответствии с таблицей

```
VPCS> ip 10.10.1.98/27 10.10.1.97
Checking for duplicate address...
VPCS : 10.10.1.98 255.255.255.224 gateway 10.10.1.97

VPCS> ip 2001:db8:1:1::2/64
PC1 : 2001:db8:1:1::2/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> █
```

Рисунок 120: IPv4 и IPv6-адресов оконечному устройству PC1

5.127 Задание для самостоятельного выполнения

Присвоим IPv4 и IPv6-адресов оконечному устройству PC2 в соответствии с таблицей

```
VPCS> ip 10.10.1.66/28 10.10.1.65
Checking for duplicate address...
VPCS : 10.10.1.66 255.255.255.240 gateway 10.10.1.65

VPCS> ip 2001:db8:1:6::2/64
PC1 : 2001:db8:1:6::2/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> █
```

Рисунок 121: IPv4 и IPv6-адресов оконечному устройству PC2

5.128 Задание для самостоятельного выполнения

Настроим IPv4-адреса на интерфейсах маршрутизатора msk-antoychubekova-gw-01.

```
frr# config terminal
frr(config)# hostname msk-antoychubekova-gw-01
msk-antoychubekova-gw-01(config)# interface eth0
msk-antoychubekova-gw-01(config-if)# ip address 10.10.1.97/27
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth1
msk-antoychubekova-gw-01(config-if)# ip address 10.10.1.5/30
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth2
msk-antoychubekova-gw-01(config-if)# ip address 10.10.1.9/30
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
```

Рисунок 122: Настройка IPv4-адреса на msk-antoychubekova-gw-01

5.129 Задание для самостоятельного выполнения

Настроим IPv4-адреса на интерфейсах маршрутизатора msk-antoychubekova-gw-02.

```
frr# configure terminal
frr(config)# hostname msk-antoychubekova-gw-02
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# ip address 10.10.1.65/28
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth1
msk-antoychubekova-gw-02(config-if)# ip address 10.10.1.18/30
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth2
msk-antoychubekova-gw-02(config-if)# ip address 10.10.1.33/30
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02# █
```

Рисунок 123: Настройка IPv4-адреса на msk-antoychubekova-gw-02

5.130 Задание для самостоятельного выполнения

Настроим IPv4-адреса на интерфейсах маршрутизатора msk-antoychubekova-gw-03.

```
frr# configure terminal
frr(config)# hostname msk-antoychubekova-gw-03
msk-antoychubekova-gw-03(config)# interface eth0
msk-antoychubekova-gw-03(config-if)# ip address 10.10.1.6/30
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth1
msk-antoychubekova-gw-03(config-if)# ip address 10.10.1.17/30
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 124: Настройка IPv4-адреса на msk-antoychubekova-gw-03

5.131 Задание для самостоятельного выполнения

Настроим IPv4-адреса на интерфейсах маршрутизатора msk-antoychubekova-gw-04.

```
frr# configure terminal
frr(config)# hostname msk-antoychubekova-gw-04
msk-antoychubekova-gw-04(config)# interface eth0
msk-antoychubekova-gw-04(config-if)# ip address 10.10.1.10/30
msk-antoychubekova-gw-04(config-if)# no shutdown
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# interface eth1
msk-antoychubekova-gw-04(config-if)# ip address 10.10.1.34/30
msk-antoychubekova-gw-04(config-if)# no shutdown
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 125: Настройка IPv4-адреса на msk-antoychubekova-gw-04

5.132 Задание для самостоятельного выполнения

Настроим IPv6-адреса на интерфейсах маршрутизатора msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# config terminal
msk-antoychubekova-gw-01(config)# ipv6 forwarding
msk-antoychubekova-gw-01(config)# interface eth0
msk-antoychubekova-gw-01(config-if)# ipv6 address 2001:db8:1:1::1/64
msk-antoychubekova-gw-01(config-if)# no ipv6 nd suppress-ra
msk-antoychubekova-gw-01(config-if)# ipv6 nd prefix 2001:db8:1:1::/64
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth1
msk-antoychubekova-gw-01(config-if)# ipv6 address 2001:db8:1:2::1/64
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth2
msk-antoychubekova-gw-01(config-if)# ipv6 address 2001:db8:1:3::1/64
msk-antoychubekova-gw-01(config-if)# no shutdown
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 126: Настройка IPv6-адреса на msk-antoychubekova-gw-01

5.133 Задание для самостоятельного выполнения

Настроим IPv6-адреса на интерфейсах маршрутизатора msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# ipv6 forwarding
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# ipv6 address 2001:db8:1:6::1/64
msk-antoychubekova-gw-02(config-if)# no ipv6 nd suppress-ra
msk-antoychubekova-gw-02(config-if)# ipv6 nd prefix 2001:db8:1:6::/64
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth1
msk-antoychubekova-gw-02(config-if)# ipv6 address 2001:db8:1:4::1/64
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth2
msk-antoychubekova-gw-02(config-if)# ipv6 address 2001:db8:1:5::1/64
msk-antoychubekova-gw-02(config-if)# no shutdown
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 127: Настройка IPv6-адреса на msk-antoychubekova-gw-02

5.134 Задание для самостоятельного выполнения

Настроим IPv6-адреса на интерфейсах маршрутизатора msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# ipv6 forwarding
msk-antoychubekova-gw-03(config)# interface eth0
msk-antoychubekova-gw-03(config-if)# ipv6 address 2001:db8:1:2::2/64
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth1
msk-antoychubekova-gw-03(config-if)# ipv6 address 2001:db8:1:4::2/64
msk-antoychubekova-gw-03(config-if)# no shutdown
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 128: Настройка IPv6-адреса на msk-antoychubekova-gw-03

5.135 Задание для самостоятельного выполнения

Настроим IPv6-адреса на интерфейсах маршрутизатора msk-antoychubekova-gw-04.

```
[OK]
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# ipv6 forwarding
msk-antoychubekova-gw-04(config)# interface eth0
msk-antoychubekova-gw-04(config-if)# ipv6 address 2001:db8:1:3::2/64
msk-antoychubekova-gw-04(config-if)# no shutdown
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# interface eth1
msk-antoychubekova-gw-04(config-if)# ipv6 address 2001:db8:1:5::2/64
msk-antoychubekova-gw-04(config-if)# no shutdown
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 129: Настройка IPv6-адреса на msk-antoychubekova-gw-04

5.136 Задание для самостоятельного выполнения

На маршрутизаторах настроим RIP для сетей IPv4.

msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# config terminal
msk-antoychubekova-gw-01(config)# router rip
msk-antoychubekova-gw-01(config-router)# version 2
msk-antoychubekova-gw-01(config-router)# network eth0
msk-antoychubekova-gw-01(config-router)# network eth1
msk-antoychubekova-gw-01(config-router)# network eth2
msk-antoychubekova-gw-01(config-router)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 130: Настройка RIP в msk-antoychubekova-gw-01

5.137 Задание для самостоятельного выполнения

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# router rip
msk-antoychubekova-gw-02(config-router)# version 2
msk-antoychubekova-gw-02(config-router)# network eth0
msk-antoychubekova-gw-02(config-router)# network eth1
msk-antoychubekova-gw-02(config-router)# network eth2
msk-antoychubekova-gw-02(config-router)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 131: Настройка RIP в msk-antoychubekova-gw-02

5.138 Задание для самостоятельного выполнения

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# router rip
msk-antoychubekova-gw-03(config-router)# version 2
msk-antoychubekova-gw-03(config-router)# network eth0
msk-antoychubekova-gw-03(config-router)# network eth1
msk-antoychubekova-gw-03(config-router)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 132: Настройка RIP в msk-antoychubekova-gw-03

5.139 Задание для самостоятельного выполнения

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# router rip
msk-antoychubekova-gw-04(config-router)# version 2
msk-antoychubekova-gw-04(config-router)# network eth0
msk-antoychubekova-gw-04(config-router)# network eth1
msk-antoychubekova-gw-04(config-router)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 133: Настройка RIP в msk-antoychubekova-gw-04

5.140 Задание для самостоятельного выполнения

Убедимся, что маршрутизация по RIP настроена.

```

mask-antoychubekova-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.10.1.16/30 [120/2] via 10.10.1.6, eth1, weight 1, 00:03:30
R>* 10.10.1.32/30 [120/2] via 10.10.1.10, eth2, weight 1, 00:02:19
R>* 10.10.1.64/28 [120/3] via 10.10.1.6, eth1, weight 1, 00:03:28
mask-antoychubekova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network        Next Hop        Metric From        Tag Time
C(i) 10.10.1.4/30    0.0.0.0          1 self             0
C(i) 10.10.1.8/30    0.0.0.0          1 self             0
R(n) 10.10.1.16/30   10.10.1.6        2 10.10.1.6         0 02:27
R(n) 10.10.1.32/30   10.10.1.10       2 10.10.1.10         0 02:29
R(n) 10.10.1.64/28   10.10.1.6        3 10.10.1.6         0 02:27
C(i) 10.10.1.96/27   0.0.0.0          1 self             0
mask-antoychubekova-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 14 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface        Send Recv  Key-chain
    eth0              2      2
    eth1              2      2
    eth2              2      2
  Routing for Networks:
    eth0,

```

5.141 Задание для самостоятельного выполнения

С PC1 пропингуем PC2. Пакет проходит через маршрутизатор msk-antoychubekova-gw-03.

```
VPCS> ping 10.10.1.66

84 bytes from 10.10.1.66 icmp_seq=1 ttl=61 time=86.858 ms
84 bytes from 10.10.1.66 icmp_seq=2 ttl=61 time=42.891 ms
84 bytes from 10.10.1.66 icmp_seq=3 ttl=61 time=18.608 ms
84 bytes from 10.10.1.66 icmp_seq=4 ttl=61 time=16.032 ms
84 bytes from 10.10.1.66 icmp_seq=5 ttl=61 time=16.968 ms

VPCS> trace 10.10.1.66 -P 6
trace to 10.10.1.66, 8 hops max (TCP), press Ctrl+C to stop
 1  10.10.1.97   13.622 ms   6.459 ms   3.115 ms
 2  10.10.1.6   36.379 ms  14.112 ms  17.345 ms
 3  10.10.1.18  39.707 ms  15.937 ms  19.319 ms
 4  10.10.1.66  16.210 ms  18.085 ms  10.885 ms

VPCS> █
```

Рисунок 135: Эхо-запрос от PC1 к PC2

5.142 Задание для самостоятельного выполнения

Отключим интерфейс msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure termina  
msk-antoychubekova-gw-03(config)# interface eth0  
msk-antoychubekova-gw-03(config-if)# shutdown  
msk-antoychubekova-gw-03(config-if)#
```

Рисунок 136: Отключение интерфейса

5.143 Задание для самостоятельного выполнения

Проверим метрики протокола RIP.

```

msk-antoychubekova-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.10.1.16/30 [120/2] via 10.10.1.6, eth1, weight 1, 00:08:33
R>* 10.10.1.32/30 [120/2] via 10.10.1.10, eth2, weight 1, 00:07:22
R>* 10.10.1.64/28 [120/3] via 10.10.1.6, eth1, weight 1, 00:08:31
msk-antoychubekova-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network        Next Hop        Metric From        Tag Time
C(i) 10.10.1.4/30    0.0.0.0          1 self             0
C(i) 10.10.1.8/30    0.0.0.0          1 self             0
R(n) 10.10.1.16/30   10.10.1.6        2 10.10.1.6         0 01:37
R(n) 10.10.1.32/30   10.10.1.10       2 10.10.1.10         0 02:39
R(n) 10.10.1.64/28   10.10.1.6        3 10.10.1.6         0 01:37
C(i) 10.10.1.96/27   0.0.0.0          1 self             0
msk-antoychubekova-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 11 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface        Send  Recv  Key-chain
    eth0              2    2
    eth1              2    2
    eth2              2    2
  Routing for Networks:
    eth0

```

5.144 Задание для самостоятельного выполнения

С PC1 пропинговали PC2. Теперь пакет проходит через маршрутизатор msk-antoychubekova-gw-04.

```
VPCS> ping 10.10.1.66

84 bytes from 10.10.1.66 icmp_seq=1 ttl=61 time=47.816 ms
84 bytes from 10.10.1.66 icmp_seq=2 ttl=61 time=30.434 ms
84 bytes from 10.10.1.66 icmp_seq=3 ttl=61 time=32.098 ms
84 bytes from 10.10.1.66 icmp_seq=4 ttl=61 time=48.561 ms
84 bytes from 10.10.1.66 icmp_seq=5 ttl=61 time=29.321 ms

VPCS> trace 10.10.1.66 -P 6
trace to 10.10.1.66, 8 hops max (TCP), press Ctrl+C to stop
 1  10.10.1.97   13.563 ms  3.367 ms  3.926 ms
 2  10.10.1.10   33.290 ms  8.692 ms  7.620 ms
 3  10.10.1.33   62.159 ms  15.387 ms 12.189 ms
 4  10.10.1.66   20.876 ms  11.717 ms 29.578 ms

VPCS> █
```

Рисунок 138: Эхо-запрос от PC1 к PC2

5.145 Задание для самостоятельного выполнения

На маршрутизаторах настроим RIPng для сетей IPv6.
msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# configure terminal
msk-antoychubekova-gw-01(config)# router ripng
msk-antoychubekova-gw-01(config-router)# network eth0
msk-antoychubekova-gw-01(config-router)# network eth1
msk-antoychubekova-gw-01(config-router)# network eth2
msk-antoychubekova-gw-01(config-router)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
% Unknown command: write memory
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 139: Настройка RIPng в msk-antoychubekova-gw-01

5.146 Задание для самостоятельного выполнения

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02 (config)# router ripng
msk-antoychubekova-gw-02 (config-router)# network eth0
msk-antoychubekova-gw-02 (config-router)# network eth1
msk-antoychubekova-gw-02 (config-router)# network eth2
msk-antoychubekova-gw-02 (config-router)# exit
msk-antoychubekova-gw-02 (config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 140: Настройка RIPng в msk-antoychubekova-gw-02

5.147 Задание для самостоятельного выполнения

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03(config)# router ripng
msk-antoychubekova-gw-03(config-router)# network eth0
msk-antoychubekova-gw-03(config-router)# network eth1
msk-antoychubekova-gw-03(config-router)# network eth2
msk-antoychubekova-gw-03(config-router)# no network eth2
msk-antoychubekova-gw-03(config-router)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 141: Настройка RIPng в msk-antoychubekova-gw-03

5.148 Задание для самостоятельного выполнения

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# router ripng
msk-antoychubekova-gw-04(config-router)# network eth0
msk-antoychubekova-gw-04(config-router)# network eth1
msk-antoychubekova-gw-04(config-router)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 142: Настройка RIPng в msk-antoychubekova-gw-04

5.149 Задание для самостоятельного выполнения

Проверим метрики протокола RIPvng.

```
msk-antoychubekova-gw-01# show ipv6 ripng
Codes: R - RIPvng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface, (a/S) - aggregated/Suppressed

  Network      Next Hop      Via      Metric Tag Time
C(i) 2001:db8:1:1::/64
      ::        self        1      0
C(i) 2001:db8:1:2::/64
      ::        self        1      0
C(i) 2001:db8:1:3::/64
      ::        self        1      0
R(n) 2001:db8:1:4::/64
      fe80::e13:7bff:feb0:0 eth1      2      0 02:25
R(n) 2001:db8:1:5::/64
      fe80::e11:aaff:fec7:0 eth2      2      0 02:43
R(n) 2001:db8:1:6::/64
      fe80::e13:7bff:feb0:0 eth1      3      0 02:43
msk-antoychubekova-gw-01#
```

Рисунок 143: Метрики RIPvng

5.150 Задание для самостоятельного выполнения

С PC1 пропинговали PC2. Пакет проходит через маршрутизатор msk-antoychubekova-gw-03.

```
VPCS> ping 2001:db8:1:6::2

2001:db8:1:6::2 icmp6_seq=1 ttl=58 time=62.463 ms
2001:db8:1:6::2 icmp6_seq=2 ttl=58 time=24.060 ms
2001:db8:1:6::2 icmp6_seq=3 ttl=58 time=15.261 ms
2001:db8:1:6::2 icmp6_seq=4 ttl=58 time=13.966 ms
2001:db8:1:6::2 icmp6_seq=5 ttl=58 time=28.518 ms

VPCS> trace 2001:db8:1:6::2

trace to 2001:db8:1:6::2, 64 hops max
 1 2001:db8:1:1::1    11.355 ms   3.372 ms   10.326 ms
 2 2001:db8:1:2::2    13.146 ms   15.304 ms   8.242 ms
 3 2001:db8:1:4::1    21.055 ms   15.649 ms   11.398 ms
 4 2001:db8:1:6::2    12.863 ms   15.398 ms   21.106 ms

VPCS> █
```

Рисунок 144: Эхо-запрос от PC1 к PC2

5.151 Задание для самостоятельного выполнения

Отключим интерфейс msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# interface eth0
msk-antoychubekova-gw-03(config-if)# shutdown
msk-antoychubekova-gw-03(config-if)#
```

Рисунок 145: Отключение интерфейса

5.152 Задание для самостоятельного выполнения

Проверим метрики протокола RIPng.

```
msk-antoychubekova-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

      Network      Next Hop      Via      Metric Tag Time
C(i) 2001:db8:1:1::/64
      ::          self        1      0
C(i) 2001:db8:1:2::/64
      ::          self        1      0
C(i) 2001:db8:1:3::/64
      ::          self        1      0
R(n) 2001:db8:1:4::/64
      fe80::e13:7bff:feb0:0 eth1      2      0 00:31
R(n) 2001:db8:1:5::/64
      fe80::e11:aaff:fec7:0 eth2      2      0 02:38
R(n) 2001:db8:1:6::/64
      fe80::e13:7bff:feb0:0 eth1      3      0 02:38
msk-antoychubekova-gw-01#
```

Рисунок 146: Метрики RIPng

5.153 Задание для самостоятельного выполнения

С PC1 пропингум PC2. Мы видим, что путь не восстановился.

```
VPCS> ping 2001:db8:1:6::2

2001:db8:1:6::2 icmp6_seq=1 timeout
2001:db8:1:6::2 icmp6_seq=2 timeout
2001:db8:1:6::2 icmp6_seq=3 timeout
2001:db8:1:6::2 icmp6_seq=4 timeout
2001:db8:1:6::2 icmp6_seq=5 timeout

VPCS> trace 2001:db8:1:6::2

trace to 2001:db8:1:6::2, 64 hops max
 1 2001:db8:1:1::1    9.162 ms  2.421 ms  2.331 ms
 2 * * *
 3 *2001:db8:1:1::1  100.482 ms (ICMP type:1, code:3, Address unreachable)

VPCS> 
```

Рисунок 147: Эхо-запрос от PC1 к PC2

5.154 Задание для самостоятельного выполнения

На маршрутизаторах настроили OSPFv2 для сетей IPv4.
msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# configure terminal
msk-antoychubekova-gw-01(config)# router ospf
msk-antoychubekova-gw-01(config-router)# network 10.10.1.64/28 area 0.0.0.0
msk-antoychubekova-gw-01(config-router)# network 10.10.1.4/28 area 0.0.0.0
msk-antoychubekova-gw-01(config-router)# network 10.10.1.8/30 area 0.0.0.0
msk-antoychubekova-gw-01(config-router)# network 10.10.1.4/30 area 0.0.0.0
msk-antoychubekova-gw-01(config-router)# no network 10.10.1.4/28 area 0.0.0.0
msk-antoychubekova-gw-01(config-router)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 148: Настройка OSPFv2 в msk-antoychubekova-gw-01

5.155 Задание для самостоятельного выполнения

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# router ospf
msk-antoychubekova-gw-02(config-router)# network 10.10.1.64/28 area 0.0.0.0
msk-antoychubekova-gw-02(config-router)# network 10.10.1.16/30 area 0.0.0.0
msk-antoychubekova-gw-02(config-router)# network 10.10.1.32/30 area 0.0.0.0
msk-antoychubekova-gw-02(config-router)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 149: Настройка OSPFv2 в msk-antoychubekova-gw-02

5.156 Задание для самостоятельного выполнения

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03(config)# router ospf
msk-antoychubekova-gw-03(config-router)# network 10.10.1.4/30 area 0.0.0.0
msk-antoychubekova-gw-03(config-router)# network 10.10.1.16/30 area 0.0.0.0
msk-antoychubekova-gw-03(config-router)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 150: Настройка OSPFv2 в msk-antoychubekova-gw-03

5.157 Задание для самостоятельного выполнения

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# router ospf
msk-antoychubekova-gw-04(config-router)# network 10.10.1.8/30 area 0.0.0.0
msk-antoychubekova-gw-04(config-router)# network 10.10.1.32/30 area 0.0.0.0
msk-antoychubekova-gw-04(config-router)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 151: Настройка OSPFv2 в msk-antoychubekova-gw-04

5.158 Задание для самостоятельного выполнения

С PC1 пропингум PC2. Пакет проходит через маршрутизатор msk-antoychubekova-gw-04.

```
VPCS> ping 10.10.1.6

34 bytes from 10.10.1.6 icmp_seq=1 ttl=63 time=27.944 ms
34 bytes from 10.10.1.6 icmp_seq=2 ttl=63 time=9.693 ms
34 bytes from 10.10.1.6 icmp_seq=3 ttl=63 time=10.378 ms
34 bytes from 10.10.1.6 icmp_seq=4 ttl=63 time=17.721 ms
34 bytes from 10.10.1.6 icmp_seq=5 ttl=63 time=13.141 ms

VPCS> trace 10.10.1.6 -P 6
Trace to 10.10.1.6, 8 hops max (TCP), press Ctrl+C to stop
 1  10.10.1.97   15.428 ms  5.282 ms  5.328 ms
 2  10.10.1.6   14.909 ms  23.971 ms  21.165 ms

VPCS> █
```

Рисунок 152: Эхо-запрос от PC1 к PC2

5.159 Задание для самостоятельного выполнения

Проверим таблицу маршрутизации протокола OSPFv2.

```
(OK)
msk-antoychubekova-gw-01# show ip ospf neighbor

Neighbor ID      Pri State          Dead Time Address        Interface
  RXmtL RqstL DBsmL
10.10.1.17        1 Full/Backup    37.929s 10.10.1.6        eth1:10.10.1.5
      0      0      0
10.10.1.34        1 Full/Backup    39.769s 10.10.1.10       eth2:10.10.1.9
      0      0      0

msk-antoychubekova-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.10.1.4/30      [100] area: 0.0.0.0
      directly attached to eth1
N   10.10.1.8/30      [100] area: 0.0.0.0
      directly attached to eth2
N   10.10.1.16/30     [200] area: 0.0.0.0
      via 10.10.1.6, eth1
N   10.10.1.32/30     [200] area: 0.0.0.0
      via 10.10.1.10, eth2
N   10.10.1.64/28     [300] area: 0.0.0.0
      via 10.10.1.6, eth1
      via 10.10.1.10, eth2

===== OSPF router routing table =====
===== OSPF external routing table =====

msk-antoychubekova-gw-01#
```

Рисунок 153: Таблица маршрутизации протокола OSPFv2

5.160 Задание для самостоятельного выполнения

Отключим интерфейс на msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04# configure terminal
msk-antoychubekova-gw-04(config)# interface eth0
msk-antoychubekova-gw-04(config-if)# shutdown
msk-antoychubekova-gw-04(config-if)#
```

Рисунок 154: Отключение интерфейса

5.161 Задание для самостоятельного выполнения

Проверим таблицу маршрутизации протокола OSPFv2.

```
msk-antoychubekova-gw-01# show ip ospf neighbor

Neighbor ID      Pri State          Dead Time Address      Interface
  RXmtL RqstL DBsmL
10.10.1.17        1 Full/Backup    32.061s 10.10.1.6     eth1:10.10.1.5
      0      0      0

msk-antoychubekova-gw-01# show ip ospf route

===== OSPF network routing table =====
N   10.10.1.4/30      [100] area: 0.0.0.0
    directly attached to eth1
N   10.10.1.8/30      [100] area: 0.0.0.0
    directly attached to eth2
N   10.10.1.16/30     [200] area: 0.0.0.0
    via 10.10.1.6, eth1
N   10.10.1.32/30     [300] area: 0.0.0.0
    via 10.10.1.6, eth1
N   10.10.1.64/28     [300] area: 0.0.0.0
    via 10.10.1.6, eth1

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-antoychubekova-gw-01#
msk-antoychubekova-gw-01#
```

Рисунок 155: Таблица маршрутизации протокола OSPFv2

5.162 Задание для самостоятельного выполнения

С PC1 пропинговали PC2. Теперь пакет проходит через маршрутизатор msk-antoychubekova-gw-03.

```
VPCS> ping 10.10.1.6

84 bytes from 10.10.1.6 icmp_seq=1 ttl=63 time=6.321 ms
84 bytes from 10.10.1.6 icmp_seq=2 ttl=63 time=11.492 ms
84 bytes from 10.10.1.6 icmp_seq=3 ttl=63 time=8.595 ms
84 bytes from 10.10.1.6 icmp_seq=4 ttl=63 time=8.544 ms
84 bytes from 10.10.1.6 icmp_seq=5 ttl=63 time=13.101 ms

VPCS> trace 10.10.1.6 -P 6
trace to 10.10.1.6, 8 hops max (TCP), press Ctrl+C to stop
 1  10.10.1.97   5.550 ms   3.910 ms   2.992 ms
 2  10.10.1.6    4.928 ms   11.135 ms   5.536 ms

VPCS> █
```

Рисунок 156: Эхо-запрос от PC1 к PC2

5.163 Задание для самостоятельного выполнения

На маршрутизаторах настроим OSPFv3 для сетей IPv6.

msk-antoychubekova-gw-01.

```
msk-antoychubekova-gw-01# configure terminal
msk-antoychubekova-gw-01(config)# router ospf6
msk-antoychubekova-gw-01(config-ospf6)# ospf6 router-id 1.1.1.1
msk-antoychubekova-gw-01(config-ospf6)# exit
msk-antoychubekova-gw-01(config)# interface eth0
msk-antoychubekova-gw-01(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth1
msk-antoychubekova-gw-01(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth3
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# interface eth2
msk-antoychubekova-gw-01(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-01(config-if)# exit
msk-antoychubekova-gw-01(config)# exit
msk-antoychubekova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-01#
```

Рисунок 157: Настройка OSPFv3 в msk-antoychubekova-gw-01

5.164 Задание для самостоятельного выполнения

msk-antoychubekova-gw-02.

```
msk-antoychubekova-gw-02# configure terminal
msk-antoychubekova-gw-02(config)# router ospf6
msk-antoychubekova-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-antoychubekova-gw-02(config-ospf6)# exit
msk-antoychubekova-gw-02(config)# interface eth0
msk-antoychubekova-gw-02(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth1
msk-antoychubekova-gw-02(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# interface eth2
msk-antoychubekova-gw-02(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-02(config-if)# exit
msk-antoychubekova-gw-02(config)# exit
msk-antoychubekova-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-02#
```

Рисунок 158: Настройка OSPFv3 в msk-antoychubekova-gw-02

5.165 Задание для самостоятельного выполнения

msk-antoychubekova-gw-03.

```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# router ospf6
msk-antoychubekova-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-antoychubekova-gw-03(config-ospf6)# exit
msk-antoychubekova-gw-03(config)# interface eth0
msk-antoychubekova-gw-03(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# interface eth1
msk-antoychubekova-gw-03(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-03(config-if)# exit
msk-antoychubekova-gw-03(config)# exit
msk-antoychubekova-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-03#
```

Рисунок 159: Настройка OSPFv3 в msk-antoychubekova-gw-03

5.166 Задание для самостоятельного выполнения

msk-antoychubekova-gw-04.

```
msk-antoychubekova-gw-04(config)# router ospf6
msk-antoychubekova-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-antoychubekova-gw-04(config-ospf6)# exit
msk-antoychubekova-gw-04(config)# interface eth0
msk-antoychubekova-gw-04(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# interface eth1
msk-antoychubekova-gw-04(config-if)# ipv6 ospf6 area 0
msk-antoychubekova-gw-04(config-if)# exit
msk-antoychubekova-gw-04(config)# exit
msk-antoychubekova-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-antoychubekova-gw-04#
```

Рисунок 160: Настройка OSPFv3 в msk-antoychubekova-gw-04

5.167 Задание для самостоятельного выполнения

С PC1 пропингуем PC2. Пакет проходит через маршрутизатор msk-antoychubekova-gw-03.

```
VPCS> ping 2001:db8:1:6::2

2001:db8:1:6::2 icmp6_seq=1 ttl=58 time=63.344 ms
2001:db8:1:6::2 icmp6_seq=2 ttl=58 time=21.117 ms
2001:db8:1:6::2 icmp6_seq=3 ttl=58 time=16.764 ms
2001:db8:1:6::2 icmp6_seq=4 ttl=58 time=20.927 ms
2001:db8:1:6::2 icmp6_seq=5 ttl=58 time=17.323 ms

VPCS> trace 2001:db8:1:6::2

trace to 2001:db8:1:6::2, 64 hops max
 1 2001:db8:1:1::1    27.992 ms   9.660 ms   4.503 ms
 2 2001:db8:1:3::2    12.377 ms  10.548 ms   7.915 ms
 3 2001:db8:1:5::1    18.749 ms  11.032 ms  21.692 ms
 4 2001:db8:1:6::2    35.325 ms  15.628 ms  15.676 ms

VPCS> 
```

Рисунок 161: Эхо-запрос от PC1 к PC2

5.168 Задание для самостоятельного выполнения

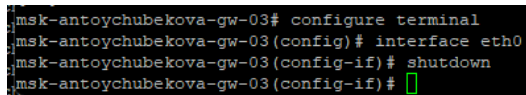
Проверим таблицу маршрутизации протокола OSPFv3.

```
[OK]
msk-antoychubekova-gw-01# show ipv6 ospf6 neighbor
Neighbor ID      Pri   DeadTime   State/IfState      Duration I/F[State]
3.3.3.3          1     00:00:38   Full/BDR           00:05:08 eth1[DR]
4.4.4.4          1     00:00:39   Full/BDR           00:02:08 eth2[DR]
msk-antoychubekova-gw-01# show ipv6 ospf6 route
*N IA 2001:db8:1:1::/64      ::                eth0 00:09:47
*N IA 2001:db8:1:2::/64      ::                eth1 00:05:15
*N IA 2001:db8:1:3::/64      ::                eth2 00:02:15
*N IA 2001:db8:1:4::/64      fe80::e13:7bff:feb0:0 eth1 00:05:00
*N IA 2001:db8:1:5::/64      fe80::e11:aaff:fec7:0 eth2 00:02:03
*N IA 2001:db8:1:6::/64      fe80::e13:7bff:feb0:0 eth1 00:02:03
msk-antoychubekova-gw-01#
```

Рисунок 162: Таблица маршрутизации протокола OSPFv3

5.169 Задание для самостоятельного выполнения

Отключим интерфейс msk-antoychubekova-gw-03.



```
msk-antoychubekova-gw-03# configure terminal
msk-antoychubekova-gw-03(config)# interface eth0
msk-antoychubekova-gw-03(config-if)# shutdown
msk-antoychubekova-gw-03(config-if)#
```

Рисунок 163: Отключение интерфейса

5.170 Задание для самостоятельного выполнения

Проверим таблицу маршрутизации протокола OSPFv3.

```
msh-antoychubekova-gw-01# show ipv6 ospf6 neighbor
Neighbor ID Pri DeadTime State/IfState Duration I/F[State]
4.4.4.4 1 00:00:37 Full/BDR 00:04:30 eth2[DR]
msh-antoychubekova-gw-01# show ipv6 ospf6 route
*N IA 2001:db8:1:1::/64 :: eth0 00:00:35
*N IA 2001:db8:1:2::/64 :: eth1 00:07:32
*N IA 2001:db8:1:3::/64 :: eth2 00:04:32
*N IA 2001:db8:1:4::/64 fe80::ell:aaff:fec7:0 eth2 00:01:14
*N IA 2001:db8:1:5::/64 fe80::ell:aaff:fec7:0 eth2 00:04:20
*N IA 2001:db8:1:6::/64 fe80::ell:aaff:fec7:0 eth2 00:01:14
msh-antoychubekova-gw-01#
```

Рисунок 164: Таблица маршрутизации протокола OSPFv3

5.171 Задание для самостоятельного выполнения

С PC1 пропингуем PC2. Теперь пакет проходит через маршрутизатор msk-antoychubekova-gw-04.

```
VPCS> ping 2001:db8:1:6::2

2001:db8:1:6::2 icmp6_seq=1 ttl=58 time=39.575 ms
2001:db8:1:6::2 icmp6_seq=2 ttl=58 time=53.129 ms
2001:db8:1:6::2 icmp6_seq=3 ttl=58 time=27.690 ms
2001:db8:1:6::2 icmp6_seq=4 ttl=58 time=17.316 ms
2001:db8:1:6::2 icmp6_seq=5 ttl=58 time=15.697 ms

VPCS> trace 2001:db8:1:6::2

trace to 2001:db8:1:6::2, 64 hops max
 1 2001:db8:1:1::1      6.750 ms   5.819 ms   2.889 ms
 2 2001:db8:1:3::2      9.859 ms   7.543 ms   8.188 ms
 3 2001:db8:1:5::1     12.054 ms   9.438 ms  18.603 ms
 4 2001:db8:1:6::2     16.785 ms  24.714 ms  19.176 ms

VPCS> 
```

Рисунок 165: Эхо-запрос от PC1 к PC2

Раздел 6

6. Выводы

6.1 Выводы

В ходе выполнения лабораторной работы №8 я изучила принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования.